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ASSIGNMENT 1: 2D FRAME CASE STUDY OF BOX- GIRDER BRIDGE

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Chapter 1

INTRODUCTION

1.1 Theory of Elasticity

The linear elastic analysis is the basis for formulating our frame structural system. A generic 3D continuum, as sketched in Figure 1.1, characterised by a volume V , a surface with imposed displacements S_u , and a surface with imposed forces S_σ , is referred to for the description of the system.

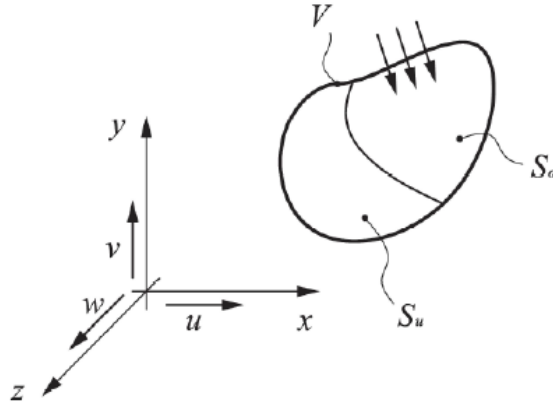


Figure 1.1: 3D continuum

The displacement components u , v , and w , as can be noted from the Figure, are dependent on the coordinates x , y , and z and are assumed to be small and continuous. This assumption enables the definition of linear compatibility equations relating the displacements to the strain components given as:

Normal strain components:

$$\varepsilon_x = \frac{\partial u}{\partial x}$$

$$\varepsilon_y = \frac{\partial v}{\partial y}$$

$$\varepsilon_z = \frac{\partial w}{\partial z}$$

Shear strain components:

$$\gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}$$

$$\gamma_{yz} = \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}$$

$$\gamma_{zx} = \frac{\partial w}{\partial x} + \frac{\partial u}{\partial z}$$

For symmetry, the shear components $\gamma_{ij} = \gamma_{ji}$, where $i, j = x, y, z$. The strains are thus stored in vectorial form as:

$$\boldsymbol{\varepsilon} = [\varepsilon_x \quad \varepsilon_y \quad \varepsilon_z \quad | \quad \gamma_{xy} \quad \gamma_{yz} \quad \gamma_{zx}]^T$$

In addition to this, imposed thermal strains also have to be accounted for in the analysis. For an isotropic material with a coefficient of thermal expansion α , subjected to a temperature differential of ΔT , the vector of imposed thermal strains is written as:

$$\boldsymbol{\varepsilon}_0 = \alpha \Delta T [1 \quad 1 \quad 1 \quad | \quad 0 \quad 0 \quad 0]^T$$

Consequently, the total strain in the continuum is expressed as:

$$\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}_e + \boldsymbol{\varepsilon}_0$$

Now, a cubic element of infinitesimal dimensions (refer to Figure 1.2) is extracted from the continuum for establishing the equilibrium relations. The element is subjected to body forces with intensity F_i in the three directions.

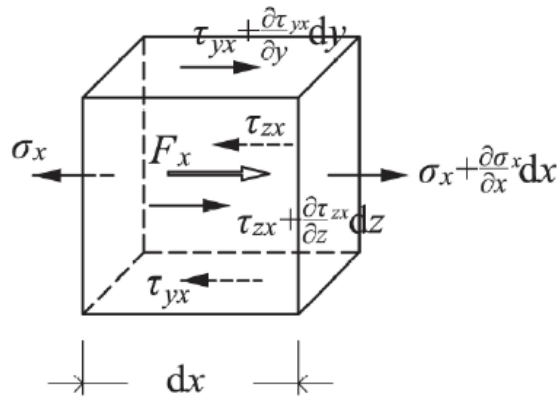


Figure 1.2: Infinitesimal cubic element for equilibrium

Equilibrating the forces along the three directions, three partial differential equations are obtained as:

$$\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} + F_x = 0$$

$$\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} + \frac{\partial \tau_{zy}}{\partial z} + F_y = 0$$

$$\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_z}{\partial z} + F_z = 0$$

Moving ahead on similar lines, an infinitesimal tetrahedral element, as shown in Figure 1.3, from the surface of the continuum is inspected for equilibrium so as to capture the boundary conditions of our system. The continuum is acted upon by surface forces of intensity f_i .

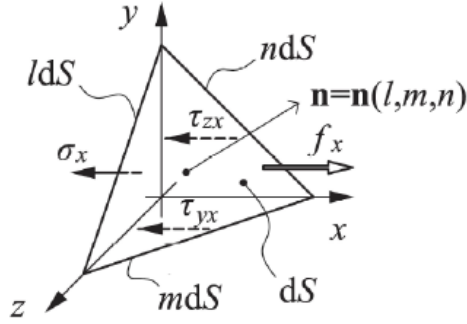


Figure 1.3: Infinitesimal tetrahedral element for equilibrium

From equilibrium in the three directions the boundary conditions in terms of the stresses, surface forces, and the direction cosines, l , m , and n , for an outward normal at the surface comes out as:

$$l\sigma_x + m\tau_{yx} + n\tau_{zx} = f_x$$

$$l\tau_{xy} + m\sigma_y + n\tau_{zy} = f_y$$

$$l\tau_{xz} + m\tau_{yz} + n\sigma_z = f_z$$

The constitutive laws correlating the stresses and the strains, under the assumption of a homogeneous, isotropic, and linear elastic material with the Young's Modulus E , and the Poisson ratio ν , are summarized as:

$$\varepsilon_x^e = \frac{1}{E} [\sigma_x - \nu(\sigma_y + \sigma_z)] \quad \gamma_{xy}^e = \frac{2(1+\nu)}{E} \tau_{xy}$$

$$\varepsilon_y^e = \frac{1}{E} [\sigma_y - \nu(\sigma_z + \sigma_x)] \quad \gamma_{yz}^e = \frac{2(1+\nu)}{E} \tau_{yz}$$

$$\varepsilon_z^e = \frac{1}{E} [\sigma_z - \nu(\sigma_x + \sigma_y)] \quad \gamma_{zx}^e = \frac{2(1+\nu)}{E} \tau_{zx}$$

The material flexibility matrix that defines the strains in terms of stresses is given as:

$$C = \begin{bmatrix} C_\varepsilon & 0 \\ 0 & C_\gamma \end{bmatrix} = \frac{1}{E} \begin{bmatrix} 1 & -\nu & -\nu & 0 & 0 & 0 \\ -\nu & 1 & -\nu & 0 & 0 & 0 \\ -\nu & -\nu & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2(1+\nu) & 0 & 0 \\ 0 & 0 & 0 & 0 & 2(1+\nu) & 0 \\ 0 & 0 & 0 & 0 & 0 & 2(1+\nu) \end{bmatrix}$$

Following this, the strains expressed through the constitutive law is given as:

$$\boldsymbol{\varepsilon}_e = \mathbf{C} \boldsymbol{\sigma}$$

The total strain, after accounting for the imposed strain as presented before can now be updated as:

$$\boldsymbol{\varepsilon} = \mathbf{C}\boldsymbol{\sigma} + \boldsymbol{\varepsilon}_0$$

The above relationship can now be inverted to formulate the stresses as a function of the strains with the introduction of the elastic stiffness matrix $\mathbf{D}$, which is the inverse of the flexibility matrix $\mathbf{C}$ reported above. This relation is given as:

$$\boldsymbol{\sigma} = \mathbf{D}(\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_0)$$

$$\mathbf{D} = \begin{bmatrix} D_\varepsilon & 0 \\ 0 & D_\gamma \end{bmatrix} = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & \nu & \nu & 0 & 0 & 0 \\ \nu & 1-\nu & \nu & 0 & 0 & 0 \\ \nu & \nu & 1-\nu & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1-2\nu}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1-2\nu}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1-2\nu}{2} \end{bmatrix}$$

Now, the formulation needs to be manipulated so as to capture the relations for beam elements that constitute the frame system that has to be analysed. Considering a predominant mono-dimensional behaviour of the beam element along the x direction, the stress and the strain vectors are reduced to:

$$\boldsymbol{\sigma} = [\sigma_x \quad \tau_{xy} \quad \tau_{xz}]^T$$

$$\boldsymbol{\varepsilon} = [\varepsilon_x \quad \gamma_{xy} \quad \gamma_{xz}]^T$$

The constitutive law in this case transforms as:

$$\mathbf{D} = E \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2(1+\nu)} & 0 \\ 0 & 0 & \frac{1}{2(1+\nu)} \end{bmatrix} = \begin{bmatrix} E & 0 & 0 \\ 0 & G & 0 \\ 0 & 0 & G \end{bmatrix}$$

In summation, for our 2D problem, there are 8 unknowns in terms of the stress, strains, and displacements with 8 equations expressed as equilibrium, compatibility, and constitutive laws. Thus, the system is statically determinate.

1.2 Principle of Virtual Work

The principle of virtual work is identified as a necessary and sufficient condition for the equilibrium of a rigid body. This formulation is developed by referring to a reference 2D deformable elastic continuum as illustrated in Figure 1.4.

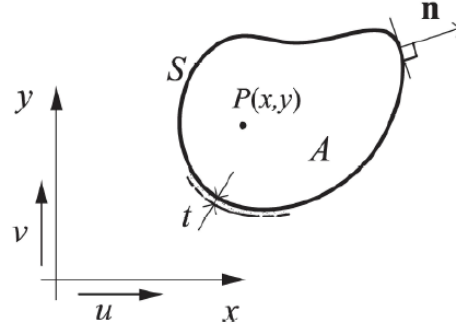


Figure 1.4: Reference 2D elastic continuum

Two admissible sets, the static and the kinematic, are identified as:

1. **Statically admissible set** (') comprises of a family of equilibrated volume forces $\mathbf{F}'$, surface forces $\mathbf{f}'$, and stresses $\boldsymbol{\sigma}'$. As a result, the equilibrium conditions are laid out as:

$$\begin{aligned}\sigma'_{ij,i} + F'_j &= 0 && \text{on } V \\ \sigma'_{ij} n_j &= f_i && \text{on } S_u\end{aligned}$$

2. **Kinematically admissible set** (") represents a family of compatible displacements $\mathbf{u}''$ and strains $\boldsymbol{\varepsilon}''$. Thus, the compatibility condition they fulfil is written as:

$$\begin{aligned}\varepsilon''_{ij} &= \frac{1}{2} (u''_{i,j} + u''_{j,i}) && \text{on } V \\ u''_{ij} &= u''_{0j} && \text{on } S_u\end{aligned}$$

The external work performed by the volume forces, the surface forces, and the reaction forces associated to the generated strains, and the internal work is evaluated as:

$$\begin{aligned}W_e &= \int_A \mathbf{u}''^T \mathbf{F} dA + \int_{S_\sigma} \mathbf{u}''^T \mathbf{f} dS + \int_{S_u} \mathbf{u}''^T \boldsymbol{\sigma}' \cdot \mathbf{n} dS \\ W_i &= \int_A \boldsymbol{\varepsilon}''^T \boldsymbol{\sigma}' dA\end{aligned}$$

Lastly, the principal of virtual work states that the internal work done by the stresses, $\boldsymbol{\sigma}'$, for the strains $\boldsymbol{\varepsilon}''$ is equal to the external work performed by the volume forces $\mathbf{F}'$, and the surface forces $\mathbf{f}'$ for the displacement $\mathbf{u}''$. Numerically, this is represented as:

$$\int_A \boldsymbol{\varepsilon}''^T \boldsymbol{\sigma}' dA = \int_A \mathbf{u}''^T \mathbf{F} dA + \int_S \mathbf{u}''^T \mathbf{f} dS$$

The consideration of static and kinematic admissible sets are the only conditions needed for the validity of the principle of virtual work. Moreover, they are arbitrary with no necessary relation to the actual stresses and strains in the body. Moreover, the principle of virtual work is independent of the constitutive law chosen.

1.3 Principle of Virtual Displacements

For developing the principle of virtual displacements two admissible sets, the static and the kinematic, are defined as:

1. **Statically admissible set** having actual stresses and forces acting on the body as $\mathbf{F}' = \mathbf{F}$ in A , $\mathbf{f}' = \mathbf{f}$ in S_σ , $\mathbf{r}' = \mathbf{r}$ in S_u , and $\boldsymbol{\sigma}' = \boldsymbol{\sigma}$.
2. **Kinematically admissible set** considering the variation of the actual displacements and the compatibility as $\mathbf{u}'' = \delta \mathbf{u}$ and $\boldsymbol{\varepsilon}'' = \delta \boldsymbol{\varepsilon}$ in A , and $\mathbf{u}'' = \delta \bar{\mathbf{u}} = 0$ in S_u .

Equating the external work done by the applied forces to the internal work we obtain:

$$\int_A \delta \boldsymbol{\varepsilon}^T \boldsymbol{\sigma} dA = \int_A \delta \mathbf{u}^T \mathbf{F} dA + \int_{S_\sigma} \delta \mathbf{u}^T \mathbf{f} dS + \int_{S_u} \delta \bar{\mathbf{u}}^T \mathbf{r} dS$$

Taking into account the null variation of the displacement on the constrained surface S_u , the principle of virtual displacement is now stated as:

$$\int_A \delta \boldsymbol{\varepsilon}^T \boldsymbol{\sigma} dA = \int_A \delta \mathbf{u}^T \mathbf{F} dA + \int_{S_\sigma} \delta \mathbf{u}^T \mathbf{f} dS$$

In closing, the stresses and the strains can be expressed through the constitutive laws and compatibility and the only unknowns of our elastic problem are the displacements $\mathbf{u}$.

$$\begin{aligned} \boldsymbol{\varepsilon} &= \mathbf{B} \mathbf{u} \\ \boldsymbol{\sigma} &= \mathbf{D}(\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_0) = \mathbf{D}(\mathbf{B} \mathbf{u} - \boldsymbol{\varepsilon}_0) \end{aligned}$$

1.4 Principle of Virtual Displacements

The 2D frame structure analysis revolves around the premise of beam elements. The Euler-Bernoulli beam theory is referenced to for a generic beam element, as presented in Figure 1.5 that is characterized with a linear distribution of strain.

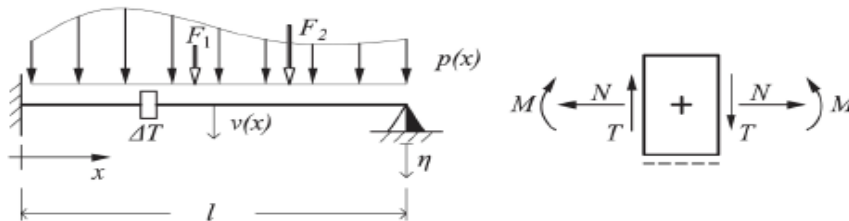
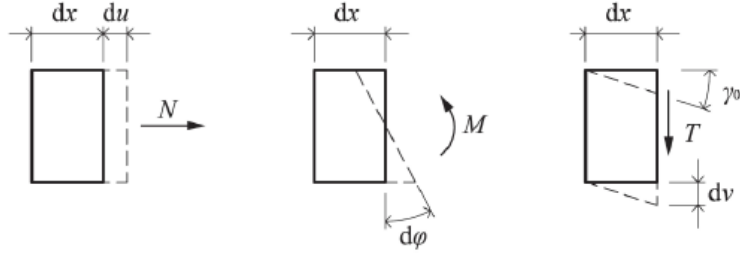


Figure 1.5: Generic beam element

The corresponding stress resultants and global strain are given as:



$$N = \int_A \sigma_x dA \quad M = - \int_A \sigma_x y dA \quad T = - \int_A \tau_{xy} dA$$

$$\varepsilon_a = \frac{du}{dx} \quad v = \frac{d\varphi}{dx} \quad \gamma_0 = - \frac{dv}{dx}$$

Vectorially, the stresses and the strains are assembled in the following way:

$$\boldsymbol{\sigma} = [N \quad T \quad M]^T$$

$$\boldsymbol{\varepsilon} = [\varepsilon_a \quad \gamma_0 \quad v]^T$$

Recalling the principle of virtual displacements stated in the previous section, the following expression is obtained:

$$\int_0^l (\delta \varepsilon_a N + \delta \gamma_0 T + \delta v M) dx = \int_0^l \delta v(x) p(x) dx + \sum_i \delta v_i$$

Lastly, employing the linear elastic constitutive laws and expressing $\boldsymbol{\sigma}$ in terms of $\boldsymbol{\varepsilon}$ the following relation is noted:

$$\begin{bmatrix} N \\ T \\ M \end{bmatrix} = \begin{bmatrix} EA & 0 & 0 \\ 0 & GA_\chi & 0 \\ 0 & 0 & EI \end{bmatrix} \cdot \begin{bmatrix} \varepsilon_a \\ \gamma_0 \\ v \end{bmatrix}$$

Where:

A_χ is the effective shear area given as $\frac{A}{\chi}$, and

χ is the shear correction factor obtained from energy – based formulation

1.5 Matrix Formulation

The matrix formulation of the principle of virtual displacements, based on the equilibrium of a general frame system is crucial for our analysis. In this framework, a generic frame system, as shown in Figure 1.6, is referred to, and the principal system is defined as:

- The degrees of kinematic indeterminacy are defined
- Several restraints equal to the degree of indeterminacy are introduced, so as to prevent the undetermined displacement
- Since the restraints introduce inconsistencies in the reaction forces, displacements correlative to the restraining forces are therefore imposed to restore equilibrium.

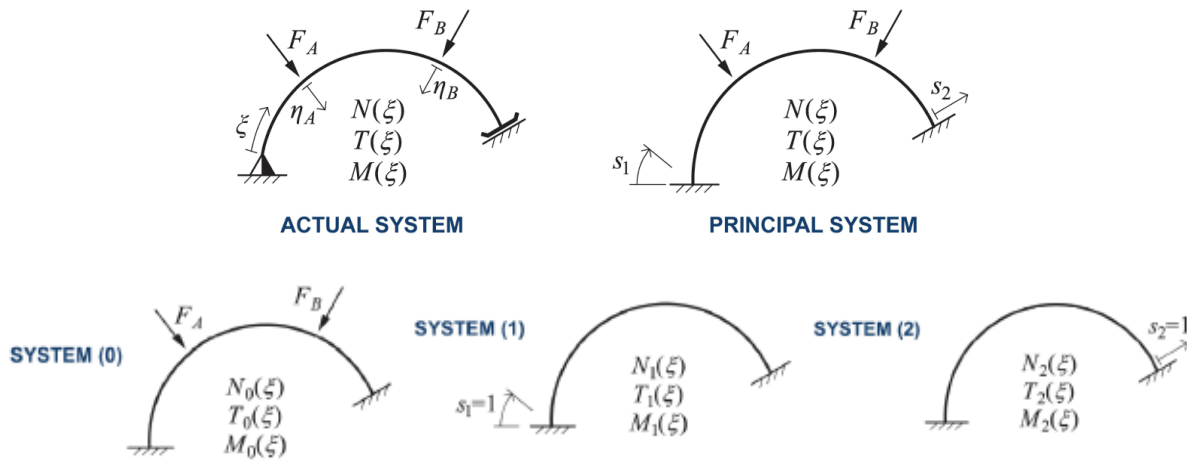


Figure 1.6: Reference frame system

As can be observed from Figure 1.6, the actual system is replaced with a principal one which is further decomposed into three systems exploiting the principle of superposition. System (0) is associated to the external applied forces while systems (1) and (2) take into account the undetermined displacements in accordance with the degree of kinematic indeterminacy. The superposition equations are thus described as:

$$N(\xi) = N_0(\xi) + N_1(\xi)s_1 + N_2(\xi)s_2$$

$$T(\xi) = T_0(\xi) + T_1(\xi)s_1 + T_2(\xi)s_2$$

$$M(\xi) = M_0(\xi) + M_1(\xi)s_1 + M_2(\xi)s_2$$

Now, depending on the number of undetermined displacements virtual systems (δs_1 and δs_2) are defined for the analysis as presented in Figure 1.7. These virtual systems are characterized with actual forces and stress resultants having virtual displacements and generalized strains.



Figure 1.7: Virtual system

Virtual System 1

$$\begin{aligned}\delta s_1 \neq 0 \quad \delta \eta_A &= \eta_{A1} \delta s_1 \\ \delta s_2 \neq 0 \quad \delta \eta_B &= \eta_{B1} \delta s_1\end{aligned}$$

The variation of the stress resultant about the virtual displacement is seen as:

$$\begin{cases} \delta N = N_1 \delta s_1 \\ \delta T = T_1 \delta s_1 \\ \delta M = M_1 \delta s_1 \end{cases}$$

The principle of virtual displacements, equating the external and the internal work, and subsequently utilizing the constitutive laws can be stated as:

$$\begin{aligned}f_1 \delta s_1 + F_A \delta \eta_A + F_B \delta \eta_B &= \int_l (N \delta \varepsilon_a + T \delta \gamma_0 + M \delta v) d\xi \\ f_1 \delta s_1 + F_A \delta \eta_{A1} \delta s_1 + F_B \delta \eta_{B1} \delta s_1 &= \int_l \left(N \frac{N_1}{EA} + T \chi \frac{T_1}{GA} + M \frac{M_1}{EI} \right) \delta s_1 d\xi\end{aligned}$$

Employing the principle of superposition and some mathematical manipulations the following expression is noted:

$$\begin{aligned}f_1 &= \int_l \left(\frac{N_1 N_0}{EA} + \chi \frac{T_1 T_0}{GA} + \frac{M_1 M_0}{EI} \right) d\xi - F_A \eta_{A1} - F_B \eta_{B1} + \int_l \left(\frac{N_1 N_1}{EA} + \chi \frac{T_1 T_1}{GA} + \frac{M_1 M_1}{EI} \right) d\xi s_1 \\ &\quad + \int_l \left(\frac{N_1 N_2}{EA} + \chi \frac{T_1 T_2}{GA} + \frac{M_1 M_2}{EI} \right) d\xi s_2\end{aligned}$$

The equilibrium equation can be represented in short-hand form as:

$$f_1 = f_{10} + k_{11} s_1 + k_{12} s_2$$

Where:

$$f_{10} = \int_l \left(\frac{N_1 N_0}{EA} + \chi \frac{T_1 T_0}{GA} + \frac{M_1 M_0}{EI} \right) d\xi - F_A \eta_{A1} - F_B \eta_{B1}$$

$$k_{11} = \int_l \left(\frac{N_1 N_1}{EA} + \chi \frac{T_1 T_1}{GA} + \frac{M_1 M_1}{EI} \right) d\xi$$

$$k_{12} = \int_l \left(\frac{N_1 N_2}{EA} + \chi \frac{T_1 T_2}{GA} + \frac{M_1 M_2}{EI} \right) d\xi$$

Virtual System 2

Following the same formulation as defined for the virtual system 1 and imposing $\delta s_2 \neq 0$, the equilibrium condition is obtained as:

$$f_2 = f_{20} + k_{21}s_1 + k_{22}s_2$$

General Matrix Formulation

For an arbitrary system, with n degrees of kinematic indeterminacy, the equilibrium equations transform as:

$$f_i - f_{i0} = \sum_{j=1}^n k_{ij}s_j \quad i = 1, \dots, n$$

Where:

$$f_{i0} = \int_l \left(\frac{N_i N_0}{EA} + \chi \frac{T_i T_0}{GA} + \frac{M_i M_0}{EI} \right) d\xi - \sum_r F_r \eta_r^F$$

$$k_{ij} = \int_l \left(\frac{N_i N_j}{EA} + \chi \frac{T_i T_j}{GA} + \frac{M_i M_j}{EI} \right) d\xi$$

Physically, f_{i0} is the reaction at the coordinate i due to the applied loads in the principal system, and k_{ij} is the reaction at the coordinate i due to a unit displacement at the coordinate j in the principal system. Now, in matricial form the equilibrium equation is defined as:

$$\mathbf{f} = \mathbf{K} \mathbf{s}$$

Here, $\mathbf{K}$ is the stiffness matrix of the system with a dimension of $n \times n$, with the property of being symmetric and having positive direct stiffness coefficients. On the other hand, the displacement vector $\mathbf{s}$ and the force vector $\mathbf{f}$ have a dimension of $n \times 1$. It must be noted that among the infinite possible compatible solutions, the solution of the previous system of linear equations provides the only equilibrated one. Numerically, the solution of the system is determined as:

$$\mathbf{s} = \mathbf{K}^{-1} \mathbf{f}$$

Chapter 2

DESCRIPTION OF THE STRUCTURAL SYSTEM

This chapter addresses the structural system to be defined for the computational modelling. Focus is laid on the geometrical and material properties of the structural system and on the characteristics of the various cross-sections to be adopted. The geometrical description is carried out by identifying the nodes, their corresponding coordinates, and their respective connectivity that constitute the element. The detailed procedure for the creation of these input parameters is laid out below.

Nodes and Coordinates

A matrix of the coordinates, **COORD**, is created that stores the x and y coordinates of the respective node. Thus, for our two-dimensional problem, the size of **COORD** is the number of nodes (**NNODE**) $\times$ 2, as illustrated below.

COORD (NNODE,2)		
Node	x	y
1	x_1	y_1
2	x_2	y_2
3	x_3	y_3
...	...	...
NNODE	x_{NNODE}	y_{NNODE}

Elements and Connectivity

A matrix of incidence, **IN**, is defined that lists the two nodes that are to be connected to form the respective element. Thus, the size of **IN** is the number of elements (**NELE**) $\times$ 2, as shown.

IN (NELE,2)		
Element	Node 1	Node 2
1	N_{11}	N_{12}
2	N_{21}	N_{22}
3	N_{31}	N_{32}
...	...	...
NELE	$N_{NELE,1}$	$N_{NELE,2}$

Cross-Section

The characteristics of the cross-sections adopted in the structural system are saved in the matrix **CSEC**. This matrix stores the Area (A), Inertia (I), Depth (h), and the Shear correction factor (χ) of the various cross-sections used. Thus, as seen below, this matrix is characterized by a size of the number of sections (**NSEC**) $\times$ 4. This classification now enables us to assign these cross-

sections to the various elements of our system. An incidence vector **ISEC** stores this assignment of the cross-section to the corresponding element.

CSEC (NSEC,4)				
Section	A	I	h	χ
1				
2				
3				
...				
NSEC				

ISEC (NELE)	
Element	Section
1	
2	
3	
...	...
NELE	...

Material

The characteristic of the different materials composing our system is collected in the matrix **CMAT**. The properties under consideration are the Young's Modulus (E), the Poisson's Ration (ν), the coefficient of thermal expansion (α), and the specific weight (γ). Thus, the resulting size of **CMAT** is the number of materials (**NMAT**) $\times$ 4. A vector of the incidences of the materials is defined allocating these various materials to the respective element as seen here.

CMAT (NMAT,4)				
Material	E	ν	α	γ
1				
2				
3				
...				
NMAT				

IMAT (NELE)	
Element	Material
1	
2	
3	
...	...
NELE	...

Chapter 3

NUMBERING OF THE DEGREES OF FREEDOM

The degrees of freedom (DOF) present in our structural system is stored in the matrix **IDOF**. Three possible DOFs are considered for each node, the horizontal displacement (u), the vertical displacement (v), and the rotation (ϕ). Thus, the size of the matrix **IDOF** is $(NNODE) \times 3$. The numbering of the DOFs is implemented in four phases as stated below.

Initialization

The matrix **IDOF** is first initialized as a *zero matrix* for all the three possible DOFs at each node of the system, as shown here.

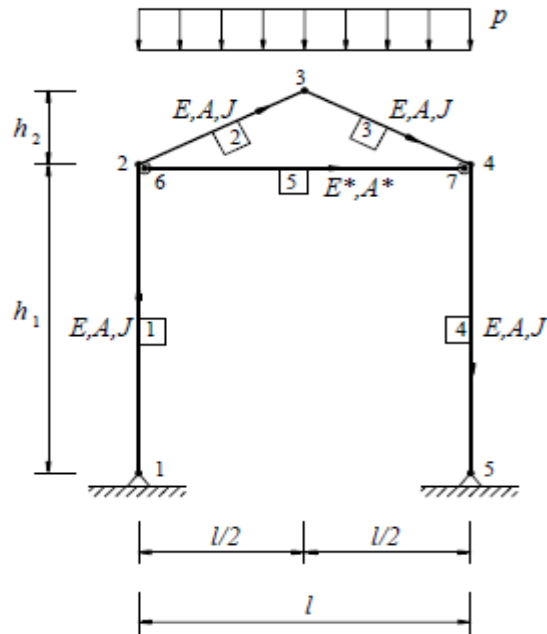
IDOF (NNODE,3)			
Node	u	v	ϕ
1	0	0	0
2	0	0	0
3	0	0	0
...	0	0	0
NNODE	0	0	0

Restraints

In order to account for the restraints a value of -1 is associated to the corresponding restrained DOFs. The following numerical example (refer to Fig. 3.1) is considered to illustrate the process of numbering of the degrees of freedom. As seen, this frame structure is restrained with a fixed end and a hinged end and the corresponding **IDOF** matrix after imposing the restraints is as shown.

Links

In case of structural systems consisting of links in between elements, one node is characterized as the master node and the other as the slave node. The node id of the master node is lower than that of the slave node. The corresponding rows of the **IDOF** matrix for the slave node are filled with the **IDNODE** of its master node in the position of the link. For the structural system under consideration, the link connecting elements 2 and 4 is modelled and the updated **IDOF** matrix is presented below.



$$p = 50 \text{ kN/m}$$

$$l = 4.00 \text{ m}$$

$$h_1 = 3.00 \text{ m}$$

$$h_2 = 1.00 \text{ m}$$

$$J = 6.75 \times 10^8 \text{ mm}^4$$

$$EA = \infty$$

$$E = 30000 \text{ N/mm}^2$$

$$A^* = 1000 \text{ mm}^2$$

$$E^* = 210000 \text{ N/mm}^2$$

$$\vartheta = \arctg \frac{2h_2}{l} = 26^\circ 33' 54'' .2$$

$$p_x = p \sin \vartheta \cos \vartheta = 20 \text{ kN/m}$$

$$p_y = p \cos^2 \vartheta = 40 \text{ kN/m}$$

Figure 3.1: Portal Frame Example

IDOF (NNODE,3)			
Node	u	v	ϕ
1	-1	-1	0
2	0	0	0
3	0	0	0
4	0	0	0
5	-1	-1	0
6	0	0	0
7	0	0	0

IDOF (NNODE,3)			
Node	u	v	ϕ
1	-1	-1	0
2	0	0	0
3	0	0	0
4	0	0	0
5	-1	-1	0
6	3	3	0
7	3	3	0

Numbering of the DOFs

The number of DOFs are now counted sequentially from the zeros in the IDOF matrix. In order to impose the link between the master and the slave node the DOF label of the slave node is to be same as that of the master node. For the frame system considered the consequent IDOF matrix is illustrated below.

The numerical implementation of the above-stated procedure is condensed as:

$$\begin{cases} d_{ij} = d_{ij,max} + 1 & \text{if component } j \text{ at node } i \text{ is not restrained} \\ d_{ij} = -1 & \text{if component } j \text{ at node } i \text{ is restrained} \\ d_{ij} = d_{kh} + 1 & \text{if component } j \text{ at node } i \text{ is linked to component } h \text{ at the node } k < i \end{cases}$$

IDOF (NNODE,3)			
Node	u	v	ϕ
1	-1	-1	1
2	2	3	4
3	5	6	7
4	8	9	10
5	-1	-1	11
6	2	3	12
7	8	9	13

Where:

i 1,..., NNODE

j 1, 2, 3

$d_{ij,max}$ $\max_{k \leq i, h \leq j} d_{kh}$

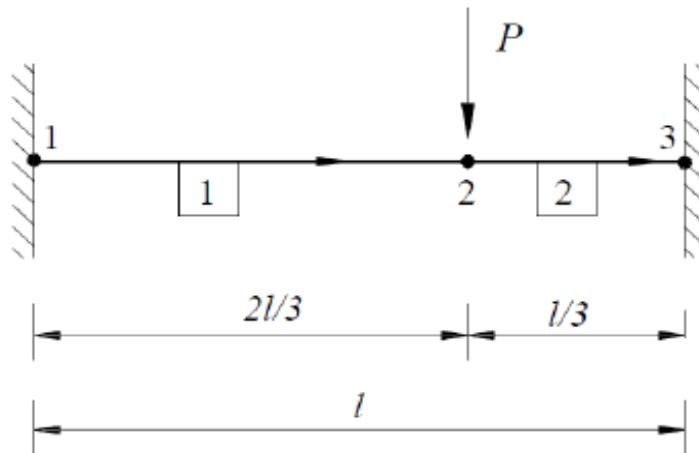
Chapter 4

NODAL LOADS IN THE GLOBAL REFERENCE SYSTEM

The loads present in the structural system in the global reference frame are detailed in this chapter. The loads are categorized into two major types: nodal loads and element loads. The parameters to be defined for the description of the load cases are detailed below.

4.1 Nodal Loads

The concentrated loads directly applied at the defined nodal points are the nodal loads. The nodal loads are stored in the vector **VLOADS** which assigns the respective load to the degree of freedom that it corresponds to. Thus, to begin with, **VLOADS** is initialized as a zero vector representing zero loads at each DOF. The id of the loaded node, the direction of application of the nodal load, and the magnitude are to be provided as input to the program. The program reads the node id and the direction and extracts the relative DOF from the matrix **IDOF** at that position. The magnitude is then noted in **VLOADS** in accordance with the DOF the applied load is associated to. For illustration purposes the following numerical example can be referred to, the vector **VLOADS** for $P = 10 \text{ kN}$ is as shown below.



VLOADS	
DOF	LOAD
1	0
2	-10
3	0

4.2 Element Loads

The distributed load acting throughout the element is the element load which is saved in the matrix **ELOADS** that associates the required parameters defined to each element. After the initialization of the matrix, the program takes five input variables listed as:

- p_x Magnitude of the distributed load along x
- p_{y1} Magnitude of the distributed load along y, at the first node
- p_{y2} Magnitude of the distributed load along y, at the second node
- ΔT_{top} Temperature differential at the top of the element
- ΔT_{bot} Temperature differential at the bottom of the element

These input variables are now assigned to the id of the element it is related to. For the case of the numerical example referred to in the previous chapter (refer to Fig. 3.1), taking $p = 10 \text{ kN/m}$, the vector **ELOADS** is as shown below.

ELOADS					
Element	p_x	p_{y1}	p_{y2}	ΔT_{top}	ΔT_{bot}
1	0	0	0	0	0
2	-20	-40	-40	0	0
3	20	-40	-40	0	0
4	0	0	0	0	0
5	0	0	0	0	0
6	0	0	0	0	0
7	0	0	0	0	0

Chapter 5

ELEMENT PROPERTIES IN THE L.R.S.

The parameters required to obtain the solution of the system have to be defined in the Local Reference System (LRS) of each element which would further be transformed and assembled to develop the global system. Therefore, three significant properties of the element are discussed hereafter.

5.1 Stiffness Matrix

The local stiffness matrix $\mathbf{K}'$ of the two-dimensional beam element under analysis is a symmetric square matrix of dimension 6. The stiffness coefficient k_{ij} is representative of the reaction at the coordinate i as a result of a unit displacement imposed at the coordinate j . The stiffness matrix incorporates the responses of two major behaviors of the element, axial and bending, and for a homogeneous, isotropic, elastic material the effects of axial and bending deformations are uncoupled. This results in the following modification of the stiffness matrix.

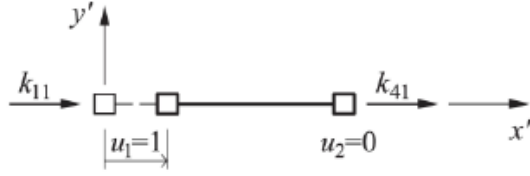
$$\mathbf{K}' = \begin{bmatrix} k_{11} & k_{12} & k_{13} & k_{14} & k_{15} & k_{16} \\ k_{21} & k_{22} & k_{23} & k_{24} & k_{25} & k_{26} \\ k_{31} & k_{32} & k_{33} & k_{34} & k_{35} & k_{36} \\ k_{41} & k_{42} & k_{43} & k_{44} & k_{45} & k_{46} \\ k_{51} & k_{52} & k_{53} & k_{54} & k_{55} & k_{56} \\ k_{61} & k_{62} & k_{63} & k_{64} & k_{65} & k_{66} \end{bmatrix}$$

Axial Behaviour

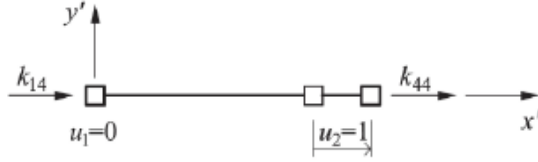
For an infinitesimal element of length dx subjected to a tensile axial force N , considering an increment du , utilising compatibility and constitutive laws, the governing equation can be written as,

$$EA \frac{du}{dx'} = N(x')$$

The response to the axial behaviour corresponds to the first and the fourth column of the stiffness matrix. Imposing a unit displacement in one node and a null displacement on the other, we obtain the stiffness coefficients relative to the former. This process is illustrated below.



$$k_{11} = \frac{EA}{l} \quad k_{41} = -\frac{EA}{l}$$



$$k_{14} = -\frac{EA}{l} \quad k_{44} = \frac{EA}{l}$$

Bending Behaviour

Considering an infinitesimal element of length dx subjected to shear and bending, the total deflection $v(x')$ is obtained as the combination of the contributions from bending and shear.

$$v(x') = v_a(x') + v_b(x')$$

On application of a bending moment, the relative rotation $d\phi$ of the two end sections lead to a bending curvature. The governing equation is given by

$$EI \frac{d^2 v_a}{dx'^2} = M(x')$$

$$\phi(x') = \frac{dv_a}{dx'}$$

The application of shear introduces a distortion of the straight segment that produces a shear strain due to pure translation of the end section. The governing equation is written as:

$$\frac{GA}{\chi} \frac{dv_b}{dx'} = -T(x')$$

Where:

χ is the shear correction factor which is formulated later in the discussion

For the general beam element analysed, the functions expressing the bending moment and the shear force are determined from equilibrium as:

$$M(x') = -m_1 + t_1 x' = m_2 + t_2 (l - x')$$

$$T(x') = t_1 = -t_2$$

The governing differential equations noted above are now solved utilizing the expressions of $M(x')$ and $T(x')$ which are then combined to write the governing equations for the total displacement v and the rotation ϕ as:

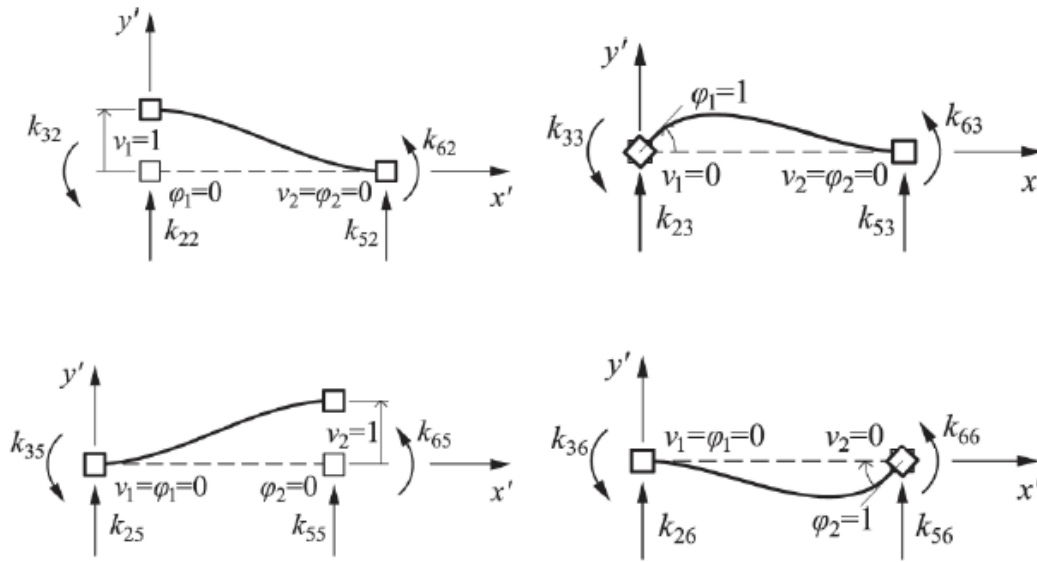
$$EIv(x') = \frac{1}{6}t_1x'^3 - \frac{1}{2}m_1x'^2 + (C_1 - t_1\Phi)x' + C_2$$

$$EI\varphi(x') = \frac{1}{2}t_1x'^2 - m_1x' + C_1$$

Where:

Φ is the shear deformability coefficient given as $\chi \frac{12EI}{l^2GA}$

The constants of these equations are eliminated with the help of the boundary conditions applicable for the case used for obtaining the respective stiffness coefficients as illustrated below.



Column of the stiffness matrix				
DOF	2	3	5	6
$v(0)$	1	0	0	0
$\varphi(0)$	0	1	0	0
$v(l)$	0	0	1	0
$\varphi(l)$	0	0	0	1

On solving these systems of equations and imposing equilibrium the stiffness matrix in the local reference system is eventually obtained as:

$$\mathbf{K}' = \begin{bmatrix} \frac{EA}{l} & 0 & 0 & -\frac{EA}{l} & 0 & 0 \\ 0 & \frac{12EI}{l^3(1+\Phi)} & \frac{6EI}{l^2(1+\Phi)} & 0 & -\frac{12EI}{l^3(1+\Phi)} & \frac{6EI}{l^2(1+\Phi)} \\ 0 & \frac{6EI}{l^2(1+\Phi)} & \frac{(4+\Phi)EI}{l(1+\Phi)} & 0 & -\frac{6EI}{l^2(1+\Phi)} & \frac{(2-\Phi)EI}{l(1+\Phi)} \\ -\frac{EA}{l} & 0 & 0 & \frac{EA}{l} & 0 & 0 \\ 0 & -\frac{12EI}{l^3(1+\Phi)} & -\frac{6EI}{l^2(1+\Phi)} & 0 & \frac{12EI}{l^3(1+\Phi)} & \frac{6EI}{l^2(1+\Phi)} \\ 0 & \frac{6EI}{l^2(1+\Phi)} & \frac{(2-\Phi)EI}{l(1+\Phi)} & 0 & -\frac{6EI}{l^2(1+\Phi)} & \frac{(4+\Phi)EI}{l(1+\Phi)} \end{bmatrix} \quad \text{Symmetric}$$

The shear correction factor noted above is introduced owing to the assumption of plane sections that neglects the warping deformation caused by transverse shear. It is found through an energy-based formulation that equates the strain energy expressed in terms of the Jourawski shear stress and the average shear stress as:

$$2E_I = \int_A \tau_{xy} \gamma_{xy} dA$$

$$2E_{II} = \int_A \bar{\tau} \gamma_{xy} dA$$

Where the respective shear stresses are defined as:

$$\tau_{xy} = G \gamma_{xy} = -\frac{TS(y)}{b(y)I}$$

$$\bar{\tau} = -\frac{1}{\chi} G \gamma_0 = -\frac{T}{A}$$

Now the formulation of the energy is rewritten as:

$$2E_I = \frac{1}{G} \int_A \tau_{xy}^2 dA = \frac{1}{G} \int_A \left[\frac{TS(y)}{b(y)I} \right]^2 dA = \frac{T^2}{GI^2} \int_A \left[\frac{S(y)}{b(y)} \right]^2 dA$$

$$2E_{II} = \bar{\tau}^2 \frac{\chi}{G} \int_A dA = \bar{\tau}^2 \frac{\chi}{G} A = \frac{T^2 \chi}{GA}$$

The two energy descriptions are now equated and the shear correction factor is obtained as:

$$E_I = E_{II} \Rightarrow \chi = \frac{A}{A_\chi} = \frac{A}{I^2} \int_A \left[\frac{S(y)}{b(y)} \right]^2 dA$$

Where

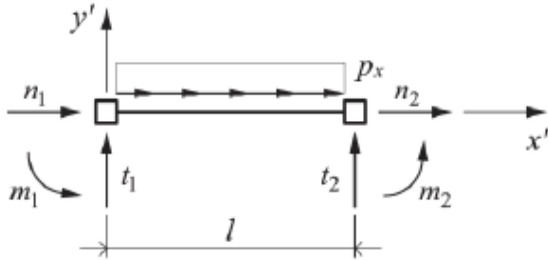
A_χ is the effective shear area

5.2 Equivalent Nodal Forces

The element loads acting on the structural system are modelled as equivalent nodal forces that produce the same effects as the element loads in terms of the nodal displacement $\mathbf{s}$. The system is solved by exploiting the superposition theorem taking into account the effects resulting from a fixed-end system and an equivalent system. The underlying principle of this approach is that the equivalent nodal forces $\hat{\mathbf{f}}$ are the reversed fixed-end forces $\mathbf{f}_0$. This relation is expressed as:

$$\hat{\mathbf{f}} = -\mathbf{f}_0$$

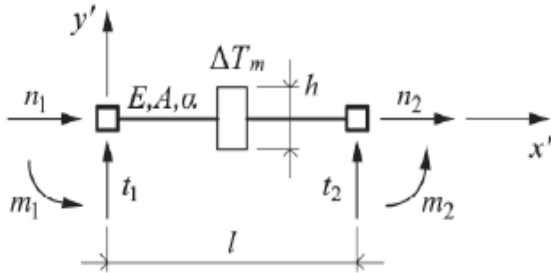
Distributed Axial Load



$$\mathbf{f}'_0 = \frac{p_x l}{2} [-1 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0]^T$$

$$\hat{\mathbf{f}}' = -\mathbf{f}'_0 = \frac{p_x l}{2} [1 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0]^T$$

Uniform Temperature Change



$$\mathbf{f}'_0 = EA \alpha \Delta T_m [1 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0]^T$$

$$\hat{\mathbf{f}}' = -\mathbf{f}'_0 = EA \alpha \Delta T_m [-1 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0]^T$$

On account of the fixed ends considered the total strain ε is zero and the thermal strain ε_T is given as:

$$\varepsilon_T = \alpha \Delta T_m$$

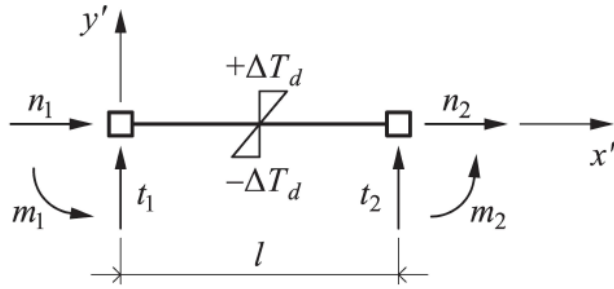
The resulting internal stress and the axial force are expressed as:

$$\sigma = E(\varepsilon - \varepsilon_0)$$

$$N = \sigma A = -EA \alpha \Delta T_m$$

The corresponding fixed-end forces and the equivalent nodal forces thus are

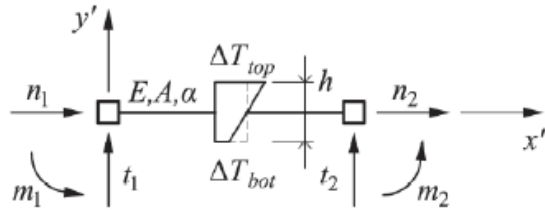
Differential Temperature Change



$$\mathbf{f}'_0 = 2 \propto \Delta T_d \frac{EI}{h} [0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 1]^T$$

$$\hat{\mathbf{f}}' = -\mathbf{f}'_0 = 2 \propto \Delta T_d \frac{EI}{h} [0 \quad 0 \quad 1 \quad 0 \quad 0 \quad -1]^T$$

Linear Temperature Differential

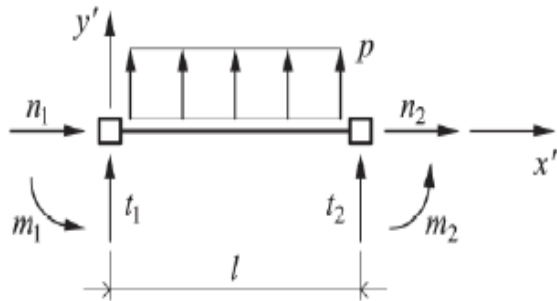


$$\Delta T_m = \frac{\Delta T_{top} + \Delta T_{bot}}{2}$$

$$\Delta T_d = \frac{\Delta T_{top} - \Delta T_{bot}}{2}$$

$$\hat{\mathbf{f}}' = \begin{bmatrix} -\propto \Delta T_m EA & 0 & 2 \propto \Delta T_d \frac{EI}{h} & \propto \Delta T_m EA & 0 & -2 \propto \Delta T_d \frac{EI}{h} \end{bmatrix}^T$$

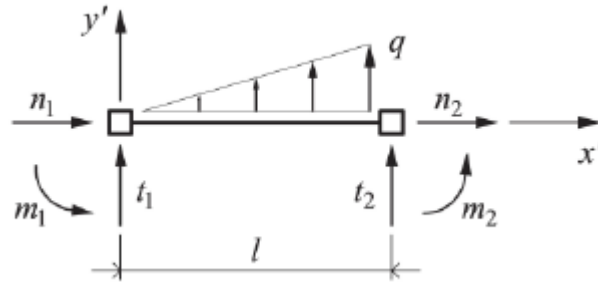
Uniformly Distributed Load



$$\mathbf{f}'_0 = \begin{bmatrix} 0 & -\frac{pl}{2} & -\frac{pl^2}{12} & 0 & -\frac{pl}{2} & \frac{pl^2}{12} \end{bmatrix}^T$$

$$\hat{\mathbf{f}}' = -\mathbf{f}'_0 = \begin{bmatrix} 0 & \frac{pl}{2} & \frac{pl^2}{12} & 0 & \frac{pl}{2} & \frac{pl^2}{12} \end{bmatrix}^T$$

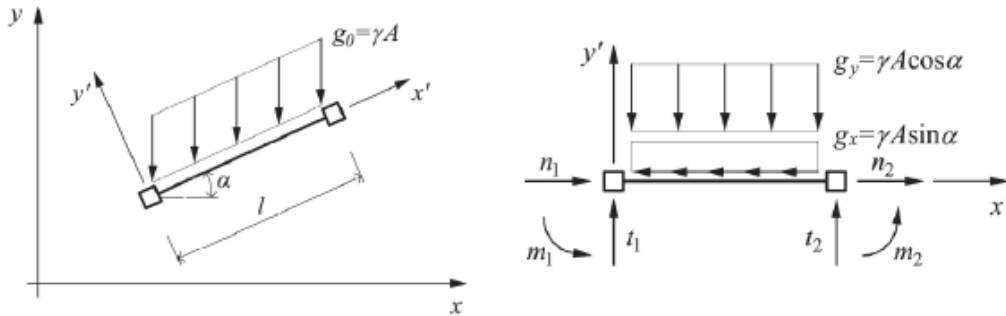
Linearly Distributed Load



$$\mathbf{f}'_0 = \begin{bmatrix} 0 & -\frac{9}{60}ql\left(1+\frac{c}{9}\right) & -\frac{ql^2}{30}\left(1+\frac{c}{4}\right) & 0 & -\frac{21}{60}ql\left(1-\frac{c}{21}\right) & -\frac{ql^2}{20}ql\left(1-\frac{c}{6}\right) \end{bmatrix}^T$$

$$\hat{\mathbf{f}}' = -\mathbf{f}'_0 = \begin{bmatrix} 0 & \frac{9}{60}ql\left(1+\frac{c}{9}\right) & \frac{ql^2}{30}\left(1+\frac{c}{4}\right) & 0 & \frac{21}{60}ql\left(1-\frac{c}{21}\right) & -\frac{ql^2}{20}ql\left(1-\frac{c}{6}\right) \end{bmatrix}^T$$

5.3 Self-weight



The self-weight acting on an arbitrary beam element is as illustrated below. The load is decomposed locally acting as distributed loads along the principal axis of the beam element and the corresponding equivalent nodal force vector is written as:

$$\hat{\mathbf{f}}' = \frac{\gamma Al}{2} \begin{bmatrix} -\sin\alpha & -\cos\alpha & -\frac{l\cos\alpha}{6} & -\sin\alpha & -\cos\alpha & \frac{l\cos\alpha}{6} \end{bmatrix}^T$$

Chapter 6

TRANSFORMATION OF THE REFERENCE SYSTEM

The parameters necessary for the solution of the system, such as the nodal force vector, the nodal displacement vector, and the stiffness matrix were defined in the local reference system of the beam element up until now. The transformation of coordinates from this local reference frame to the global one is carried out with the transformation matrix T_0 . This transformation performed for a 2D coordinate system is illustrated in Fig. 6.1.

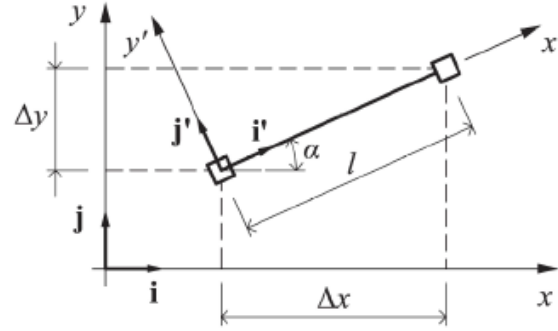


Figure 6.1: Transformation of 2D coordinate systems

For this case of our two-dimensional beam element, the transformation matrix is formed as

$$T_0 = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The global displacement vector and the nodal force vector is expressed in terms of the respective local ones through transformation as

$$\mathbf{s} = \mathbf{T} \mathbf{s}'$$

$$\mathbf{f} = \mathbf{T} \mathbf{f}'$$

Here the vectors in the local reference frame are denoted with the prime symbol and $\mathbf{T}$ is obtained as

$$\mathbf{T} = \begin{bmatrix} \mathbf{T}_0 & 0 \\ 0 & \mathbf{T}_0 \end{bmatrix}$$

The equilibrium in the local reference system is given by

$$\mathbf{K}' \mathbf{s}' = \mathbf{f}'$$

Utilizing the above relations the following operations are performed to move to the global reference system

$$\mathbf{K}'\mathbf{T}^T \mathbf{s} = \mathbf{T}^T \mathbf{f}$$

$$\mathbf{T}\mathbf{K}'\mathbf{T}^T \mathbf{s} = \mathbf{T}\mathbf{T}^T \mathbf{f} = \mathbf{f}$$

The global equilibrium of the system is thus written as

$$\mathbf{K} \mathbf{s} = \mathbf{f}$$

Where the global stiffness matrix $\mathbf{K}$ is defined as

$$\mathbf{K} = \mathbf{T}\mathbf{K}'\mathbf{T}^T$$

Chapter 7

STRUCTURAL ASSEMBLY

The equilibrium equation written for each beam element of the structural system is assembled in the global reference system in order to obtain the overall structural equilibrium equation. For a m^{th} beam element the equilibrium equation in the global reference system is

$$\mathbf{K}_m \mathbf{s}_m = \mathbf{f}_m$$

The size of the parameters involved is dependent on the degree of freedom of the system under consideration. Thus, for the beam element having 6 DOFs, the stiffness matrix is of the size 6×6 whereas the nodal displacement vector and the nodal force vector is of the size 6×1 . Thus, on similar lines, the stiffness matrix of the whole structure in the global reference frame would be of the size $n \times n$, where n is the total DOF of the structural system, and the nodal displacement vector and the nodal force vector is of the size $n \times 1$. The assembly of the equilibrium equations is thus noted as

$$\mathbf{K} \mathbf{s} = \mathbf{f}$$

The overall structural nodal displacement vector is related to the elemental one with the help of an assembly matrix as

$$\begin{aligned} \mathbf{s} &= \mathbf{Z}_m \mathbf{s}_m \\ \mathbf{s}_m &= \mathbf{Z}_m^T \mathbf{s} \end{aligned}$$

Where

$\mathbf{Z}_m$ is the assembly matrix of the m^{th} beam element defined as,

$$\mathbf{Z}_m = [z_{ij}^m] = \begin{cases} z_{ij}^m = 1 & \text{if } j_{element} \rightarrow i_{system} \\ z_{ij}^m = 0 & \text{otherwise} \end{cases}$$

Now, replacing the expression of the elemental nodal displacement vector in terms of the structural one in the elemental equilibrium equation we obtain

$$\mathbf{K}_m \mathbf{Z}_m^T \mathbf{s} = \mathbf{f}_m$$

Pre-multiplying each side by $\mathbf{Z}_m$ and then summing over the structural system we obtain the overall structural equilibrium equation as

$$\begin{aligned} \mathbf{Z}_m \mathbf{K}_m \mathbf{Z}_m^T \mathbf{s} &= \mathbf{Z}_m \mathbf{f}_m \\ \sum_m \mathbf{Z}_m \mathbf{K}_m \mathbf{Z}_m^T \mathbf{s} &= \sum_m \mathbf{Z}_m \mathbf{f}_m \\ \Rightarrow \mathbf{K} \mathbf{s} &= \mathbf{f} \end{aligned}$$

Through this way of assembly of the nodal load vector the equilibrium in the nodes is taken care of by the contributions of different elements connected to the same node. To forego of all these contributions a different approach to the assembly could be implemented. The direct assembly procedure doesn't require the assembly matrix and consists in placing the coefficients relative to the various elements directly in the global stiffness matrix. This method however is not implemented in our version of the code.

Chapter 8

SOLUTION OF THE SYSTEM OF LINEAR EQUATIONS

The algorithm implemented for solving the structural system influences the overall computational efficiency of the system and the accuracy and reliability of the results. Thus, the selection of the solution method becomes critical for our analysis. Direct inversion is a straightforward method for solving a system of equation expressed as $\mathbf{Ax}=\mathbf{B}$. However, the direct inversion of the coefficient matrix $\mathbf{A}$ is computationally inefficient and thus this method is rarely adopted. Thus, two alternative solution methods that can be employed are extensively discussed hereafter.

8.1 Solution Methods

8.1.1 Direct Elimination Methods

Direct elimination methods are based on factoring the coefficient matrix $\mathbf{A}$ into a product of a lower and an upper triangular matrix. Besides the round-off error this method generates the exact solution in a finite number of elementary arithmetic operations. However, it may be inadequate for solving a system comprised of large number of equations because of the high number of arithmetic operations involved and the accumulation of the round-off error. The procedure for the direct elimination method is as shown:

$$\begin{aligned}\mathbf{Ax} &= \mathbf{b} \\ \mathbf{A} &= \mathbf{LU} \\ \Rightarrow \mathbf{LUx} &= \mathbf{b}\end{aligned}$$

Where

$\mathbf{L}$ is the lower triangular matrix, and

$\mathbf{U}$ is the upper triangular matrix

Following this decomposition theorem, the system is rewritten as:

$$\begin{cases} \mathbf{Ly} = \mathbf{b} \\ \mathbf{Ux} = \mathbf{y} \end{cases}$$

The solution method involves two major steps:

1. Decomposition by forward or backward elimination usually through the Gauss or Cholesky method, and
2. Solution by forward and/or backward substitution

The fundamental procedure behind the forward and backward substitution is described below.

- **Forward Substitution**

For the decomposed system defined as $\mathbf{Ly} = \mathbf{b}$, the forward substitution is carried out as:

$$y_i = \frac{b_i - \sum_{k=1}^{i-1} l_{ik} y_k}{l_{ii}} \quad i = 1, \dots, n$$

Here

$$l_{ii} \neq 0$$

$$\det L \neq 0$$

- **Backward Substitution**

For the decomposed system defined as $Ux = y$, the backward substitution is carried out as:

$$x_i = \frac{b_i - \sum_{k=i+1}^n u_{ik} x_k}{u_{ii}} \quad i = 1, \dots, n$$

Here

$$u_{ii} \neq 0$$

$$\det U \neq 0$$

The detailed algorithm to be implemented for these steps are laid out in the subsequent discussion.

Gauss Elimination

In the Gauss elimination method the unknown x_j can be eliminated from any equation $i \neq k$ as:

$$a'_{ir} = a_{ir} - \frac{a_{ij}}{a_{kj}} a_{kr}$$

$$a_{kj} \neq 0$$

$$i \neq k, \text{ and}$$

$$r = 1, \dots, n$$

For the case of $r = j$, we get $a'_{ij} = 0$. On sequential elimination from x_1 to x_{n-1} after $n-1$ elimination steps the original linear equation system is reduced to an upper triangular form

$$\tilde{A} \equiv U$$

$$\tilde{b} \equiv y = L^{-1}b$$

$$\tilde{A}x = \tilde{b} \Rightarrow Ux = y$$

+ **Backward Substitution**

Comapct Procedure

The computational efficiency of the elimination procedure described above can be improved on dividing the equation i by the pivotal term a_{ii} before eliminating the unknown x_i . The corresponding algorithm is illustrated below.

$$\mathbf{Ax} = \mathbf{b} \quad \mathbf{W} = [\mathbf{A}|\mathbf{b}]$$

Forward Elimination

$$\tilde{\mathbf{A}}\mathbf{x} = \tilde{\mathbf{b}} \quad \tilde{\mathbf{W}} = [\tilde{\mathbf{A}}|\tilde{\mathbf{b}}]$$

$$\begin{cases} \tilde{w}_{ik} = \frac{w_{ik}}{w_{ii}} & i = 1, \dots, n-1 \\ \tilde{w}_{jk} = w_{jk} - \frac{w_{ik}}{w_{ii}} w_{ji} & k = i, \dots, n+1 \end{cases}$$

Backward Substitution

$$x_i = \frac{\tilde{b}_i - \sum_{k=i+1}^n \tilde{a}_{ik} x_k}{\tilde{u}_{ii}} = \tilde{b}_i - \sum_{k=i+1}^n \tilde{a}_{ik} x_k \quad i = 1, \dots, n$$

Cholesky Method

If the matrix of coefficients $\mathbf{A}$ is symmetric and positive definite, its factorization can be carried out as:

$$\mathbf{A} = \mathbf{LU} = \mathbf{LL}^T$$

Where the lower triangular matrix $\mathbf{L}$ is computed as:

$$\begin{aligned} l_{ii} &= \sqrt{a_{ii} - \sum_{k=1}^{i-1} l_{ik}^2} \\ l_{ji} &= \frac{a_{ji} - \sum_{k=1}^{i-1} l_{jk} l_{ik}}{l_{ii}} \quad j = i+1, \dots, n \end{aligned}$$

The solution by Forward and Backward substitution is now obtained as:

$$\begin{cases} \mathbf{Ly} = \mathbf{b} \\ \mathbf{Ux} = \mathbf{L}^T \mathbf{x} = \mathbf{y} \end{cases}$$

These elimination methods intrinsically differ in the way that the matrix of coefficient $\mathbf{A}$ is factored into $\mathbf{L}$ and $\mathbf{U}$. The computational efficiencies of these elimination methods could be compared for a rationalized selection. The Cholesky method is numerically stable and accurate and it requires fewer floating-point operations and less workspace relative to the Gauss elimination method. However, the computational efficiencies of these two methods are comparable when the symmetry of the matrix $\mathbf{A}$ is exploited. The compact Gauss elimination method is adopted for obtaining the solution of our structural system defined with the equation

$$\mathbf{Ks} = \mathbf{f}$$

8.1.2 Iterative Methods

Iterative methods entails the selection of an initial approximation that is progressively refined to a better approximation with the aid of a suitable algorithm, until we reach a solution within a predetermined error tolerance. This method is advantageous with respect to the direct elimination methods owing to the fact that it is relatively insensitive to the accumulation of the round-off errors and moreover, the operations involved are uniform and simple. The iterative process may however converge, or diverge slowly.

8.2 Conditioning

Conditioning reflects the sensitivity of any function $f(\mathbf{x})$ to changes in its argument $\mathbf{x}$ and it only depends on the function to be evaluated. For a system of linear equations such as $\mathbf{Ks} = \mathbf{f}$ of our structural system, if small errors in the stiffness matrix $\mathbf{K}$ or in the load vector $\mathbf{f}$ have little effects on the displacement vector $\mathbf{s}$, the system is well-conditioned. However, if the effects of these small errors are large then the system is ill-conditioned.

The condition number is a criteria that enables this classification of the system. It allows an a priori estimate of the upper bound of the relative error of the solution. The smaller the condition number, the well-conditioned the system is. The condition number is defined as:

$$\text{Cond}(\mathbf{K}) = \|\mathbf{K}\| \|\mathbf{K}^{-1}\|$$

Where

$\|\cdot\|$ *is the matrix norm*

However, artificial ill-conditioning or skewness is possible, where high condition numbers could be seen even for well-conditioned systems. This misleading error estimate is resulting from isolated stiff regions raising the condition number of $\mathbf{K}$.

With the help of appropriate scaling and preconditioning of the stiffness matrix, this issue could be averted.

Scaling and Preconditioning

Taking into consideration the system of equation of our structural problem, $\mathbf{Ks} = \mathbf{f}$, the following operations are made for scaling

$$\mathbf{D}_1 \mathbf{Ks} = \mathbf{D}_1 \mathbf{f}$$

Now, $\mathbf{s}$ and $\mathbf{D}_1 \mathbf{f}$ are substituted in this equation as:

$$\begin{aligned}
\mathbf{s} &= \mathbf{D}_2 \bar{\mathbf{s}} \\
\mathbf{D}_1 \mathbf{f} &= \bar{\mathbf{f}} \\
\Rightarrow \mathbf{D}_1 \mathbf{K} \mathbf{D}_2 \bar{\mathbf{s}} &= \bar{\mathbf{f}}
\end{aligned}$$

The scaled system can now be rewritten as:

$$\bar{\mathbf{K}} \bar{\mathbf{s}} = \bar{\mathbf{f}}$$

Here $\mathbf{D}_1$ and $\mathbf{D}_2$ are the preconditioning matrices defined as:

$$\begin{aligned}
\mathbf{D}_1 &= \mathbf{D}_2 = \mathbf{D} \\
d_{ii} &= \frac{1}{\sqrt{k_{ii}}}
\end{aligned}$$

The estimate of the maximum number of significant digits that could be lost in the solution process due to ill-conditioning can be made as:

$$\bar{m} = \log [\text{Cond}(\bar{\mathbf{K}})]$$

This manipulation makes the diagonal coefficients of the scaled stiffness matrix as unity. Moreover, the nature of the problem remains unaltered even after scaling and if a system is truly ill-conditioned, it remains so even after preconditioning. Scaling helps in preventing artificial ill-conditioning and thus the unscaled stiffness matrix can be used to solve the system.

Diagonal Decay Error Test

On factorizing a matrix into triangular form it is noted that the ratio of a diagonal term in the factorized matrix to the corresponding term in the original matrix decreases in size as one moves down the diagonal. If this ratio goes to zero the matrix is singular and if it is negative, the matrix is not positive definite. The diagonal decay of the stiffness matrix $\mathbf{K}$ can be utilized as an approximate estimate of its condition number. The procedure is as follows:

$$\delta_i = \frac{k_{ii}}{\bar{k}_{ii}}$$

Where

k_{ii} is the diagonal i th term of the original matrix, and

$\bar{k}_{ii}$ is the diagonal term just before the elimination of the i th DOF

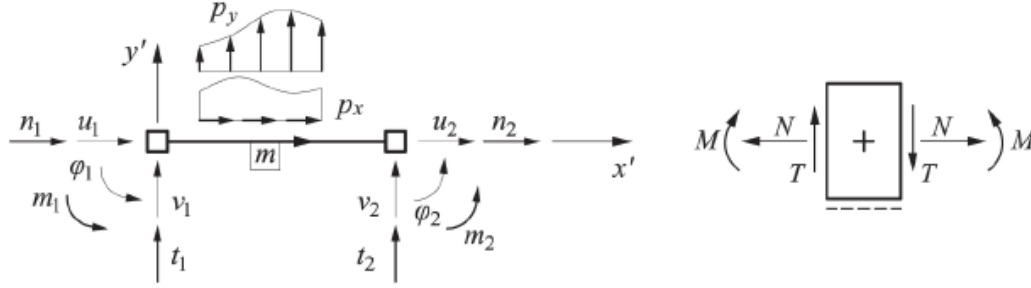
The estimate of the maximum number of significant digits that could be lost in the solution process due to ill-conditioning can be made as:

$$\begin{aligned}
\max \delta_i &= 10^m \\
\begin{cases} 0 \leq m \leq 2 & \text{well - conditioned system} \\ 3 \leq m \leq 9 & \text{possible ill - conditioning or modelling problem} \\ m \geq 10 & \text{ill - conditioned system} \end{cases}
\end{aligned}$$

Chapter 9

STRESS ANALYSIS

The generic beam element of our structural system and the adopted convention for the local displacement and forces is as illustrated below.



The nodal displacements and the nodal forces of the beam element in the local reference system are:

$$\mathbf{s}'_m = [u_1 \quad v_1 \quad \varphi_1 \quad u_2 \quad v_2 \quad \varphi_2]^T$$

$$\mathbf{f}'_m = [n_1 \quad t_1 \quad m_1 \quad n_2 \quad t_2 \quad m_2]^T$$

These quantities are derived through transformation from the global displacement vector obtained from the solution of the system earlier, and the equivalent nodal

Force vector as

$$\mathbf{s}'_m = \mathbf{T}^T \mathbf{s}_m$$

$$\mathbf{f}'_m = \mathbf{K}'_m \mathbf{s}'_m - \hat{\mathbf{f}}'_m$$

Plotting

The plots for the deformed shape of the structural system and the internal stress distribution are obtained with the help of shape functions that are identified bearing the following hypotheses in mind:

- Unloaded elements
- No imposed deformations
- No shear deformability
- Linear elastic problem

In order to obtain the deformed configuration of the structural system the axial and the flexural behaviors are treated separately. The displacement models for these two cases are discussed below.

Axial Behaviour

The governing differential equation for axial behaviour is

$$\frac{d^2u}{dx'^2} = 0$$

Integrating the above relation we obtain the expression of the horizontal displacement function as

$$u(x') = C_1x' + C_2$$

The corresponding boundary conditions for the beam element under analysis are

$$u(x' = 0) = u_1$$

$$u(x' = l) = u_2$$

Imposing these boundary conditions we obtain the unknown coefficients of the system and the horizontal displacement function is rewritten as

$$\begin{aligned} u(x') &= (u_2 - u_1) \frac{x'}{l} + u_1 \\ &= \left(1 - \frac{x'}{l}\right) u_1 + \frac{x'}{l} u_2 \end{aligned}$$

The functions appearing with the horizontal nodal displacements here are known as the shape functions and they are given as

$$\begin{aligned} N_1(x') &= \left(1 - \frac{x'}{l}\right) \\ N_4(x') &= \frac{x'}{l} \end{aligned}$$

Thus, the horizontal displacement function can now be modified as

$$u(x') = N_1(x')u_1 + N_4(x')u_2$$

Flexural Behaviour

The governing differential equation for flexural behaviour is given by

$$\frac{d^4u}{dx'^4} = 0$$

Integrating the above relation we obtain the expression of the transverse displacement function as

$$v(x') = C_1 \frac{x'^3}{6} + C_2 \frac{x'^2}{2} + C_3x' + C_4$$

The corresponding boundary conditions for the beam element under analysis are

$$v(x' = 0) = v_1$$

$$v(x' = l) = v_2$$

$$v'(x' = 0) = \varphi_1$$

$$v'(x' = l) = \varphi_2$$

Imposing these boundary conditions we obtain the unknown coefficients of the system and the transverse displacement function is rewritten and the shape functions are identified as

$$\begin{aligned} v(x') &= \left[1 - 3\left(\frac{x'}{l}\right)^2 + 2\left(\frac{x'}{l}\right)^3 \right] v_1 + \left[\frac{x'}{l} - 2\left(\frac{x'}{l}\right)^2 + \left(\frac{x'}{l}\right)^3 \right] l\varphi_1 + \left[3\left(\frac{x'}{l}\right)^2 - 2\left(\frac{x'}{l}\right)^3 \right] v_2 \\ &\quad + \left[-\left(\frac{x'}{l}\right)^2 + \left(\frac{x'}{l}\right)^3 \right] l\varphi_2 \\ &= N_2(x')v_1 + N_3(x')\varphi_1 + N_5(x')v_2 + N_6(x')\varphi_2 \end{aligned}$$

The displacement function can now be assembled from the axial and the flexural responses as

$$\mathbf{U} = [u(x') \quad v(x')]$$

$$\mathbf{N} = \begin{bmatrix} N_1 & 0 & 0 & N_4 & 0 & 0 \\ 0 & N_2 & N_3 & 0 & N_5 & N_6 \end{bmatrix}$$

$$\mathbf{U}(x') = \mathbf{N}(x')\mathbf{s}'$$

The equilibrium in any section of the generic beam element considered provides us with the distribution functions of the internal stresses as

$$N(x') = -n_1 - \int_0^{x'} p_x(\xi) d\xi$$

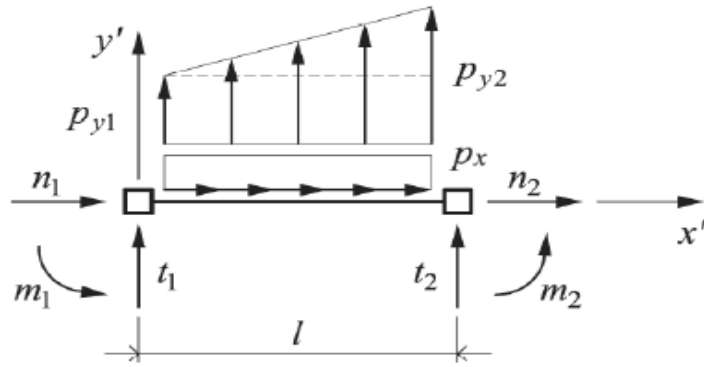
$$T(x') = t_1 + \int_0^{x'} p_y(\xi) d\xi$$

$$M(x') = -m_1 + t_1 x' + \int_0^{x'} p_y(\xi) d\xi$$

Where

ξ is a dimensionless coordinate given as $\frac{x'}{l}$

For our analysis, we consider a generic case of a linearly distributed transverse load and a uniformly distributed axial load acting on the beam element as shown.



The internal force distribution functions are obtained by utilizing the relations noted above as

$$N(x') = -n_1 - p_x x'$$

$$T(x') = t_1 + p_{y1} x' + (p_{y2} - p_{y1}) \frac{x'^2}{2l}$$

$$M(x') = -m_1 + t_1 x' + p_{y1} \frac{x'^2}{2} + (p_{y2} - p_{y1}) \frac{x'^3}{6l}$$

Chapter 10

SPECIALIZED PROBLEMS

Not all structures can be modeled properly with the base code only. Many specialized problems have to be taken into account when designing, for this reason some changes must be implemented in the code so that the results can better reflect the behavior of the problem at hand.

10.1 Elastic Constraints

In some cases structures might not be restrained to solid, immovable objects but to softer ones that behave like springs; these might be other flexible and extensible structures or soil. To account for their contribution we must introduce their stiffness into the system.

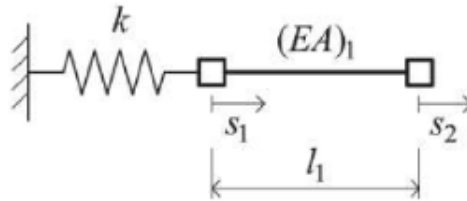
The following is the stiffness matrix for a 1D spring, the 2 degrees of freedom are related to the 2 nodes connected by the spring.

$$\mathbf{K}_e = \begin{bmatrix} k & -k \\ -k & k \end{bmatrix}$$

Beams work much the same way:

$$\mathbf{K}_1 = \begin{bmatrix} k_1 & -k_1 \\ -k_1 & k_1 \end{bmatrix} \quad \text{where } k_1 = \frac{(EA)_1}{l}$$

Therefore the stiffness coefficients for beams and springs can easily be assembled.

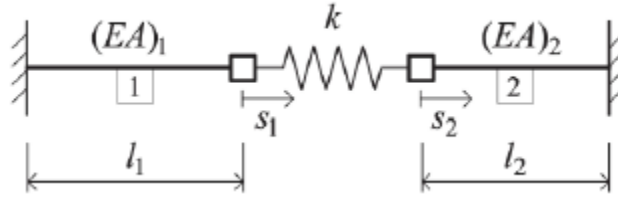


$$\mathbf{K} = \begin{bmatrix} k_1 + k & -k_1 \\ -k_1 & k_1 \end{bmatrix}$$

If the spring is an external constraint then one of its degrees of freedom will be restrained and the stiffness coefficient will only factor in in the global stiffness matrix of the structure in the diagonal.

$$k_{ii}^* = k_{ii} + k$$

If the spring is internal however both of the degrees of freedom take effect and the



Assembly procedure must include both diagonal terms and coupling.

$$\mathbf{K}_1 = \begin{bmatrix} k_1 & -k_1 \\ -k_1 & k_1 \end{bmatrix} \quad \mathbf{K}_2 = \begin{bmatrix} k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix}$$

Where

$$k_1 = \frac{(EA)_1}{l_1} \quad \text{and} \quad k_2 = \frac{(EA)_2}{l_2}$$

- Case (I): $k = 0$

$$\mathbf{K} = \begin{bmatrix} k_1 & 0 \\ 0 & k_2 \end{bmatrix}$$

- Case (II): $k \neq 0$

$$\mathbf{K} = \begin{bmatrix} k_1 + k & -k \\ -k & k_2 + k \end{bmatrix}$$

The degrees of freedom of the springs are always assigned at the nodes therefore in the assembly procedure it's possible to account for all the elastic constraints and restraints with the following equations to be applied in the global reference frame.

$$k_{ii}^* = k_{ii} + k$$

$$k_{jj}^* = k_{jj} + k$$

$$k_{ij}^* = k_{ij} - k$$

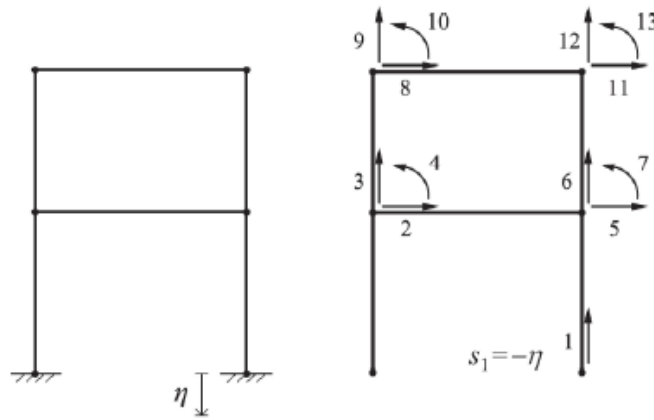
$$k_{ji}^* = k_{ji} - k$$

10.2 Imposed Displacements

The structural system under analysis might be subjected to imposed displacements that can be either absolute or relative. The imposed displacements have to be accounted for in the compatibility equations rather than the linear equilibrium equations. The imposed displacements generate reaction forces in the system and are to be accounted for in the analysis. The formulations pertaining to the two classes of imposed displacements as stated earlier are noted in the discussion that follows.

10.2.1 Imposed Absolute Displacements

The structural model, as shown below, could be subjected to an imposed absolute displacement in a particular DOF of the system. A typical example of this case would be foundation settlement. The degree of freedom on which the displacement is imposed is freed in the analysis. The structural system is characterized with n equilibrium equations corresponding to the n degrees of freedom, and m compatibility equations relating to the m number of imposed absolute displacements. Thus, effectively, the system has $(n+m)$ linear equations, equilibrium and compatibility, with $(n+m)$ unknowns, displacements and reactions. For the case illustrated below, $n = 13$ and $m = 1$.



The corresponding displacement vector, reaction vector, force vector, and the stiffness matrix are partitioned as:

$$\mathbf{s} = \begin{bmatrix} \mathbf{s}_1 \\ \mathbf{s}_2 \end{bmatrix} \quad \mathbf{r} = \begin{bmatrix} \mathbf{0} \\ \mathbf{r}_2 \end{bmatrix} \quad \mathbf{f} = \begin{bmatrix} \mathbf{f}_1 \\ \mathbf{f}_2 \end{bmatrix} \quad \mathbf{K} = \begin{bmatrix} \mathbf{K}_{11} & \mathbf{K}_{12} \\ \mathbf{K}_{21} & \mathbf{K}_{22} \end{bmatrix}$$

The compatibility and equilibrium equations, respectively, are written as:

$$\mathbf{s}_2 = \bar{\mathbf{s}} \\ \mathbf{f} + \mathbf{r} = \mathbf{K} \mathbf{s}$$

Now, substituting the compatibility condition in the equilibrium equations we obtain the following set of equations:

$$\mathbf{f}_1 = \mathbf{K}_{11} \mathbf{s}_1 + \mathbf{K}_{12} \bar{\mathbf{s}} \\ \mathbf{f}_2 + \mathbf{r}_2 = \mathbf{K}_{12}^T \mathbf{s}_1 + \mathbf{K}_{22} \bar{\mathbf{s}}$$

The unknowns of the system $\mathbf{s}$ and $\mathbf{r}_2$ are evaluated as:

$$\mathbf{s}_1 = \mathbf{K}_{11}^{-1} (\mathbf{f}_1 + \hat{\mathbf{f}}_1) \\ \mathbf{r}_1 = \mathbf{K}_{12}^T \mathbf{s}_1 + \mathbf{K}_{22} \bar{\mathbf{s}} - \mathbf{f}_2$$

Where $\hat{f}_1$ represents the nodal forces that would equivalently result in similar response of the structure as the imposed displacements, given by:

$$\hat{f}_1 = -K_{12}\bar{s}$$

The algorithm implemented is noted below, highlighting the modifications carried out in the vector of nodal loads and on the stiffness matrix.

$$f_j^* = f_j + k_{ji}\eta_i \quad j \neq i$$

$$f_i^* = \bar{k} \eta_i$$

$$k_{ij}^* = k_{ji}^* = 0 \quad j \neq i$$

$$k_{ii}^* = \bar{k}$$

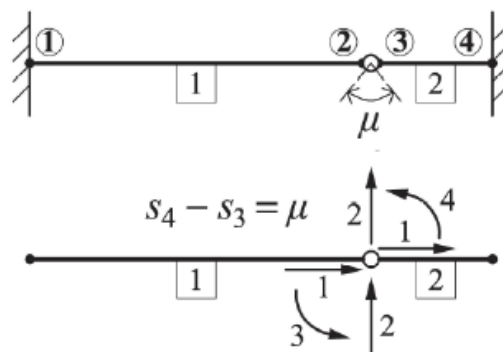
Where:

i is associated to the DOF on which the absolute displacement is imposed

η is the imposed absolute displacement

10.2.2 Imposed Relative Displacements

The structural model, as shown below, could be subjected to an imposed relative displacement between two degrees of freedom of the system. An example of this case could be flexural distortion as in this case. The degrees of freedom on which the relative displacement is imposed is freed in the analysis. The structural system is characterized with n equilibrium equations corresponding to the n degrees of freedom, and m compatibility equations relating to the m number of imposed relative displacements. Thus, effectively the system has $(n+m)$ linear equations with $(n+m)$ unknowns. For the case illustrated below, $n = 4$ and $m = 1$.



The corresponding displacement vector, reaction vector, force vector, and the stiffness matrix are partitioned as:

$$\mathbf{s} = \begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix} \quad \mathbf{r} = \begin{bmatrix} 0 \\ r_2 \\ r_3 \end{bmatrix} \quad \mathbf{f} = \begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix} \quad \mathbf{K} = \begin{bmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{bmatrix}$$

The compatibility and equilibrium equations, respectively, are written as:

$$\mathbf{s}_3 - \mathbf{s}_2 = \Delta \bar{\mathbf{s}}$$

$$\mathbf{f} + \mathbf{r} = \mathbf{K} \mathbf{s}$$

Now, replacing the compatibility equations in the equilibrium ones we identify the relations as:

$$\mathbf{f}_1 = \mathbf{K}_{11}\mathbf{s}_1 + \mathbf{K}_{12}\mathbf{s}_2 + \mathbf{K}_{13}(\mathbf{s}_1 + \Delta \bar{\mathbf{s}})$$

$$\mathbf{f}_2 + \mathbf{r}_2 = \mathbf{K}_{12}^T\mathbf{s}_1 + \mathbf{K}_{22}\mathbf{s}_2 + \mathbf{K}_{23}(\mathbf{s}_2 + \Delta \bar{\mathbf{s}})$$

$$\mathbf{f}_3 + \mathbf{r}_3 = \mathbf{K}_{13}^T\mathbf{s}_1 + \mathbf{K}_{23}^T\mathbf{s}_2 + \mathbf{K}_{33}(\mathbf{s}_2 + \Delta \bar{\mathbf{s}})$$

The system is solved to determine the unknowns of the problem, $\mathbf{s}$ and $\mathbf{r}_3$ as:

$$\begin{bmatrix} s_1 \\ s_2 \end{bmatrix} = \begin{bmatrix} \mathbf{K}_{11} & \mathbf{K}_{12} + \mathbf{K}_{13} \\ \mathbf{K}_{11}^T + \mathbf{K}_{13}^T & \mathbf{K}_{22} + \mathbf{K}_{23}^T + \mathbf{K}_{23} + \mathbf{K}_{33} \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{f}_1 - \mathbf{K}_{13} \Delta \bar{\mathbf{s}} \\ \mathbf{f}_2 + \mathbf{f}_3 - \mathbf{K}_{23} \Delta \bar{\mathbf{s}} \end{bmatrix}$$

$$\mathbf{r}_3 = \mathbf{K}_{13}^T\mathbf{s}_1 + (\mathbf{K}_{23}^T + \mathbf{K}_{33})\mathbf{s}_2 + \mathbf{K}_{33}\Delta \bar{\mathbf{s}} - \mathbf{f}_3$$

The algorithm implemented is noted below, highlighting the modifications carried out in the vector of nodal loads and on the stiffness matrix.

$$f_h^* = f_h + k_{hj}\mu_{ij} \quad h \neq i, \quad h \neq j$$

$$f_i^* = f_i + f_j - (k_{ij} + k_{jj} + \bar{k})\mu_{ij}$$

$$f_j^* = \bar{k} \mu_{ij}$$

$$k_{ih}^* = k_{ih} + k_{jh} = k_{hi}^* \quad h \neq i, \quad h \neq j$$

$$k_{hj}^* = k_{jh}^* = 0 \quad h \neq i, \quad h \neq j$$

$$k_{ii}^* = k_{ii} + 2k_{ij} + k_{jj} + \bar{k}$$

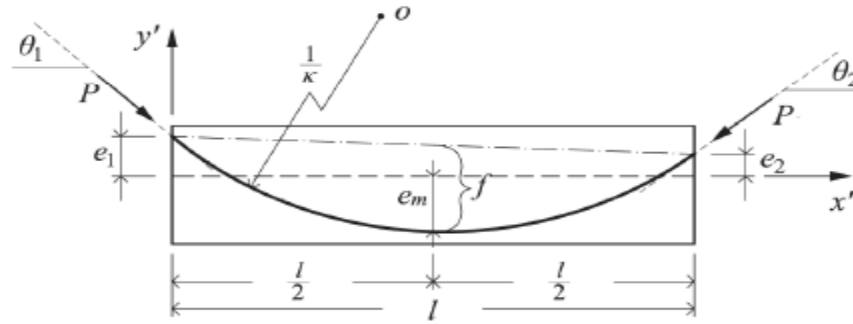
$$k_{jj}^* = \bar{k}$$

$$k_{ij}^* = k_{ji}^* = -\bar{k}$$

Here i and j are associated to the DOFs on which the relative displacement is imposed.

10.3 Prestressing

Prestressing is adopted in concrete structures in order to counterbalance the effects of the applied loading with the help of tensioned steel wires, cables or tendons.



The objective of prestressing is to minimize the tensile stresses acting on concrete so as to avoid cracking. A typical concrete beam element with prestressing is illustrated below.

Here

$e = e(x')$ is the eccentricity of the tendon profile

$\theta = \theta(x')$ is the slope of the tendon profile

$\kappa = \kappa(x')$ is the curvature of the tendon profile

Several hypothesis are considered for the analysis that concern the prestressing forces, the tendon profile, and the tendon curvature as listed below.

- **Constant Prestressing Force**

The prestressing force acting on the tendon is assumed to be constant.

$$P_1 = P_2 = P = \text{constant}$$

- **Parabolic Tendon Profile**

The eccentricity of the tendon profile is considered to be parabolic.

$$e(x') = ax'^2 + bx' + c$$

Imposing the boundary conditions of known eccentricities at the extremes and at the center, the eccentricity of is determined as:

$$e(0) = e_1 \quad e\left(\frac{l}{2}\right) = e_m \quad e(l) = e_2$$

$$e(x') = 2(e_1 + e_2 - 2e_m)\left(\frac{x'}{l}\right)^2 - (3e_1 + e_2 - 4e_m)\left(\frac{x'}{l}\right) + e_1$$

The slope of the tendon profile is now evaluated as the derivative of the eccentricity as:

$$\theta(x') = e'(x') = \frac{4}{l}(e_1 + e_2 - 2e_m)\left(\frac{x'}{l}\right) - \frac{1}{l}(3e_1 + e_2 - 4e_m)$$

$$\theta_1 = \theta(0) = -\frac{1}{l}(3e_1 + e_2 - 4e_m)$$

$$\theta_2 = \theta(l) = \frac{1}{l}(e_1 + 3e_2 - 4e_m)$$

The tendon sag f , as represented in the figure above is computed as

$$e''(x') = \frac{4}{l^2}(e_1 + e_2 - 2e_m) = \frac{8f}{l^2}$$

$$f = \frac{e_1 + e_2}{2} - e_m$$

- **Small Tendon Slope**

The slope of the tendon profile, and as a result, the derivative of the eccentricity, is assumed to be small, leading to the following relations:

$$\theta(x') = e'(x') \ll 1$$

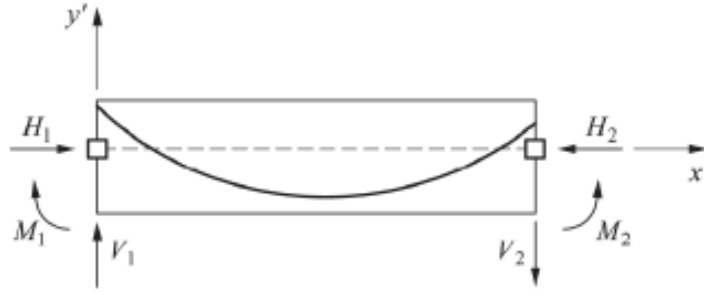
$$\sin\theta(x') \cong \theta(x')$$

$$\cos\theta(x') \cong 1$$

As a result, the curvature of the tendon profile is expressed as

$$\kappa(x') = \frac{e''(x')}{[1 + e''(x')^2]^{\frac{3}{2}}} \cong e''(x') = \frac{8f}{l^2} = \text{constant}$$

The application of prestressing, taking into account the above-stated hypotheses, results in anchorage nodal forces as sketched below and they are computed as:



$$H_1 = P \cos \theta_1 \cong P$$

$$H_2 = P \cos \theta_2 \cong P$$

$$V_1 = P \sin \theta_1 \cong P \theta_1$$

$$V_2 = P \sin \theta_2 \cong P \theta_2$$

$$M_1 = H_1 e_1 \cong P e_1$$

$$M_2 = H_2 e_2 \cong P e_2$$

The resulting anchorage nodal force vector $\hat{\mathbf{f}}'_a$ is:

$$\hat{\mathbf{f}}'_a = [H_1 \quad V_1 \quad -M_1 \quad | \quad -H_2 \quad -V_2 \quad M_2]$$

If a balancing load w is applied to the tendon a distributed transverse force acts on the beam element due to the curvature of the cable and through cable equilibrium we obtain the following relation:

$$w = 2P \sin \frac{\kappa}{2} \cong 2P \frac{\kappa}{2} = P \kappa = P \frac{8f}{l^2} = P \frac{4}{l^2} (e_1 + e_2 - 2e_m)$$

The equivalent nodal forces generated by the application of this loading condition is expressed as:

$$\hat{\mathbf{f}}'_b = \left[0 \quad \frac{wl}{2} \quad \frac{wl^2}{12} \quad | \quad 0 \quad \frac{wl}{2} \quad -\frac{wl^2}{12} \right]^T$$

The overall equivalent nodal forces incorporating the contributions from the anchorage nodal forces and the nodal forces due to the balancing loads is given by:

$$\hat{\mathbf{f}}'_p = P \left[\left[1 \quad -\frac{e_1 - e_2}{l} \quad \frac{1}{3}(-2e_1 + e_2 - 2e_m) \quad | \quad -1 \quad \frac{e_1 - e_2}{l} \quad -\frac{1}{3}(-2e_1 + e_2 - 2e_m) \right]^T \right]$$

The equilibrium of the system is now modified and the forces at the beam end $\mathbf{f}'^*$ is recognized as:

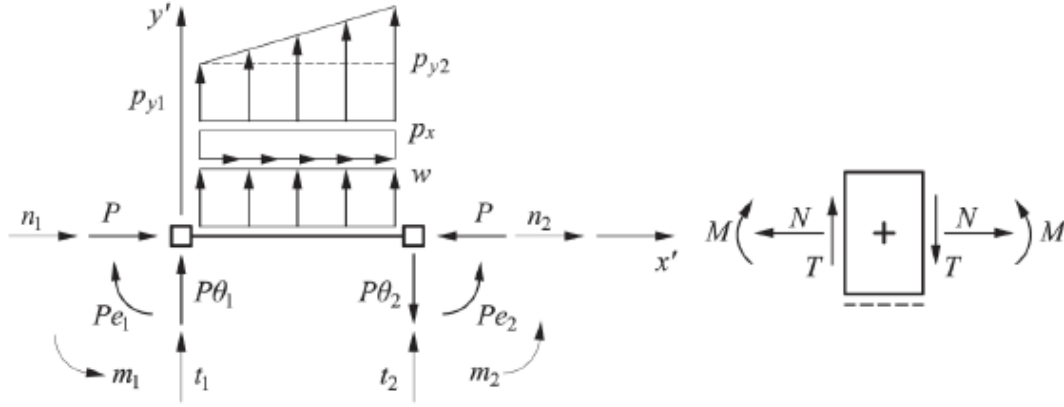
$$\mathbf{f}' + \hat{\mathbf{f}}'_p = \mathbf{K}' \mathbf{s}'$$

$$\mathbf{f}' + \hat{\mathbf{f}}'_a + \hat{\mathbf{f}}'_b = \mathbf{K}' \mathbf{s}'$$

$$\mathbf{f}'^* = \mathbf{f}' + \hat{\mathbf{f}}'_a = \mathbf{K}'\mathbf{s}' - \hat{\mathbf{f}}'_b$$

Stress Analysis

The internal stress resultants for a beam element subjected to prestressing forces as presented below can be formulated as:



$$N(x') = -n_1 - p_x x' - P$$

$$T(x') = t_1 - P\theta_1 + (p_{y1} + w)x' + (p_{y2} - p_{y1})\frac{x'^2}{2l}$$

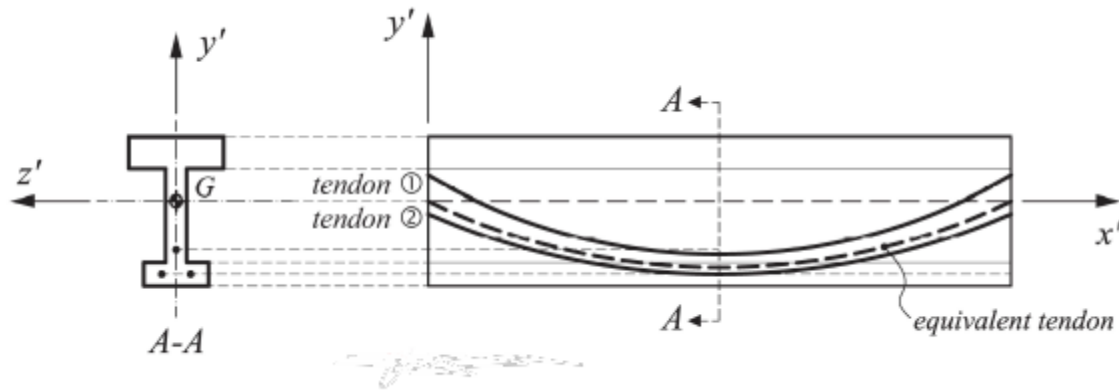
$$M(x') = -m_1 + Pe_1 + (t_1 + P\theta_1)x' + (p_{y1} + w)\frac{x'^2}{2} + (p_{y2} - p_{y1})\frac{x'^3}{6l}$$

Equivalent Tendon

A structural member could be composed of multiple prestressed tendons as demonstrated below.

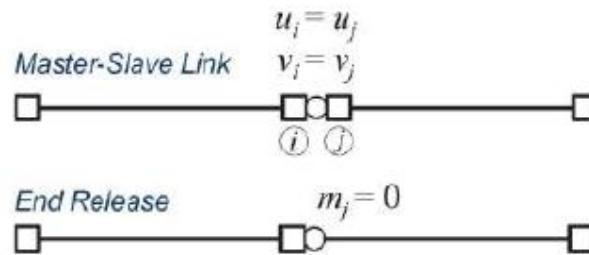
In such cases, an equivalent tendon is assumed to fictitiously replace them. The prestressing force applied to each tendon and the eccentricities of each tendon at the extremes and at the middle is requested from the user as input. The equivalent tendon profile is then determined, characterized with eccentricity estimated as:

$$e(x') = \frac{1}{P} \sum_{i=1}^n P_i e_i(x')$$

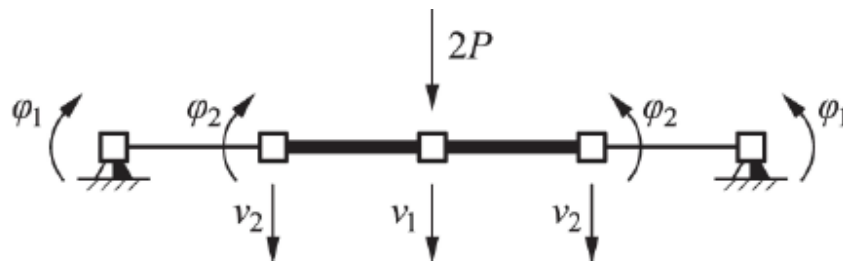


10.4 End Releases

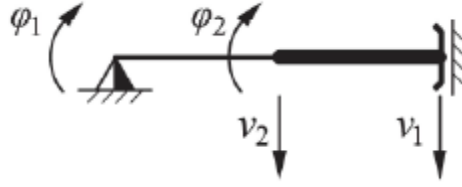
An alternate way to account for internal links in structures is through the implementation of end releases. Instead of using the master-slave dynamic, that sometimes requires the introduction of additional nodes in the input, it is possible to release the displacement of the degree of freedom related to the internal constraint between multiple elements.



The way this is done is by applying static condensation. This procedure allows to solve the system by disregarding the stiffness contribution of constrained elements, to the nodes interested by the release, except for one. The process is illustrated in the following example:



Due to the symmetry the problem can be simplified by applying an end release in the center node. This allows for lower computing costs and thus faster solution.



Decomposition of the displacement vector:

$$\mathbf{s}_1 = [v_1 \quad \varphi_1]^T \quad \mathbf{s}_2 = [v_2 \quad \varphi_2]^T$$

Decomposition of the nodal load vector:

$$\mathbf{f}_1 = \mathbf{K}_{11}\mathbf{s}_1 + \mathbf{K}_{12}\mathbf{s}_2$$

$$\mathbf{f}_2 = \mathbf{K}_{12}^T\mathbf{s}_1 + \mathbf{K}_{22}\mathbf{s}_2$$

The displacement vector $\mathbf{s}_2$ is taken from the above equation and substituted to extract the expression for $\mathbf{s}_1$ as:

$$\mathbf{s}_2 = \mathbf{K}_{12}^{-1}(\mathbf{f}_2 - \mathbf{K}_{12}^T\mathbf{s}_1)$$

$$\mathbf{s}_1 = \mathbf{K}_{12}^{*-1} \mathbf{f}_1^*$$

Where:

$$\mathbf{f}_1^* = \mathbf{f}_1 - \mathbf{K}_{12} \mathbf{K}_{22}^{-1} \mathbf{f}_2$$

$$\mathbf{K}_{11}^* = \mathbf{K}_{11} - \mathbf{K}_{12} \mathbf{K}_{22}^{-1} \mathbf{K}_{12}^T$$

The problem is thus solved on the local level for any type of end release. Matrices and vectors are subdivided as follows:

$$\begin{bmatrix} f_a \\ f_b \end{bmatrix} + \begin{bmatrix} \hat{f}_a \\ \hat{f}_b \end{bmatrix} = \begin{bmatrix} K_a & K_c \\ K_c^T & K_b \end{bmatrix} \begin{bmatrix} s_a \\ s_b \end{bmatrix}$$

$$\begin{bmatrix} t_1 \\ m_1 \\ t_2 \\ m_2 \end{bmatrix} + \begin{bmatrix} \hat{t}_1 \\ \hat{m}_1 \\ \hat{t}_2 \\ \hat{m}_2 \end{bmatrix} = \begin{bmatrix} k_{11} & k_{12} & k_{13} & k_{14} \\ k_{21} & k_{22} & k_{23} & k_{24} \\ k_{31} & k_{32} & k_{33} & k_{34} \\ k_{41} & k_{42} & k_{43} & k_{44} \end{bmatrix} \begin{bmatrix} v_1 \\ \varphi_1 \\ v_2 \\ \varphi_2 \end{bmatrix}$$

And the following transformations to the local stiffness matrix and nodal load vector must be applied:

$$\begin{cases} f_a = K_a s_a + K_c s_b - \hat{f}_a = 0 \\ f_b + \hat{f}_b = K_c^T s_a + K_b s_b \end{cases} \Rightarrow \begin{cases} s_a = -K_a^{-1} K_c s_b + K_a^{-1} \hat{f}_a \\ f_b + f_b^* = K_b^* s_b \end{cases}$$

Where

$$\begin{aligned} f_b^* &= \hat{f}_b - K_c^T K_a^{-1} \hat{f}_a \\ K_b^* &= K_b - K_c^T K_a^{-1} K_c \end{aligned}$$

$$\begin{bmatrix} t_1 \\ m_1 \\ t_2 \\ m_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \hat{f}_b \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & & & \\ 0 & & K_b^* & \\ 0 & & & \end{bmatrix} \begin{bmatrix} v_1 \\ \varphi_1 \\ v_2 \\ \varphi_2 \end{bmatrix}$$

Or in simplified form:

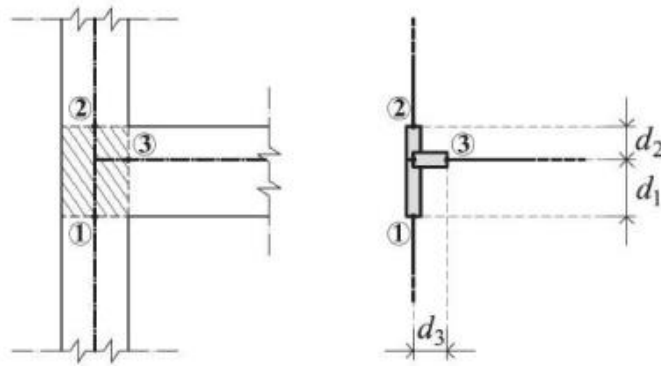
$$\mathbf{K}^* = \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{K}_b^* \end{bmatrix}$$

$$\hat{\mathbf{f}}^* = \begin{bmatrix} \mathbf{0} \\ \mathbf{f}_b^* \end{bmatrix}$$

$$\mathbf{f} + \hat{\mathbf{f}}^* = \mathbf{K}^* \mathbf{s}$$

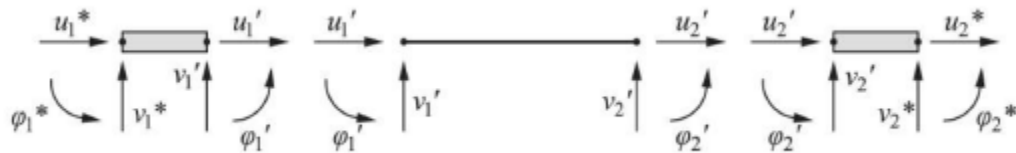
10.5 Rigid ends and rigid joints

The solution of frame problems can be further enhanced by considering part of the structure as rigid, removing them from the elastic equation and allowing for better precision where beam theory could not be applied. We implemented two different possibility for modeling of rigid structures: Rigid ends and rigid joints. Both of them work by changing data about the ends of the flexible beams in order to account for compatibility, through matrix transformations.



10.5.1 Rigid ends

Rigid ends allow for the user to design beams that are not entirely flexible but include rigid sections at its sides.



The following are the compatibility equations for such beams:

$$\begin{cases} u_1' = u_1^* \\ v_1' = v_1^* + d_1 \phi_1^* \\ \phi_1' = \phi_1^* \end{cases} \quad \begin{cases} u_2' = u_2^* \\ v_2' = v_2^* - d_2 \phi_2^* \\ \phi_2' = \phi_2^* \end{cases}$$

Where d_1 and d_2 are the respective lengths of the rigid ends in node 1 and node 2. This reasoning can be applied to any element in the structure and thus the transformation can be applied at the local level for each beam involved.

$$\mathbf{s}'_1 = \mathbf{T}_{d_1} \mathbf{s}_1^*$$

$$\mathbf{s}'_2 = \mathbf{T}_{d_2} \mathbf{s}_2^*$$

Where

$$\mathbf{T}_{d_1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & d_1 \\ 0 & 0 & 1 \end{bmatrix} \quad \mathbf{T}_{d_2} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -d_2 \\ 0 & 0 & 1 \end{bmatrix}$$

These transformation matrices are created and applied inside the **MKK** subroutine are easily assembled for each element by utilizing the **ECC** matrix, created during the **GEOMET** subroutine that stores all the lengths of rigid elements.

ECC (NELE,2)		
Element	d_1	d_2
1		
2		
3		
...		
NNODE		

The matrix transformations applied to the displacements must be applied element by elements after the return to the local reference system.

$$\begin{bmatrix} s'_1 \\ s'_2 \end{bmatrix} = \begin{bmatrix} T_{d1} & 0 \\ 0 & T_{d2} \end{bmatrix} \begin{bmatrix} s_1^* \\ s_2^* \end{bmatrix}$$

$$\Rightarrow \mathbf{s}' = \mathbf{T}_d \mathbf{s}^*$$

The local stresses are also transformed as follows:

$$\begin{cases} n_1^* = n'_1 \\ t_1^* = t'_1 \\ m_1^* = m'_1 + d_1 t'_1 \end{cases} \quad \begin{cases} n_2^* = n'_2 \\ t_2^* = t'_2 \\ m_2^* = m'_2 + d_2 t'_2 \end{cases}$$

$$\mathbf{f}_1^* = \mathbf{T}_{d1}^T \mathbf{f}'_1$$

$$\mathbf{f}_2^* = \mathbf{T}_{d2}^T \mathbf{f}'_2$$

$$\mathbf{T}_{d1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & d_1 & 1 \end{bmatrix} \quad \mathbf{T}_{d2} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -d_2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} f_1^* \\ f_2^* \end{bmatrix} = \begin{bmatrix} \mathbf{T}_{d1}^T & \mathbf{0} \\ \mathbf{0} & \mathbf{T}_{d2}^T \end{bmatrix} \begin{bmatrix} f'_1 \\ f'_2 \end{bmatrix}$$

$$\Rightarrow \mathbf{f}^* = \mathbf{T}_d^T \mathbf{f}'$$

And lastly though the following equations the transformations needed to the local stiffness matrices and the nodal load vector can be obtained to solve the system:

$$\mathbf{K}' \mathbf{s}' = \mathbf{f}' + \hat{\mathbf{f}}'$$

$$\mathbf{K}' \mathbf{T}_d \mathbf{s}^* = (\mathbf{T}_d^T)^{-1} \mathbf{f}^*$$

$$\mathbf{T}_d^T \mathbf{K}' \mathbf{T}_d \mathbf{s}^* = \mathbf{f}' + \hat{\mathbf{f}}^*$$

$$\mathbf{K}^* \mathbf{s}^* = \mathbf{f}^* + \hat{\mathbf{f}}^*$$

Where

$$\mathbf{K}^* = \mathbf{T}_d^T \mathbf{K}' \mathbf{T}_d \quad \hat{\mathbf{f}}^* = \mathbf{T}_d^T \hat{\mathbf{f}}'$$

The following equations are then used for the stress analysis:

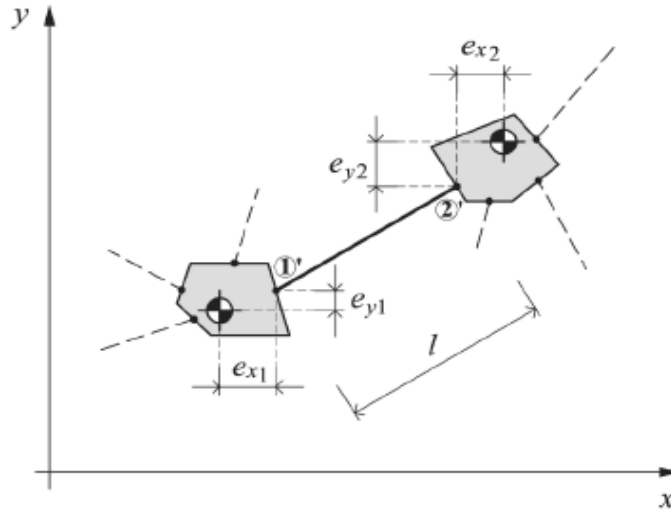
$$\begin{cases} \mathbf{s}_m^* = \mathbf{T}^T \mathbf{s}_m \\ \mathbf{s}_m' = \mathbf{T}_d \mathbf{s}_m^* \end{cases}$$

$$\mathbf{f}_m' = \mathbf{K}_m' \mathbf{s}_m' - \hat{\mathbf{f}}_m'$$

10.5.2 Rigid joints

Rigid joints, also called offsets, work analogously to rigid ends. This time disregarding elements like d-regions and joints between different beams. This way, in the case of multi-connected beams, a centroid for the connection can be manually computed by the user and the offset from it taken

into account, inducing both bending moments and axial forces in the surrounding beams. This last problem specifically can arise when two elements with different sections are connected in succession and their centroids are misaligned.



Compatibility equations:

$$\begin{cases} u_1^* = u_1' - e_{y1}\varphi_1^* \\ v_1^* = v_1' + e_{x1}\varphi_1^* \\ \varphi_1^* = \varphi_1' \end{cases} \quad \begin{cases} u_2^* = u_2' - e_{y2}\varphi_2^* \\ v_2^* = v_2' + e_{x2}\varphi_2^* \\ \varphi_2^* = \varphi_2' \end{cases}$$

Equilibrium equations:

$$\begin{cases} n_1^* = n_1' \\ t_1^* = t_1' \\ m_1^* = m_1' + e_{x1}t_1' - e_{y1}n_1' \end{cases} \quad \begin{cases} n_2^* = n_2' \\ t_2^* = t_2' \\ m_2^* = m_2' + e_{x2}t_2' - e_{y2}n_2' \end{cases}$$

Transformations:

$$\mathbf{s}_1' = \mathbf{T}_{e1}\mathbf{s}_1^*$$

$$\mathbf{f}_1^* = \mathbf{T}_{e1}^T\mathbf{f}_1'$$

Where:

$$\mathbf{T}_{e1} = \begin{bmatrix} 1 & 0 & -e_{y1} \\ 0 & 1 & e_{x1} \\ 0 & 0 & 1 \end{bmatrix}$$

The transformations must be applied in the global reference system and the matrices are generated element by element using the **OFF** matrix that contains all the information about eccentricities.

OFF (NELE,4)				
Element	e_{x1}	e_{y1}	e_{x2}	e_{y2}
1				
2				
3				
...				
NNODE				

The following are all the transformations to be applied, first in the **MKK** subroutine when determining the global stiffness matrix and nodal load vector, second in the stress subroutine when computing the stresses after the elastic problem is solved.

$$\begin{aligned} \mathbf{s}' &= \mathbf{T}_e \mathbf{s}^* & \mathbf{s}' &= \mathbf{T}_e^T \mathbf{f}' \\ \mathbf{K}^* &= \mathbf{T}_e^T \mathbf{K}' \mathbf{T}_e & \hat{\mathbf{f}}^* &= \mathbf{T}_e^T \hat{\mathbf{f}}' \end{aligned}$$

Where:

$$\mathbf{T}_e = \begin{bmatrix} \mathbf{T}_{e1} & 0 \\ 0 & \mathbf{T}_{e2} \end{bmatrix}$$

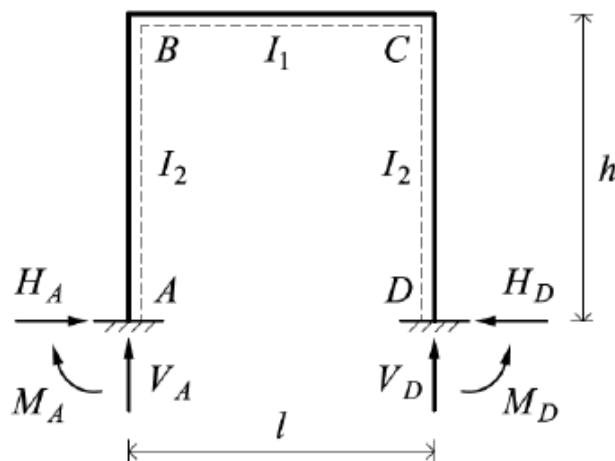
Chapter 11

BENCHMARKS

To ensure that the code works properly in all subroutines and specialized problems the program must be tested. The results can be compared to benchmarks obtained with analytical formulas to gauge accuracy and precision.

11.1 Portal frame

The first benchmark is a general purpose one to check if the program works and can handle different sections and materials for elements, internal and external restraints, distributed and point loads.



E	3.00E+07	kN/m ²
I₁	1.04166E-02	m ⁴
I₂	5.20830E-02	m ⁵
l	6.00	m
h	4.00	m
EA	∞	kN

Figure 11.1: Portal frame

We'll study this frame under 3 different loading conditions:

1. Distributed vertical load (Figure 11.3 & Table 11.1)
2. Distributed lateral load (Figure 11.4 & Table 11.2)
3. Lateral point load (Figure 11.5 & Table 11.3)

In Figure 11.2: $q = 50$ kN/m, $P = 200$ kN and the area of every cross section is $A = 10^7 \times (i_1/l^2)$.

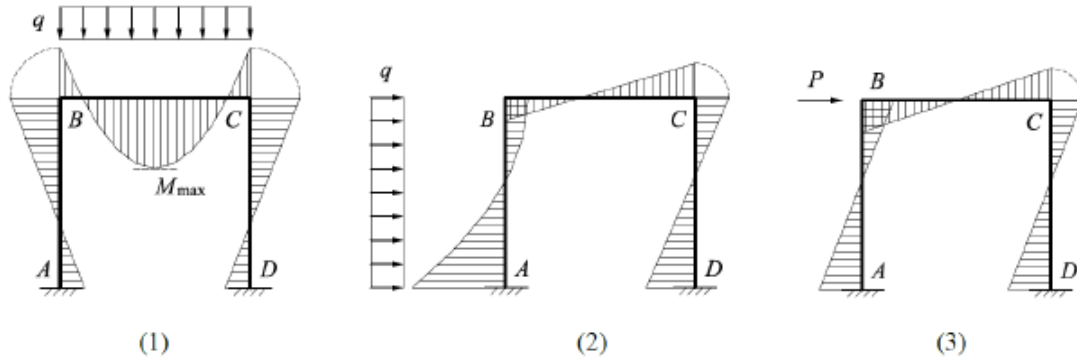
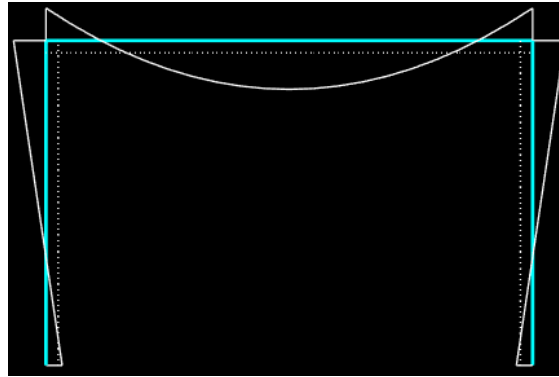


Figure 11.2: Loading conditions and resulting moment diagrams

The program outputs a *.txt* and a *.dat* file containing all the information related to the solution of the problem. Through the *.dat* file it is possible to plot the deformed configuration and the internal forces diagrams, through the *.txt* file it is possible to confront both displacement and internal forces in all the nodes of the frame. The results of the benchmarks are contained in the following pictures and tables.

11.1.1 Loading case 1



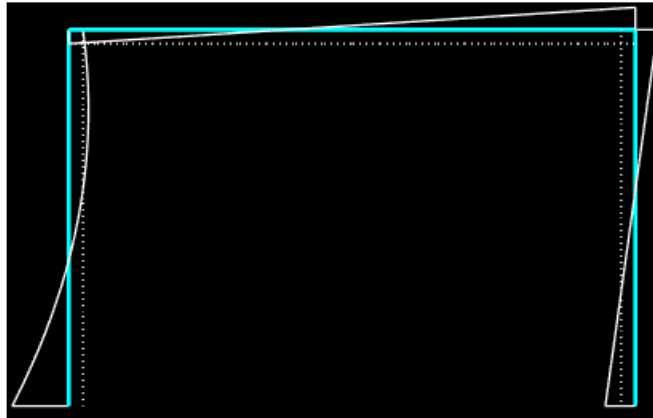
Bending Moment Diagram

Figure 11.3: Distributed Vertical Load on Beam

Output Parameter	Numerical	Reference	Error [%]
H_A [kN]	0.3375×10^2	0.3375×10^2	0.0
H_D [kN]	0.3375×10^2	0.3375×10^2	0.0
V_A [kN]	0.15×10^2	0.15×10^2	0.0
V_D [kN]	0.15×10^2	0.15×10^2	0.0
M_A [kN. m]	0.45×10^2	0.45×10^2	0.0
M_B [kN. m]	-0.9×10^2	-0.9×10^2	0.0
M_{max} [kN. m]	0.135×10^2	0.135×10^2	0.0
M_C [kN. m]	-0.9×10^2	-0.9×10^2	0.0
M_D [kN. m]	0.45×10^2	0.45×10^2	0.0

Table 11.1: Distributed vertical load results

11.1.2 Loading case 2



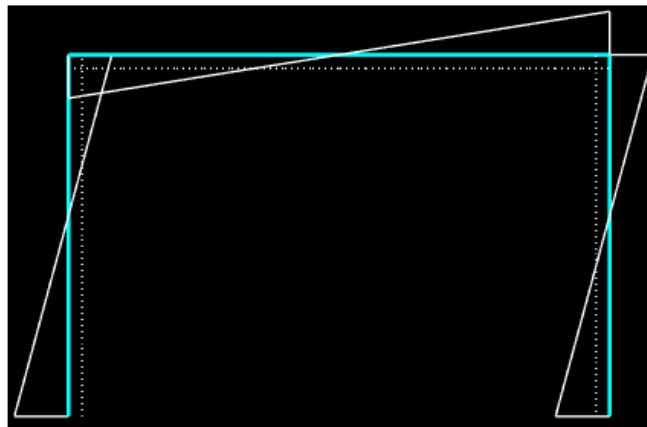
Bending Moment Diagram

Figure 11.4: Distributed Vertical Load on column

Output Parameter	Numerical	Reference	Error [%]
H_A [kN]	-0.1575×10^2	-0.1575×10^2	0.0
H_D [kN]	0.425×10^2	0.425×10^2	0.0
V_A [kN]	-0.19753×10^2	-0.19753×10^2	0.0
V_D [kN]	0.19753×10^2	0.19753×10^2	0.0
M_A [kN. m]	-0.18407×10^2	-0.18407×10^2	0.0
M_B [kN. m]	0.45926×10^2	0.45926×10^2	0.0
M_C [kN. m]	-0.72593×10^2	-0.72593×10^2	0.0
M_D [kN. m]	0.97407×10^2	0.97407×10^2	0.0

Table 11.2: Distributed lateral load results

11.1.3 Loading case 3



Bending Moment Diagram

Figure 11.5: Lateral point load

Output Parameter	Numerical	Reference	Error [%]
H_A [kN]	-0.1×10^2	-0.1×10^2	0.0
H_D [kN]	0.1×10^2	0.1×10^2	0.0
V_A [kN]	-0.59259×10^2	-0.59259×10^2	0.0
V_D [kN]	0.59259×10^2	0.59259×10^2	0.0
M_A [kN.m]	-0.22222×10^2	-0.22222×10^2	0.0
M_B [kN.m]	0.17778×10^2	0.17778×10^2	0.0
M_C [kN.m]	0.17778×10^2	0.17778×10^2	0.0
M_D [kN.m]	0.22222×10^2	0.22222×10^2	0.0

Table 11.3: Lateral point load results

The numerical results match perfectly and no error is reported.

11.2 Clamped beam with nodal load: Influence of shear deformability

The first benchmark is a double clamped beam with a concentrated nodal load applied at $2/3$ length. The aim is to investigate the influence of the shear deformability coefficient.

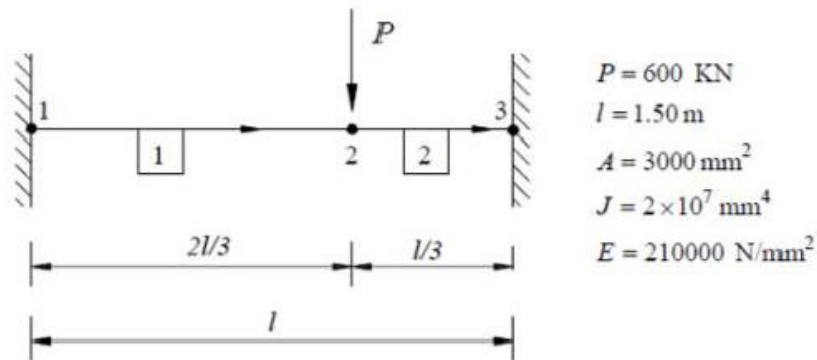
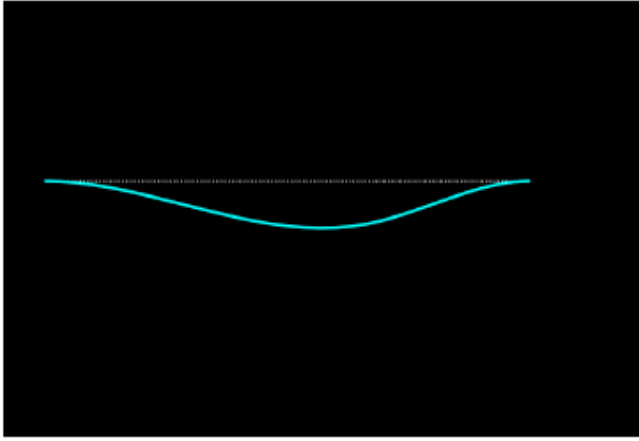


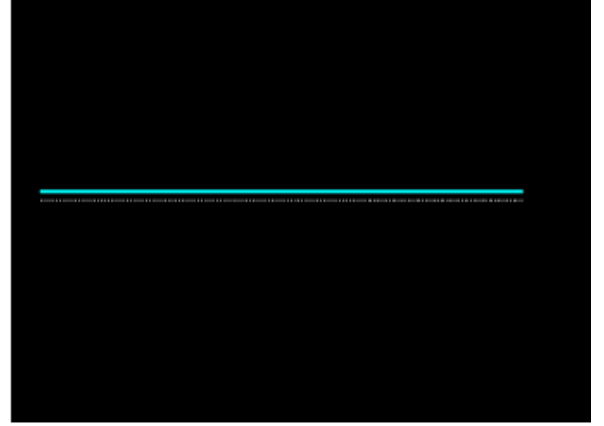
Figure 11.6: Doubly-Clamped beam

11.2.1 Case 1: Shear Deformability Coefficient $\chi = 0$

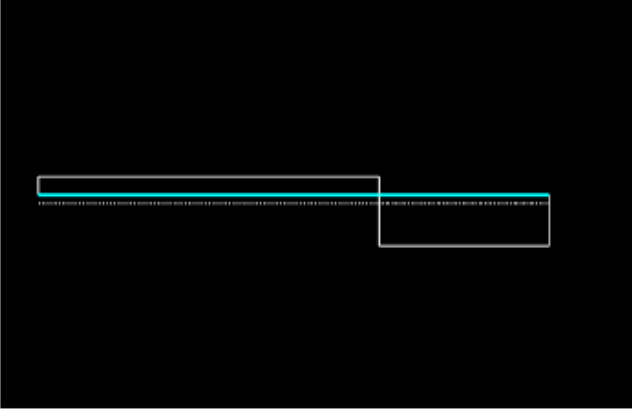
We firstly consider the structural problem with a shear deformability coefficient $\chi=0$. The displacement results for node 2 are presented in the table below.



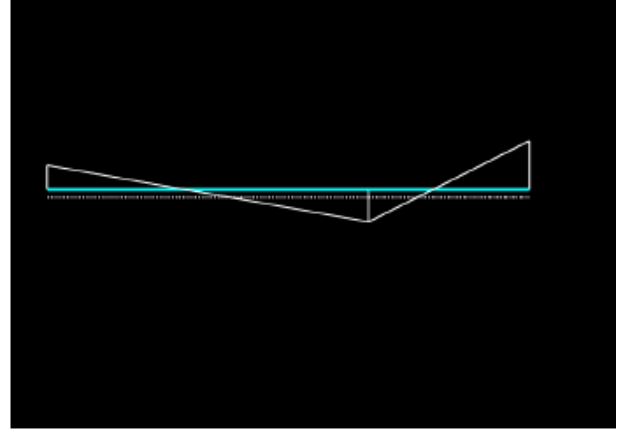
(a) Deflection Diagram



(b) Axial Diagram



(c) Shear Force Diagram



(d) Bending Moment Diagram

Figure 11.7: Doubly-clamped beam with $\chi = 0$

Output Parameter	Numerical	Reference	Error [%]
$v_2[\text{mm}]$	-1.764	-1.764	0.0
$\varphi_2[\text{rad}]$	2.646 E-03	2.646 E-03	0.0

Table 11.4: Doubly-clamped beam with $\chi = 0$ results

11.2.2 Case 2: Shear Deformability Coefficient $\chi = 2.5$

We next consider the same problem with the shear deformability coefficient set to $\chi=2.5$ and the poisons coefficient $\nu=0.3$.

Output Parameter	Numerical	Reference	Error [%]
$v_2[\text{mm}]$	-3.869	-3.869	0.0
$\varphi_2[\text{rad}]$	2.149 E-03	2.149 E-03	0.0

Table 11.5: Doubly-clamped beam with $\chi = 2.5$ results

11.3 Frame with uniform temperature change

The next benchmark is a two element frame, clamped at both ends and subject to a uniform temperature change on both elements.

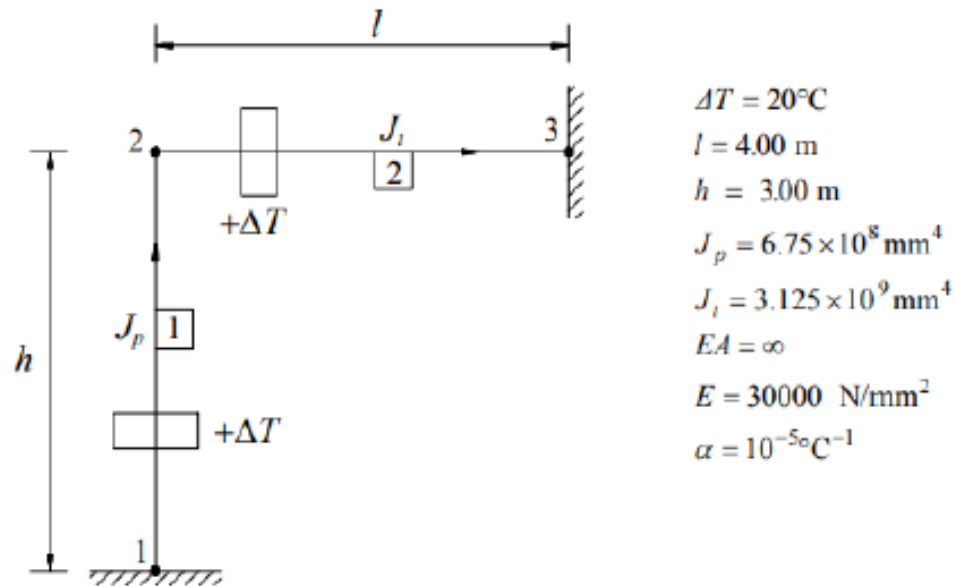


Figure 11.8: Frame under uniform temperature change

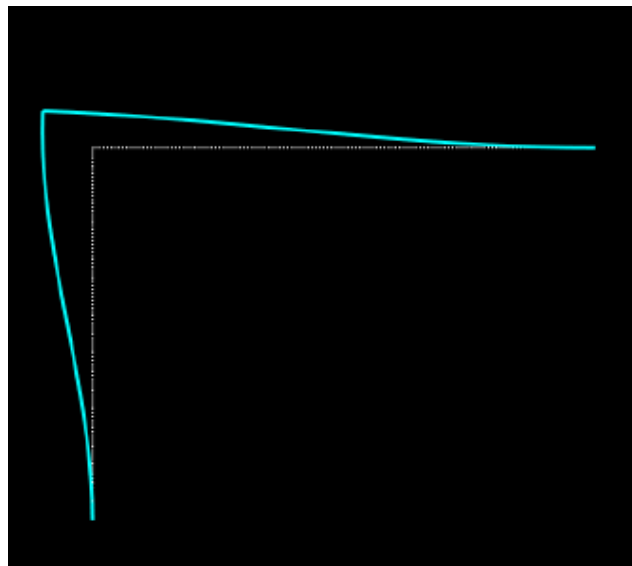
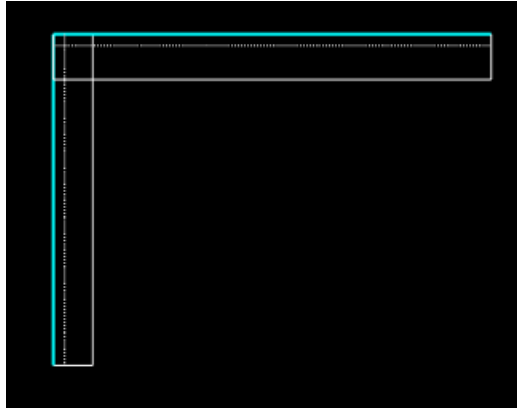
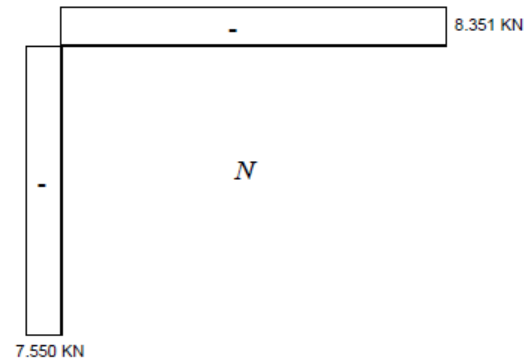


Figure 11.9: Deformed configuration

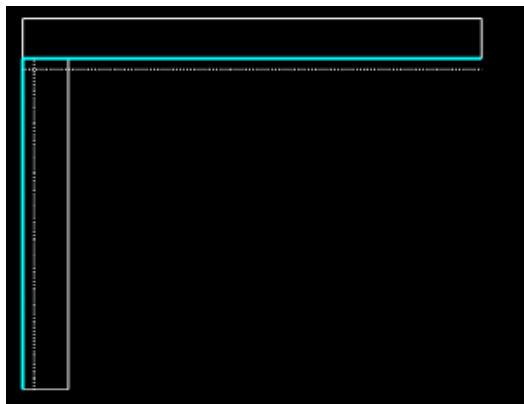


(a) Numerical solution

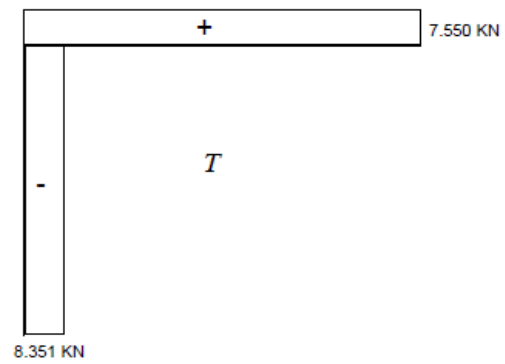


(b) Analytical solution

Figure 11.10: Axial Force

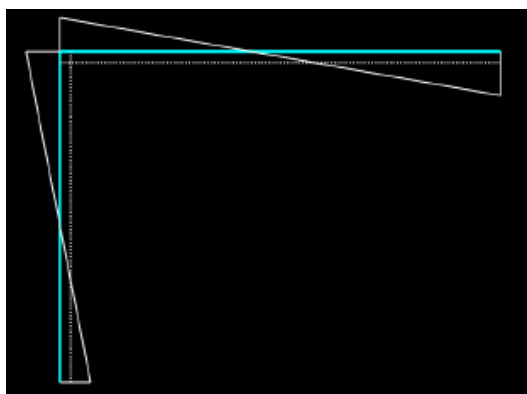


(a) Numerical solution

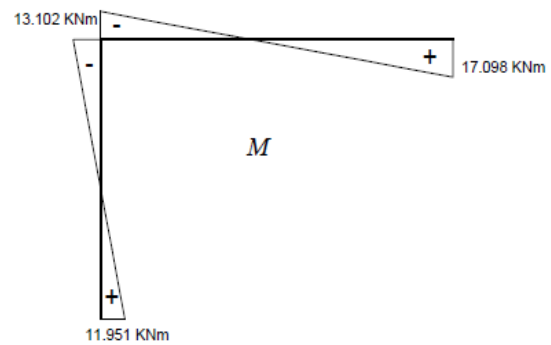


(b) Analytical solution

Figure 11.11: Shear Force



(a) Numerical solution



(b) Analytical solution

Figure 11.12: Bending Moment

The numerical results can be read in the output and are as follows in Table 11.6, they match the analytical solution exactly.

Output Parameter	Numerical	Reference	Error [%]
N_1 [kN]	7.550	7.550	0.0
T_1 [kN]	8.351	8.351	0.0
M_1 [kN. m]	11.951	11.951	0.0
N_2 [kN]	8.351	8.351	0.0
T_2 [kN]	7.550	7.550	0.0
M_2 [kN. m]	13.102	13.102	0.0
M_3 [kN. m]	17.098	17.098	0.0

Table 11.6: Frame under uniform temperature change results

11.4 Frame with foundation settlement and internal constraint

This benchmark is to test the imposed displacement functionality, with this expansion to the code the problem of imposed displacements can be solved and as a result, the internal forces and deformations are given as output.

Another functionality tested with this benchmark is the end releases, foregoing the assignment of master and slave relations between the nodes. The internal hinge is introduced by connecting 2 different nodes with links and by releasing the rotation at the end of the element. With this test the results show an error smaller than 1%. (Separate identical results not shown in the report)

The problem is defined as described in Figure 11.6.

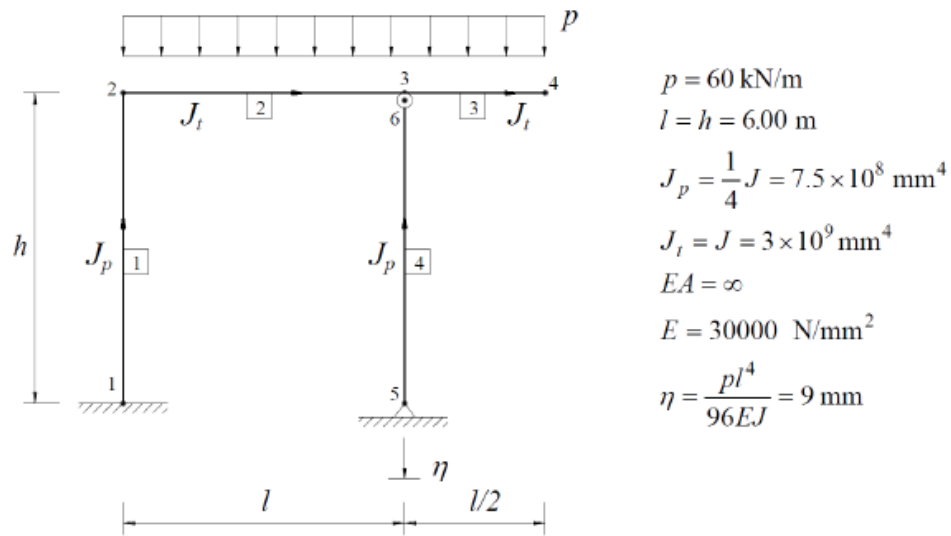


Figure 11.13: Frame with Foundation settlement

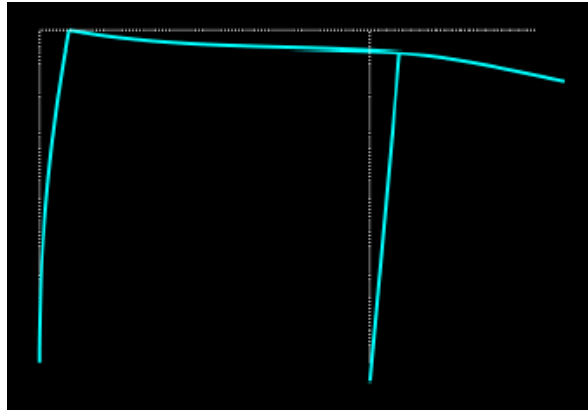
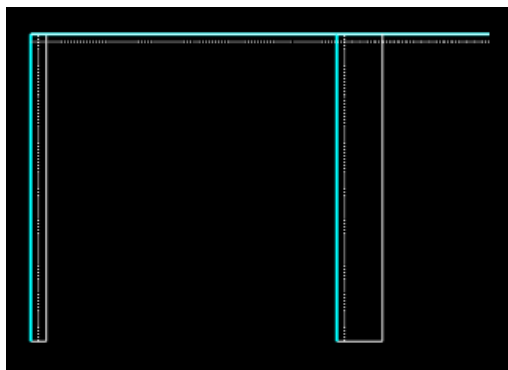
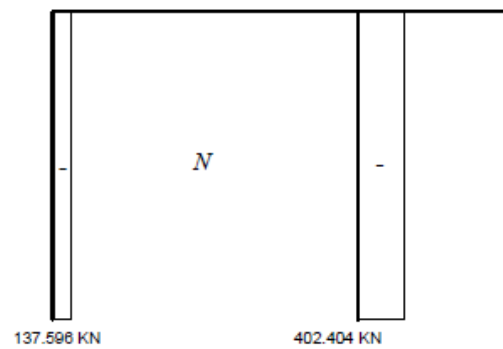


Figure 11.14: Deformed configuration

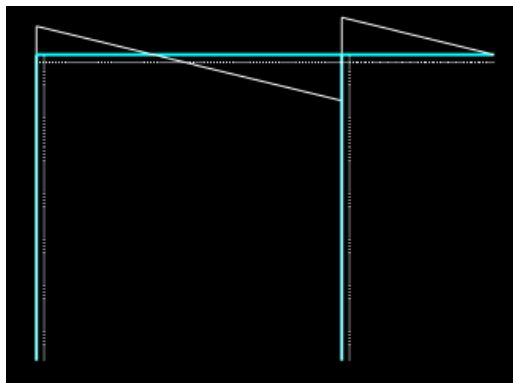


(a) Numerical solution

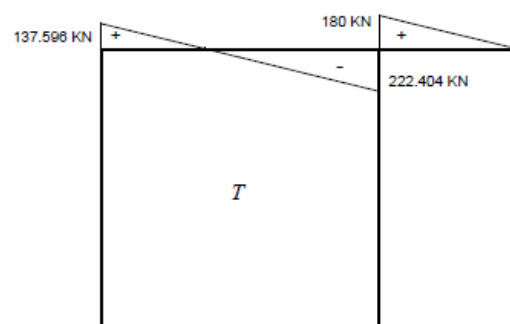


(b) Analytical solution

Figure 11.15: Axial Force

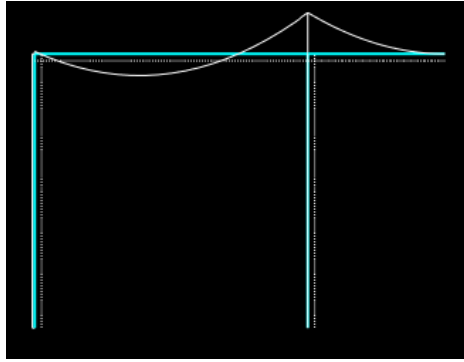


(a) Numerical solution

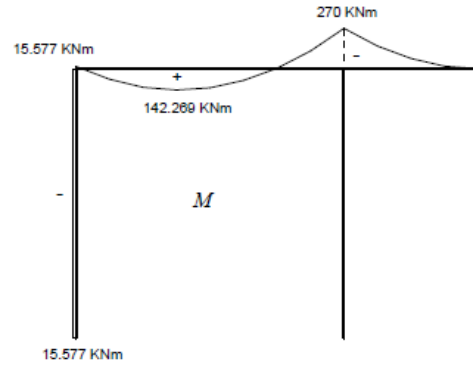


(b) Analytical solution

Figure 11.16: Shear Force



(a) Numerical solution



(b) Analytical solution

Figure 11.17: Bending Moment

Element	1	2	3	4
N_1 [kN]	137.569	0	0	402.404
T_1 [kN]	0	137.569	180	0
M_1 [kN.m]	15.577	15.577	270	0
N_2 [kN]	-137.569	0	0	-402.404
T_2 [kN]	0	222.404	0	0
M_2 [kN.m]	-15.577	-270	0	0

Table 11.7: Frame with foundation settlement results

The results can be observed from the files mentioned in the previous chapter. A comparison between the analytical solution and numerical solution is displayed in the following figures.

11.5 Frame with elastic restraints and temperature change

This benchmark is to test the functionality of both imposed internal stains due to temperature change and external elastic restraints. With this expansion to the code the problem of thermal expansion and elastic restraints can be solved and as a result the internal forces and deformations are given as output. The problem is defined as described in Figure 11.11.

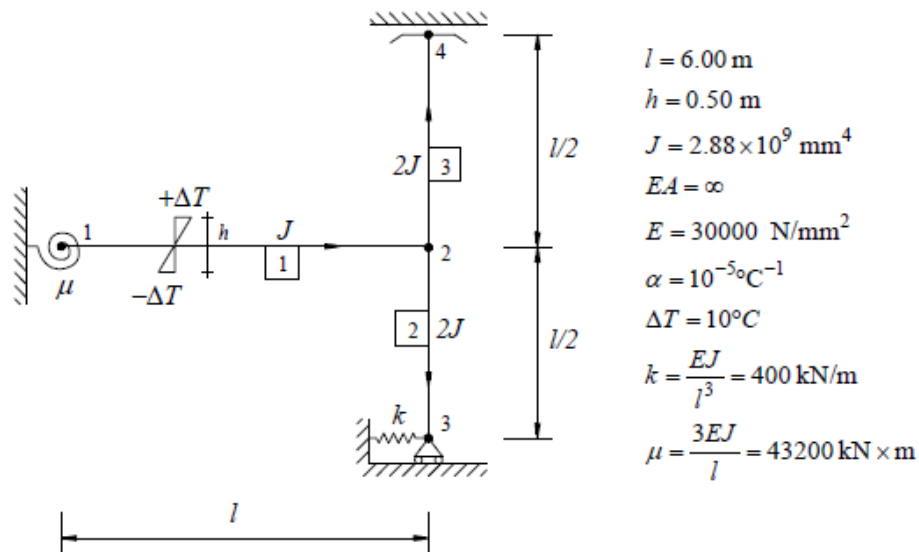


Figure 11.18: Frame with elastic restraints and temperature change

The results can be observed from the files mentioned in the previous chapter. A comparison between the analytical solution and numerical solution is displayed in the following figures.

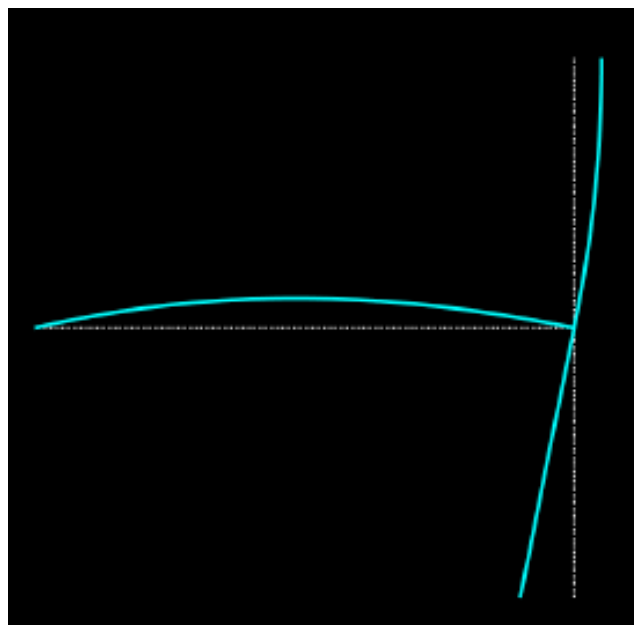
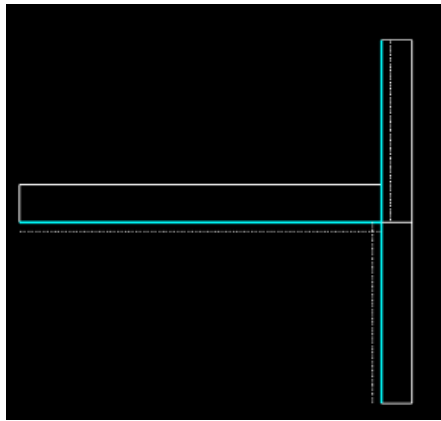
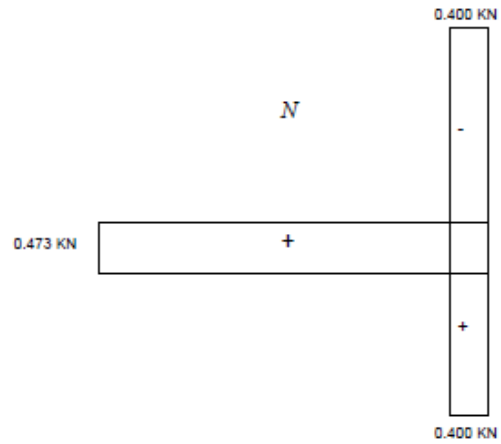


Figure 11.19: Deformed configuration

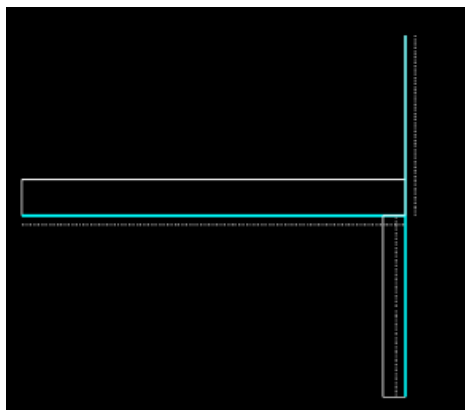


(a) Numerical solution

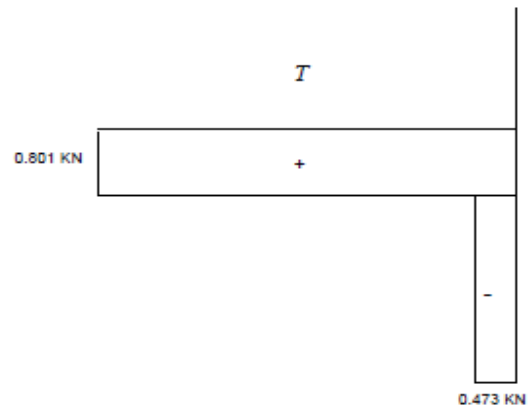


(b) Analytical solution

Figure 11.20: Axial Force

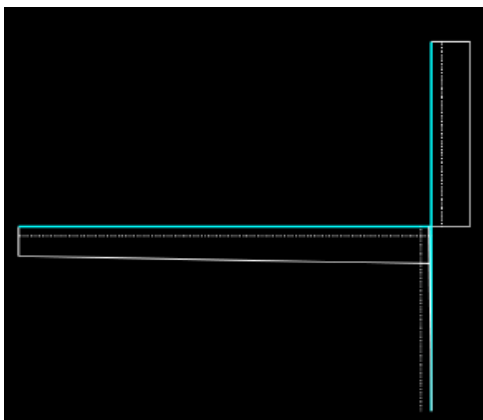


(a) Numerical solution

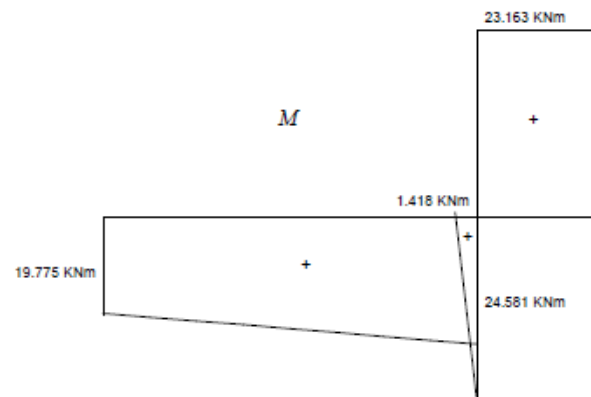


(b) Analytical solution

Figure 11.21: Shear Force



(a) Numerical solution



(b) Analytical solution

Figure 11.22: Bending Moment

The numerical results can be read in the output and are as follows in Table 11.5, they match the analytical solution exactly.

Element	1	2	3
N_1 [kN]	-0.473	-0.4	0.4
T_1 [kN]	0.801	-0.473	0
M_1 [kN. m]	-19.775	-1.418	-23.163
N_2 [kN]	0.473	0.4	-0.4
T_2 [kN]	-0.801	0.473	0
M_2 [kN. m]	24.581	0	23.163

Table 11.8: Frame with elastic restraints and temperature change results

11.6 Prestressing

This benchmark is to test the prestressing functionality, with this expansion to the code it's possible to add internal forces in the frame due to prestressing tendons inside any element. The problem is defined as follows (Figure 11.16), it will contain just one unloaded element, an horizontal beam of length $l = 1\text{m}$ with only 1 prestressing tendon of force $P = 1\text{ kN}$.

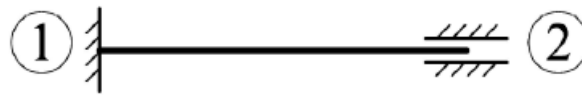
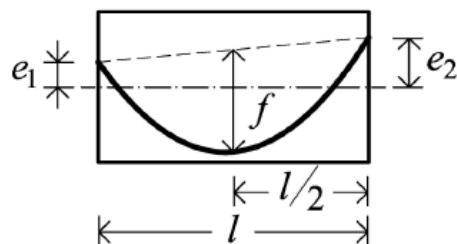


Figure 11.23: Prestressed beam

The indices at the sides indicate the analytical solutions to the internal stress problem depending on the shape of the prestressing tendons inside the beam. Two different examples will be discussed each with the relative data, analytical formulation and numerical solution.

11.6.1 Tendon layout 1

In this case the shape of the prestressing is a simple parabola that starts from one extremity of the beam and ends at the other. A visualization and data for the benchmark problem are contained in Figure 11.17.



(a) Problem definition

ECC.	[m]
e_1	0
e_2	0.2
f	0.3

(b) Benchmark data

Figure 11.24: Prestressing problem 1

The following formulas govern the moment in the edges of the beam.

End	Moment
1	$\frac{2}{3}f$
2	$\frac{2}{3}f$

Table 11.9: Analytical solution

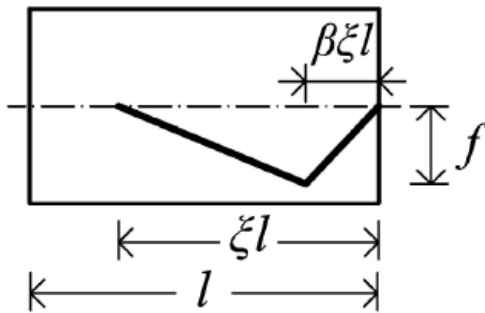
The numerical solution contained in Table 11.7 as well as the axial force match the values in Table 11.6 exactly.

Element	N_1 [kN]	T_1 [kN]	M_1 [kN.m]	N_2 [kN]	T_2 [kN]	M_2 [kN.m]
1	1.00	-0.120	-0.200	-1.000	-0.120	-0.200

Table 11.10: Prestressing Benchmark 1 results

11.6.2 Tendon layout 2

In this case the shape of the prestressing is a combination of 2 straight tendons that don't have any eccentricity at the edges. A visualization and data for the benchmark problem are contained in Figure 11.18.



(a) Problem definition

ECC.	
f [m]	0.2
ξ	0.8
β	0.25

(b) Benchmark data

Figure 11.25: Prestressing problem 2

The following formulas govern the moment in the edges of the beam.

End	Moment
1	$[\xi(1 + \beta) - 1]\xi f$
2	$[2 - \xi(1 + \beta)]\xi f$

Table 11.10: Analytical solution

The numerical solution contained in Table 11.9 as well as the axial force match the values in Table 11.8 exactly. In this benchmark the beam had to be divided into 3 elements to describe the complex prestressing layout.

Element	N_1 [kN]	T_1 [kN]	M_1 [kN.m]	N_2 [kN]	T_2 [kN]	M_2 [kN.m]
1	0.000	0.016	0.000	0.000	-0.016	0.032
2	1.000	-0.017	-0.032	-1.000	0.017	-0.072
3	1.000	0.116	0.072	-1.000	-0.116	0.160

Table 11.11: Prestressing Benchmark 2 results

11.7 Determinant beam with inclined support

This benchmark is studied to assess the code for the case of inclined constraint. For this purpose, a benchmark frame has been adopted from the textbook. The structural model of the benchmark is shown below:

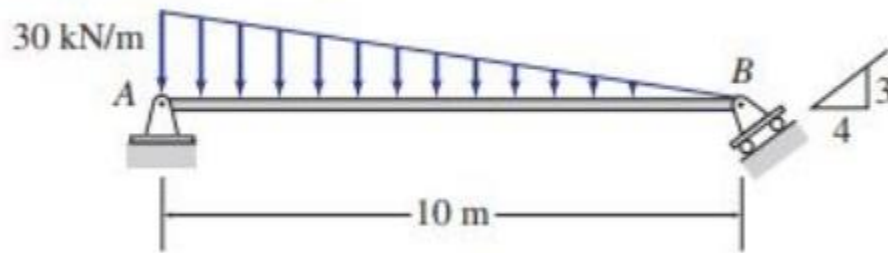


Figure 11.26: Beam with inclined support

The inclined roller is modelled as **INCLINED RESTRAINT** in the code. The results obtained from the code are tabulated below along with a comparison with the solution given in the subject textbook for the problem.

Output Parameter	Numerical	Reference	Error (%)
Axial Force[kN]	37.5	37.5	0.0
Shear at left support [kN]	100	100	0.0
Roller Reaction[kN]	62.5	62.5	0.0

Table 11.12: Beam with inclined support results

11.8 Indeterminate beam with rigid end

This benchmark is studied to check the ability of code to model and analyze the structure with rigid ends. A benchmark problem was created which was studied with two approaches. The first approach utilizes the standard **RIGID END** functionality of the code. The second approach has been adopted to compare the results of the first approach. It models the rigid end as an element with a significantly high moment of inertia (i.e., rigid in bending as the axial loading is not expected in the subject problem). The structural model of the benchmark is shown below ($F = 100$ KN):

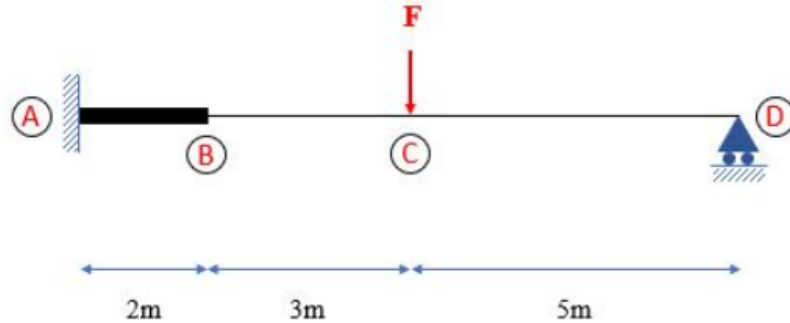


Figure 11.27: Beam with rigid end

A finite part of the beam with size shown in the figure is modelled as RIGID END. The results from both approaches are tabulated below:

Output Parameter	Numerical	Reference	Error (%)
N_{ABC} [kN]	0	0	0.0
N_{CD} [kN]	0	0	0.0
$T_{ABC}(1)$ [kN]	81.45	81.45	0.0
$T_{ABC}(2)$ [kN]	-81.45	-81.45	0.0
$T_{CD}(1)$ [kN]	-18.46	-18.46	0.0
$T_{CD}(2)$ [kN]	18.46	18.46	0.0
$M_{ABC}(1)$ [kN. m]	315.43	315.43	0.0
$M_{ABC}(2)$ [kN. m]	92.29	92.29	0.0
$M_{CD}(1)$ [kN. m]	-92.29	-92.29	0.0
$M_{CD}(2)$ [kN. m]	0	0	0.0

Table 11.13: Beam with rigid end results

Chapter 12

CASE STUDY: BOX GIRDER BRIDGE

The program created is to be used on a specific application, the case study of a box girder bridge. This bridge is built as an infrastructure with the purpose of vehicular transportation. Here follows an overview.



Figure 12.1: Photo of the box girder bridge

Due to the purpose of the structure the appropriate testing has taken place and trucks have been loaded on it to evaluate deflection after construction. Our goal with this case study is to model the bridge in the most appropriate way as to obtain the lowest errors possible while using the code.

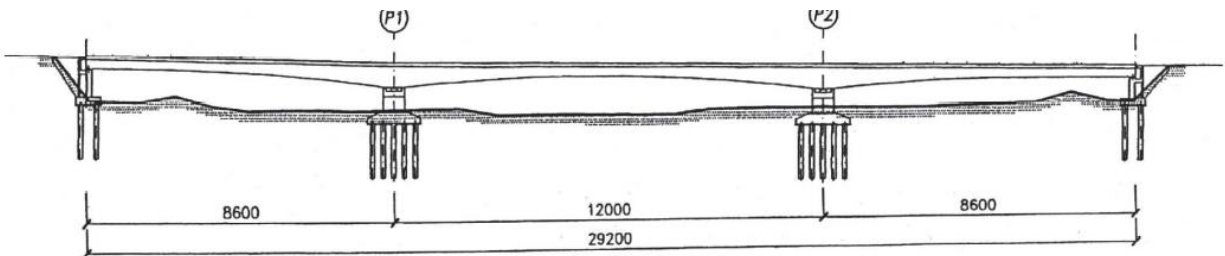
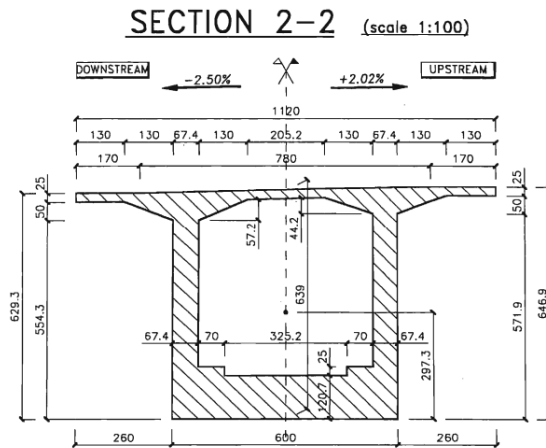


Figure 12.2: Lateral view of the bridge

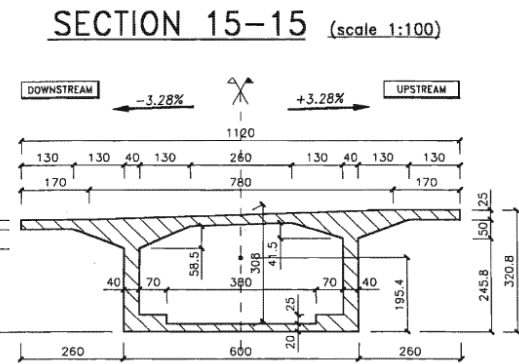
The geometrical information on the bridge is all provided and the modeling will not include the pillars nor the abutments, but consider them as external restraints.

12.1 Evaluation of the cross-sectional properties

For the bridge modeling, many cross sections have been provided. The bridge itself is thicker near the pillars and thinner in the mid-span and near the abutments. Such sections have the box shape illustrated in Figure 12.3 and above the pillars have a fuller section illustrated in Figure 12.4.



(a) Section 2 (near pillars)



(b) Section 15 (mid-span and abutments)

Figure 12.3: Different box girder cross-sections

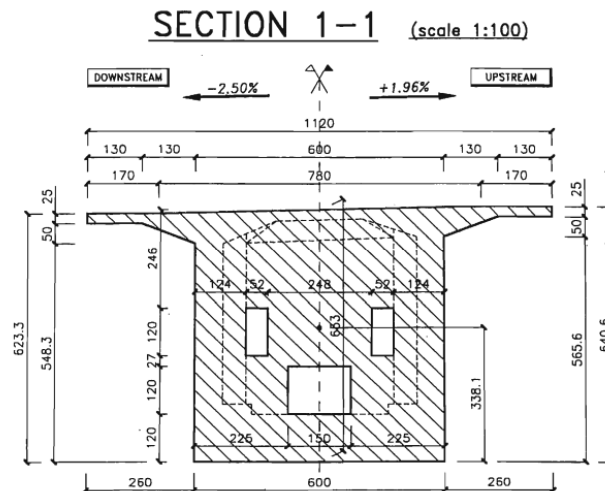


Figure 12.4: Section 1 (on the pillars)

These are just a few of the cross-sections provided for the bridge, in Figure 12.5 the reference for each cross-section is provided with the relative coordinates in Figure 12.6 (dimensions in meters).

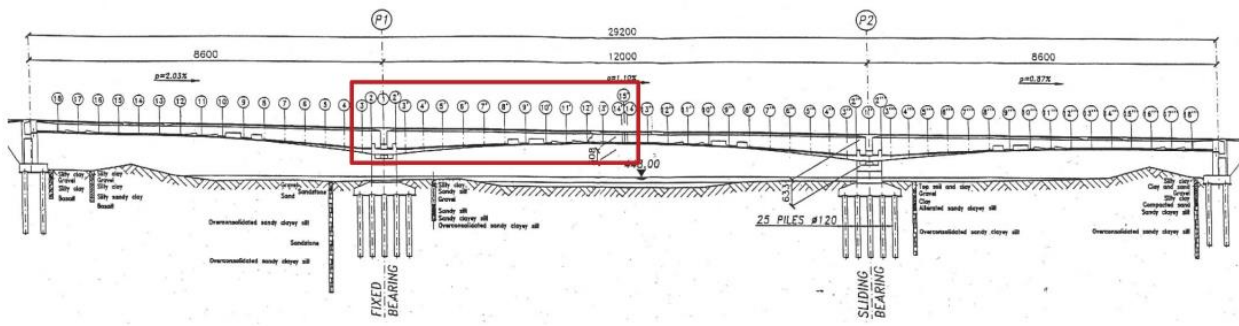


Figure 12.5: Overview of bridge sections layout

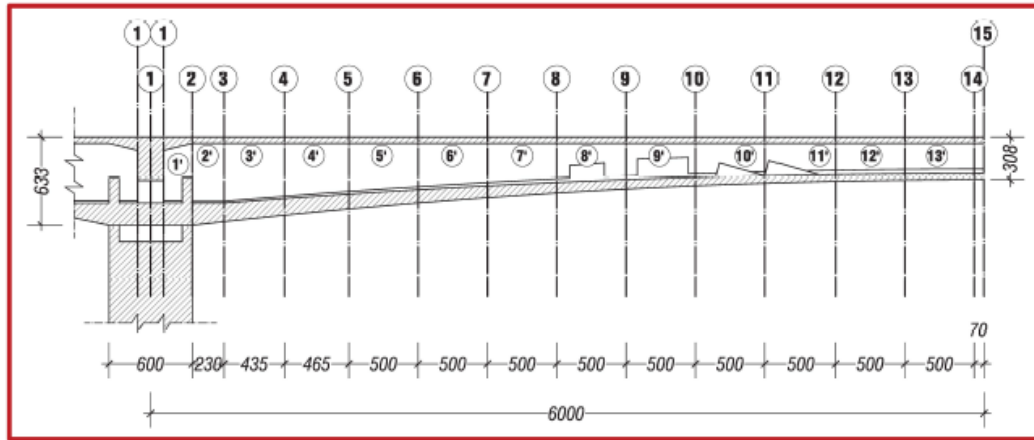


Figure 12.6: Details of the bridge deck

Different modeling assumptions can be taken regarding the properties of the cross-sections, considering their differing stiffness and geometry and their coordinates.

12.1.1 Material properties

The bridge is entirely built in reinforced concrete but for the purpose of this application the contributions of reinforced concrete and prestressing cables are momentarily neglected. Therefore the material properties are to be considered constant throughout the whole length. Moreover the self-weight of the bridge itself is not to be taken into account but to be considered separately as a uniform dead load distribution that includes both self-weight and permanent loads such as roadbed, curbs, sidewalk and guard-rails.

E_c [MPa]	ν	$\alpha \left[\frac{1}{^\circ\text{C}} \right]$	ρ
37000	0.18	0	0

Because of all the factors that are neglected the Young's modulus taken in consideration will need to be updated by a significant scale factor.

12.1.2 Geometrical properties extraction

Through AutoCAD it is possible to extract all the relevant properties from each section after redrawing them. While redrawing the sections the base and sides were considered horizontal and vertical respectively and to simplify the acquisition the flanges were considered horizontal as well. In AutoCAD the regions must be defined and mass properties can be exported. The data in Table 12.1 was gathered with this procedure.

Section	Area [m ²]	Height [m]	Centroid Y [m]	Moment of inertia [m ⁴]
1	36.819	6.406	3.378	137.5691
2	18.306	6.469	2.967	100.6464
3	16.019	6.150	2.994	82.6470
4	14.377	5.703	2.865	64.9822
5	12.754	5.311	2.804	49.5661
6	11.672	4.829	2.569	38.7738
7	9.690	4.441	2.616	25.8960
8	8.968	4.114	2.467	20.5524
9	8.334	3.838	2.343	16.5128
10	8.096	3.613	2.217	13.9645
11	7.954	3.438	2.107	12.2882
12	7.842	3.303	2.020	11.0675
13	7.778	3.228	1.971	10.4024
14	7.754	3.203	1.952	10.1585
15	7.754	3.208	1.952	10.1575

Table 12.1: Section properties

These properties will be used to model the structure, defining different frame elements that each have their own assigned section. It is notable how much the properties vary from section to section, as the height influences the moment of inertia relative to the horizontal axis greatly.

12.1.3 Shear factor

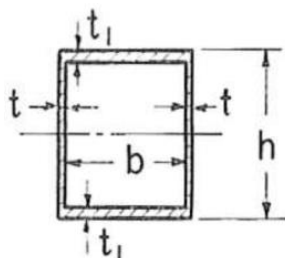
Another property that varies from section to section is the shear correction factor. To extend the applicability of the one-dimensional theory of beams and take into account transverse shear deformability the stiffness matrix of every element must be modified accordingly.

An analytical solution for the value of the shear correction factor can be obtained though the formulas described in Section 5.1, for this application however a different definition of the shear correction factor, proposed by G. R. Cowper, will be used.

Since the shear factor is a geometrical property it is possible to use the following formulas on equivalent sections, similar enough in shape to the ones relevant to the application to obtain, a reasonable approximation of the value of the shear factor.

Thin-walled box section

This section type is taken into consideration as the box girder is itself a closed section. The drawback is not being able to account for the vertical asymmetry introduced by the flanges.



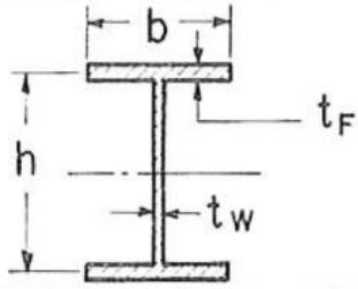
Defining

$$m = \frac{bt_1}{ht} \quad , \quad n = \frac{b}{h}$$

$$\kappa = \frac{10(1+\nu)(1+3m)^2}{(12+72m+150m^2+90m^3)+\nu(11+66m+135m^2+90m^3)+10n^2((3+\nu)m+3m^2)}$$

Thin-walled I section

This section type is taken into consideration as the mass of the cross-sections is concentrated in the flanges. The drawback is again not being able to account for the vertical asymmetry.



Defining

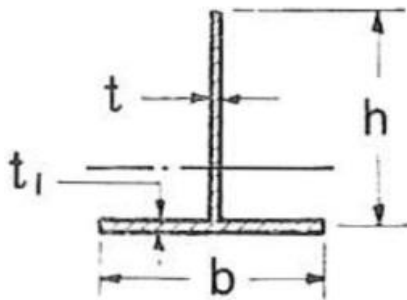
$$m = \frac{2bt_F}{ht_W}, \quad n = \frac{b}{h}$$

κ

$$= \frac{10(1+\nu)(1+3m)^2}{(12+72m+150m^2+90m^3)+\nu(11+66m+135m^2+90m^3)+30n^2(m+m^2)+5\nu n^2(8m+9m^2)}$$

Thin-walled T section

This section type is taken into consideration as the box girder becomes more and more asymmetrical as the lower part becomes thinner in the bridge mid-span. The drawback is not being able to account for the large mass in the lower part of the section.



Defining

$$m = \frac{bt_1}{ht}, \quad n = \frac{b}{h}$$

κ

$$= \frac{10(1+\nu)(1+4m)^2}{(12+96m+276m^2+192m^3)+\nu(11+88m+248m^2+216m^3)+30n^2(m+m^2)+10\nu n^2(4m+5m^2+m^3)}$$

Equivalent sections

Many design decisions have to be made while defining the parameters of the equivalent sections, in this process it is important to keep in mind the advantages and disadvantages of one model over the other. For the various shear correction factors in this application, a combination of models will be used to approximate real behaviour of the cross-sections.

The values for the shear factors computed are contained in Table 12.2, professional software can utilize the finite element method to compute the shear factor of any shape.

1. Section 1 (Figure 12.4) is approximated as a rectangle with the formula:

$$\chi = \frac{12 + 11\nu}{10(1 + \nu)}$$

2. Section 2 to section 6 will have the shear factor of an equivalent thin walled box section. This is to capture all the element that make up the box and the mostly symmetric structure.
3. Section 7 to section 15 will have an average shear factor between an I section and a T section. This is to capture both the eccentricity of the horizontal plates and the asymmetry of the flanges.

Section	Box shape	I shape	T shape	Design χ
1	1.189	1.384	1.790	1.185
2	2.259	3.580	3.099	2.259
3	2.034	3.060	2.798	2.034
4	2.082	3.144	2.859	2.082
5	2.126	3.215	2.908	2.126
6	2.166	3.283	3.162	2.166
7	2.274	3.473	3.282	3.378
8	2.473	3.807	3.625	3.716
9	2.718	4.238	4.060	4.149
10	2.836	4.525	4.316	4.420
11	2.991	4.889	4.616	4.753
12	3.078	5.096	4.787	4.942
13	3.172	5.324	4.973	5.148
14	3.196	5.384	5.022	5.203
15	3.196	5.384	5.022	5.203

12.2 Modeling of the bridge structure

After having extracted the properties of each section it is necessary to define the elements to which they will be assigned. To do so the bridge has to be modeled properly.

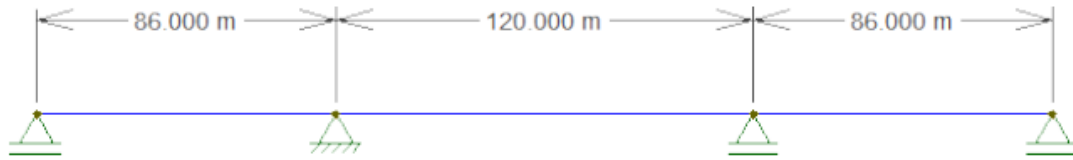


Figure 12.7: Modeling of restraints

For this case study the bridge was modeled as an entire unit, the base model is in fact a continuous beam held by several simple supports as seen in Figure 12.7, but partitioning its whole length into beam elements. This procedure is necessary for the finite element solution to be as precise as possible, moreover by partitioning the beam more details can be added to the structure, nodal loads can be applied to the correct location, offsets can be implemented between elements with different cross section properties and most importantly every beam can be characterized by the appropriate cross-section properties.

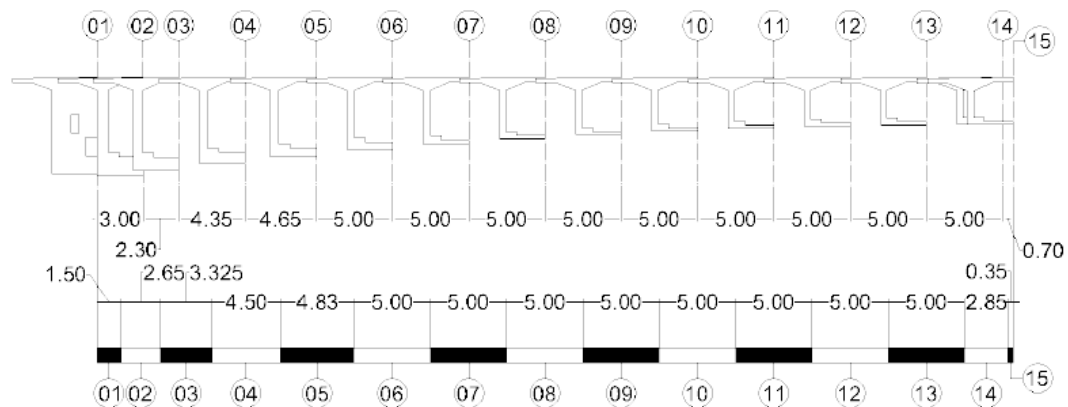


Figure 12.8: Deck discretization

For more correct modeling of the elements that constitute the bridge, the discretization does not happen in the points where the data about the cross-section is taken, but each portion of the bridge that sits in between two cross-sections is separated in half by a node. All of these nodes are connected one after the other and the elements created this way have the properties of the cross-section that sits in that range.

In Figure 12.8 the process of creating the nodes and elements is displayed for the deck portion of the bridge. In the upper side of the figure there is the data as for the problem definition, in the lower portion there is the element discretization and the nodes taken with the previously described method. This process is carried on for the whole length of the bridge.

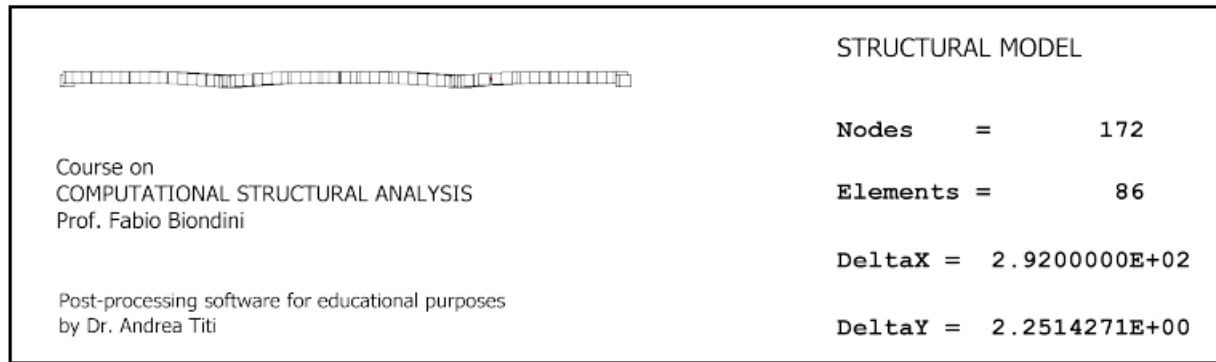


Figure 12.9: Complete bridge model

In Figure 12.9 the whole model, with 172 nodes and 86 elements, is visualized. The model contains extra nodes for the concentrated forces and most notably 2 nodes for every horizontal coordinate. This design decision is due to the introduction of offsets. Every beam is designed with a different cross-section, for this reason, because only the top portion of every element is aligned, the vertical coordinate of the centroid varies from section to section (see Figure 12.10). To better represent the internal forces produced by this misalignment, offsets equal to the eccentricity of the centroids are introduced in every node.

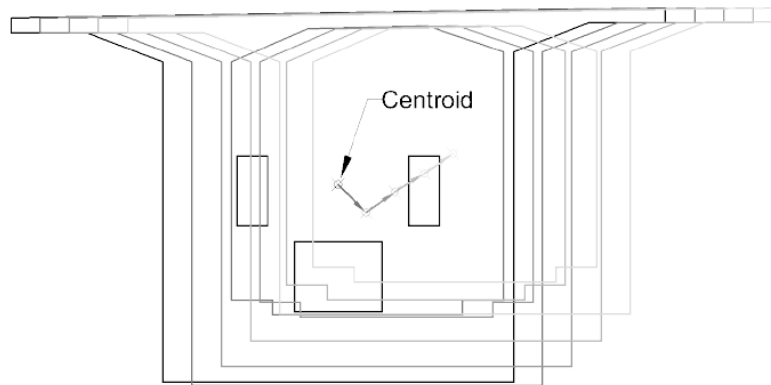


Figure 12.10: Centroid eccentricity

One more detail will be employed in the design of the bridge. The connections near the abutments and the ones on the pillars have thicker sections and for this reason, are much stiffer than the rest of the structure. In the code, they will be input as rigid ends and the displacement at the ends will be considered as a simple rotation with no deformation.

Cross-sectional properties

What follows is a summary of the properties of the various sections and their distribution throughout the whole length of the bridge.

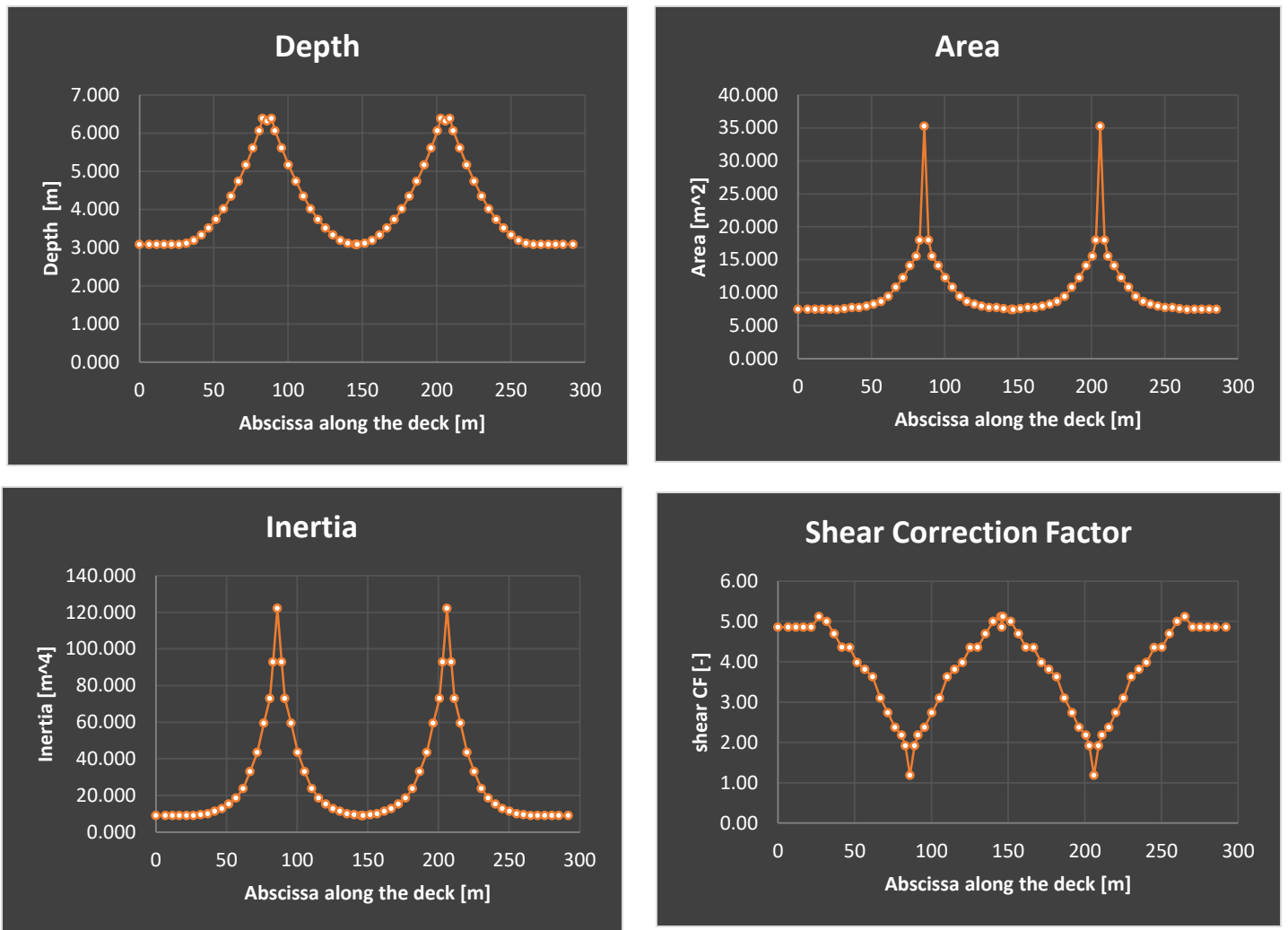


Figure 12.11: Cross-sectional properties

12.3 Stress analysis under dead and live loads

After the design is determined it is finally possible to proceed with the code. The following sections show, case after case, the different loading conditions to which the bridge can be subjected to.

12.3.1 Dead loads

The first case to examine is the case of uniform dead loads as described in Figure 12.12.

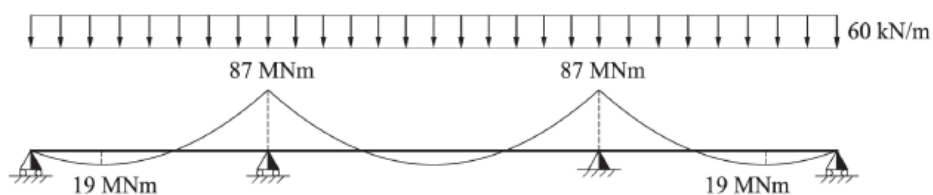
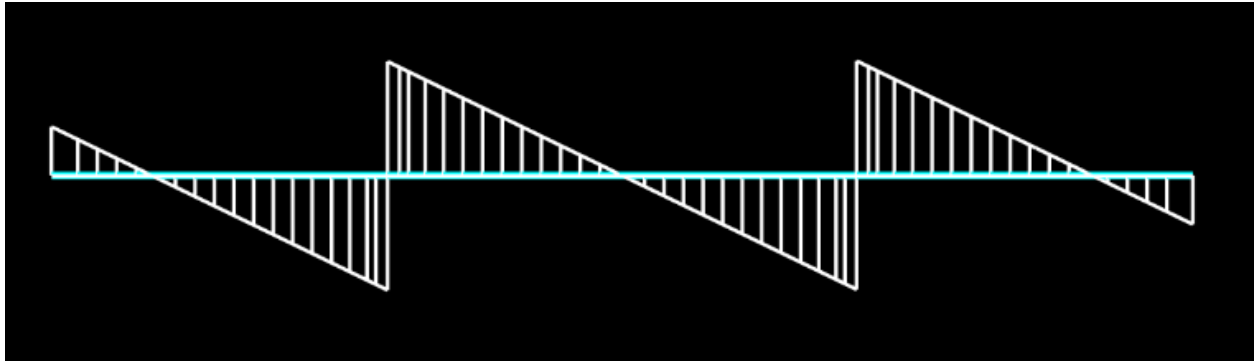
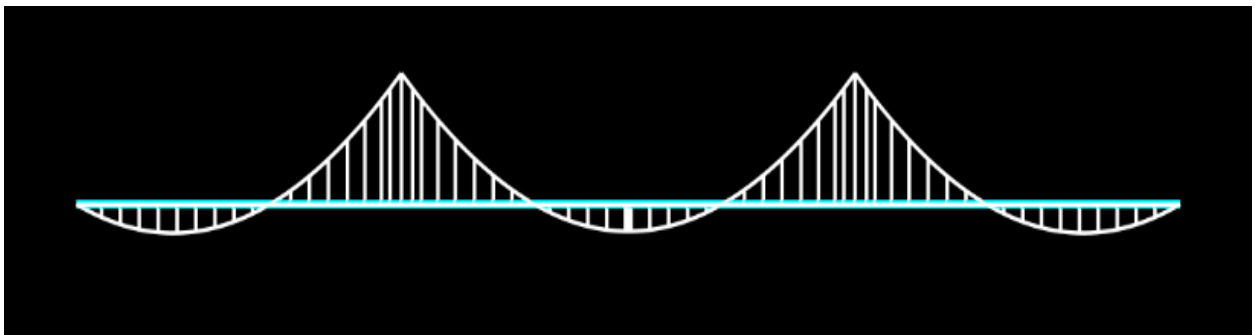


Figure 12.12: Distribution of dead load

In Figure 12.13 and Table 12.3 the response of the system is displayed.



(a) Shear force distribution



(b) Bending moment distribution

Figure 12.12: Internal force diagram for dead loads

Type	Positive	Negative
Shear force [KN]	3609	-3609

(a) Shear force

Location	Central span	Side span	Pier support
Bending moment [KN.m]	19508	20025	-83158

(b) Bending moment

Table 12.3: Maximum values

12.3.2 Phase 1 of live loads

After dead loads are considered there are different live load cases to examine. In the first one the live loads are only in the center portion of the bridge as shown in Figure 12.14.

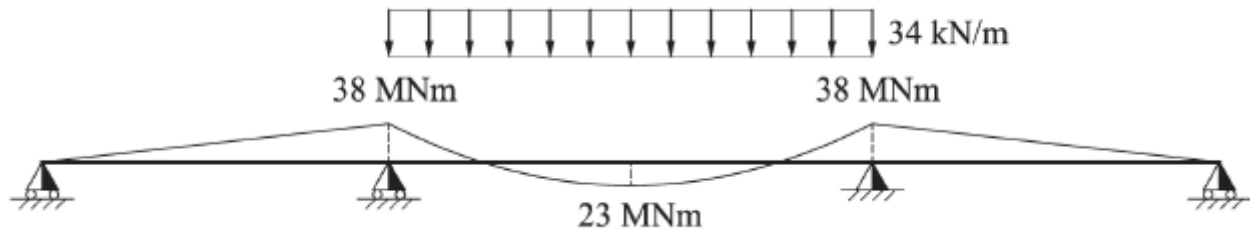
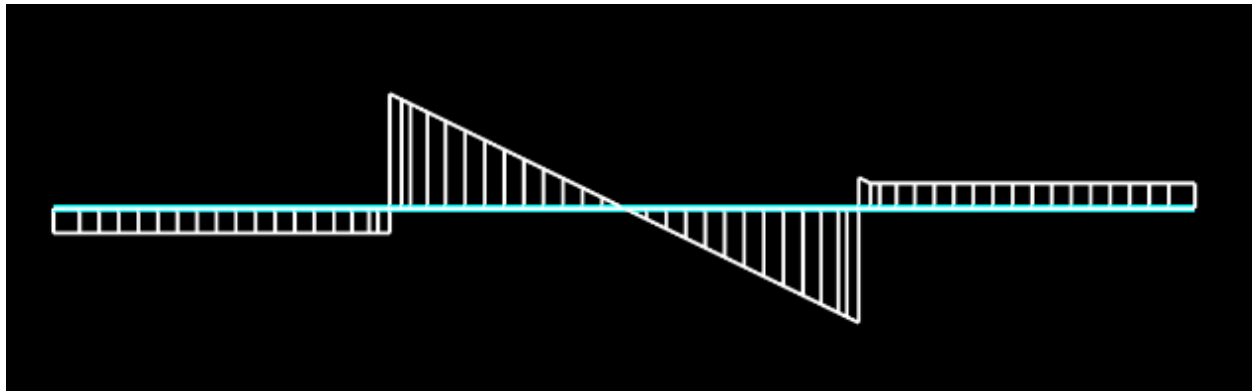
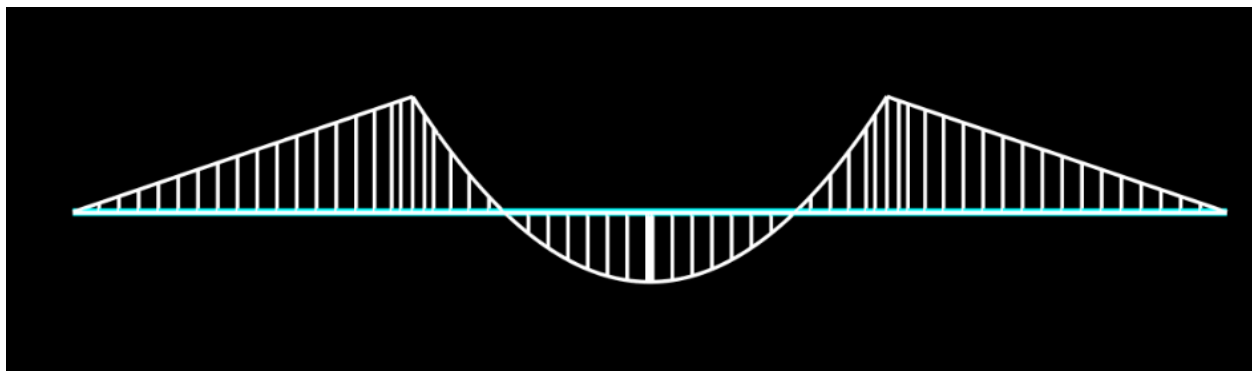


Figure 12.14: Distribution of live loads in phase 1

In Figure 12.15 and Table 12.4 the response of the system is displayed.



(a) Shear force distribution



(b) Bending moment distribution

Figure 12.15: Internal forces diagrams for phase 1 loads

Type	Positive	Negative
Shear force [KN]	2040	-2040

(a) Shear force

Location	Central span	Side span	Pier support
Bending moment [KN.m]	19508	20025	-83158

(b) Bending moment

Table 12.4: Maximum values

Concentrated live loads

Two additional distributions of loads are examined for each live load case, they are related to the testing that takes place after the design process. These tests take place with trucks, weighing 340kN each, that move on the bridge and in the modeling are considered as concentrated loads.

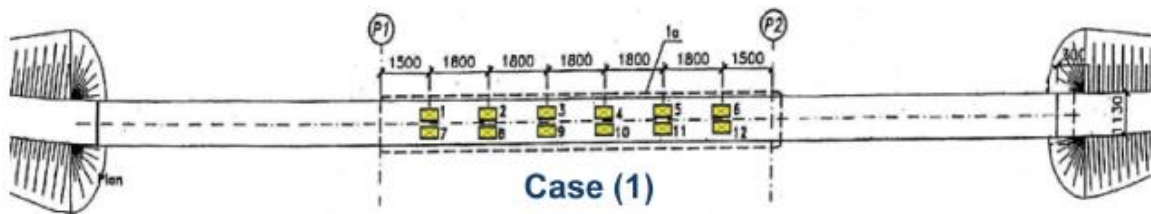
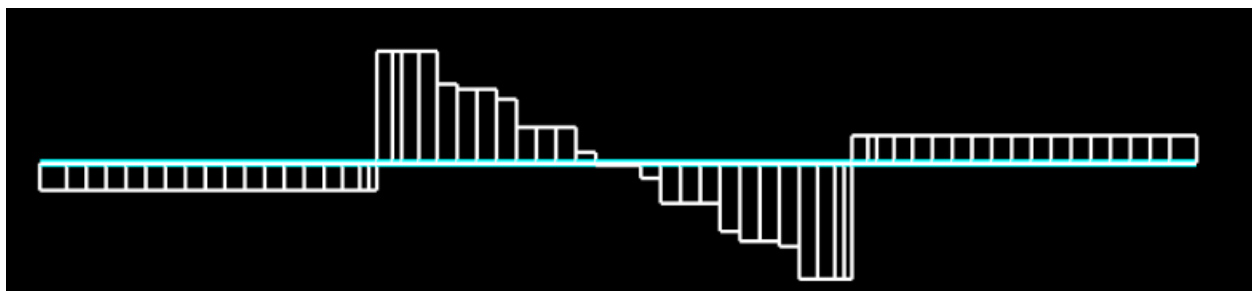


Figure 12.16: Phase 1 testing

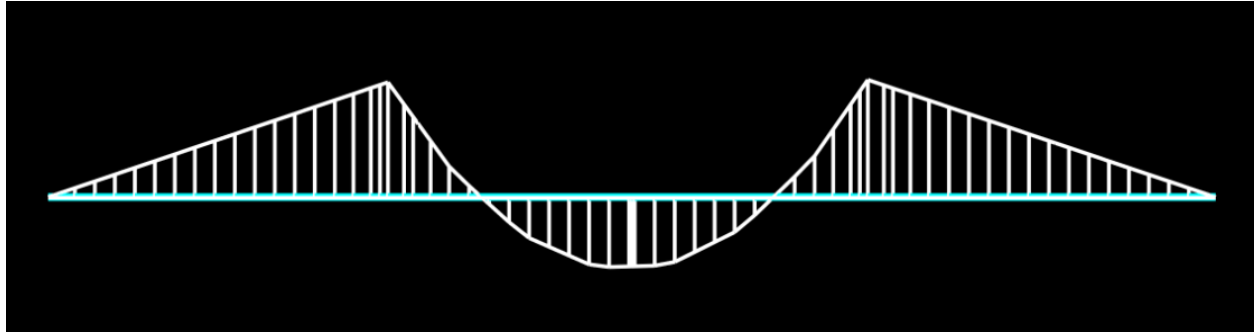
Each phase has 2 different loading conditions:

Phase 1A

In this first case only the first 6 trucks displayed in Figure 12.16 are present on the bridge. The resulting diagrams are shown in this section.



(a) Shear force distribution



(b) Bending moment distribution

Figure 12.17: Internal force diagrams for dead loads

Type	Positive	Negative
Shear force [kN]	1020	-1020

(a) Shear force

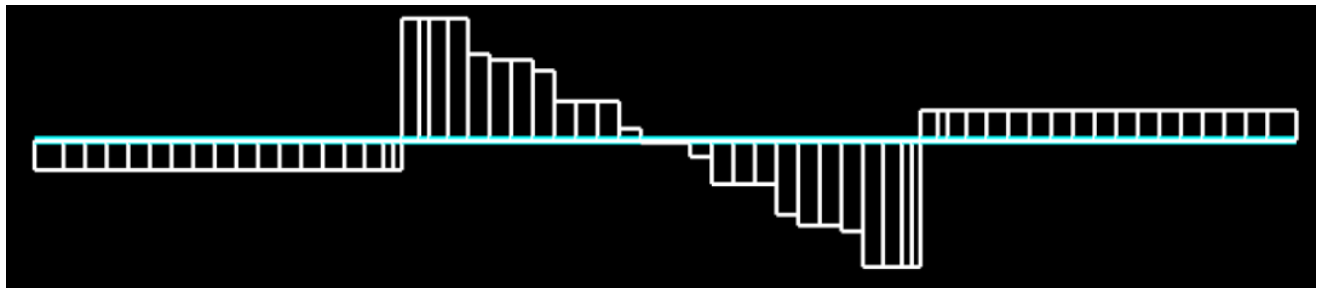
Location	Central span	Side span	Pier support
Bending moment [kN.m]	-20892	12767	-20892

(b) Bending moment

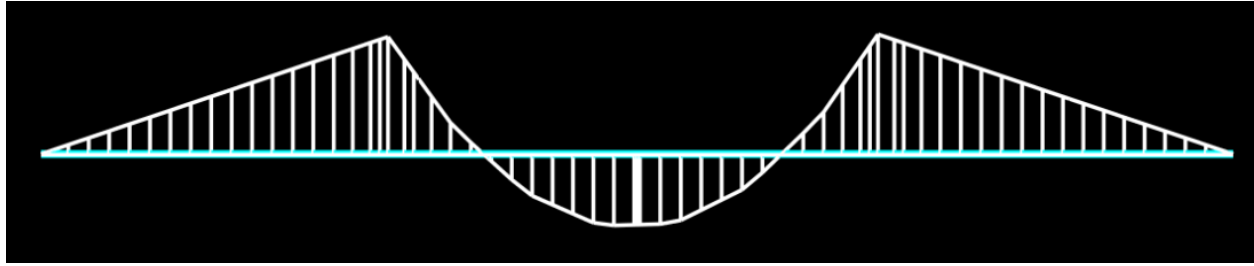
Table 12.5: Maximum values

Phase 1B

In the second case the other 6 are also loaded reaching a total of 12 trucks and a total load of 4080 kN which is equal the distributed load.



(a) Shear force distribution



(b) Bending moment distribution

Figure 12.18: Internal force diagrams for dead loads

Type	Positive	Negative
Shear force [kN]	2040	-2040

(a) Shear force

Location	Central span	Side span	Pier support
Bending moment [kN.m]	-41785	25534	-41785

(b) Bending moment

Table 12.6: Maximum values

Due to the superposition of effects and the similarity between the loading conditions the results to this case are just double the previous one.

12.3.3 Phase 2 live load

In the second phase the live loads are only in the side spans of the bridge as shown in Figure 12.19.

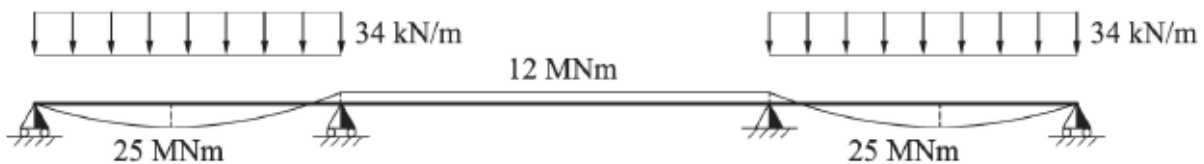
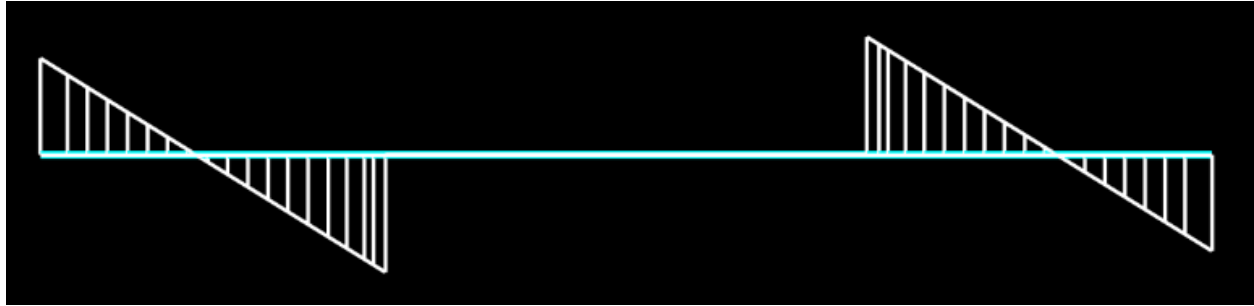


Figure 12.19: Distribution of live loads in phase 2

In Figure 12.20 and Table 12.7 the response of the systems is displaced.



(a) Shear force distribution



(b) Bending moment distribution

Figure 12.20: Distribution forces diagrams for phase 2 loads

Type	Positive	Negative
Shear force [kN]	1606	-1606

(a) Shear force

Location	Central span	Side span	Pier support
Bending moment [kN.m]	25530	-12420	-12420

(b) Bending moment

Table 12.7: Maximum values

Concentrated loads

Two additional distributions of loads are examined for each live load case, they are related to the testing that takes place after the design process. These tests take place with trucks, weighing 340kN each, that move on the bridge and in the modeling are considered as concentrated loads.

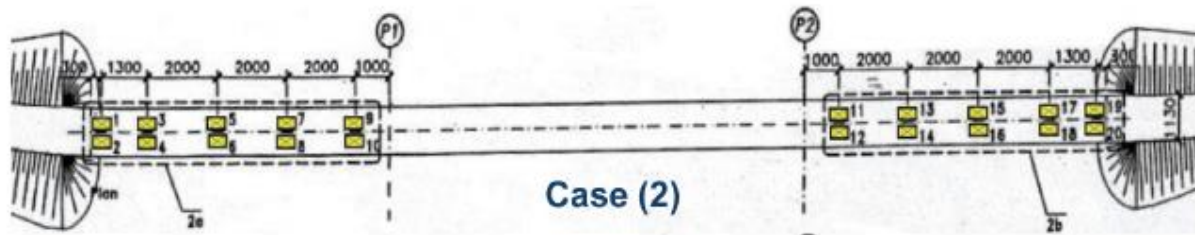
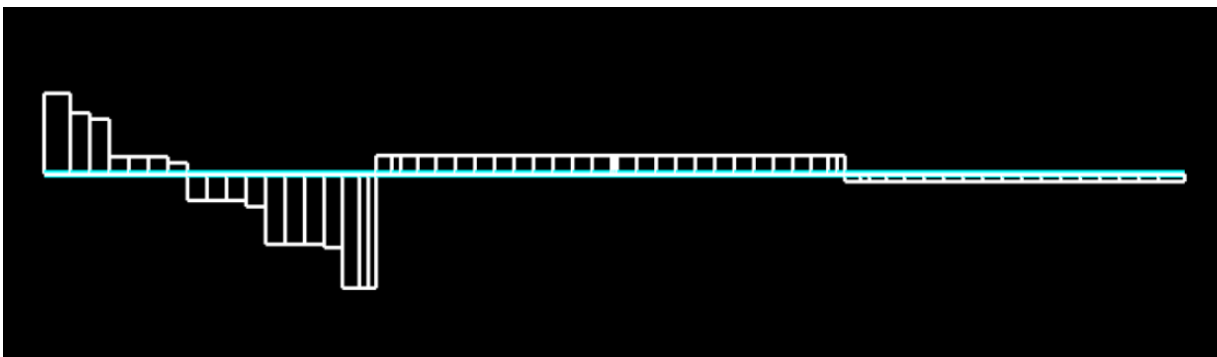


Figure 12.21: Phase 2 testing

Each phase has 2 different loading conditions:

Phase 2A

In this first case only the first 10 trucks displayed in Figure 12.21 are present on the bridge. The resulting diagrams are shown in this section.



(a) Shear force distribution



(b) Bending moment distribution

Figure 12.22: Internal forces diagrams for dead loads

Type	Positive	Negative
Shear force [KN]	1650	-1750

(a) Shear force

Location	Central span	Side span	Pier support
Bending moment [kN.m]	-23339	23360	-23339

(b) Bending moment

Table 12.8: Maximum values

Phase 2B

In the second case the other 10 are also loaded reaching a total of 20 trucks and a total load of 6800 kN which is equal the distributed load.



(a) Shear force distribution



(b) Bending moment distribution

Figure 12.23: Internal forces diagrams for dead loads

Type	Positive	Negative
Shear force [kN]	1770	-1770

(a) Shear force

Location	Central span	Side span	Pier support
Bending moment [kN.m]	-13073	27658	-13073

(b) Bending moment

Table 12.9: Maximum values

12.3.4 Phase 3 live load

In the third phase the live loads are in the center and lateral span of the bridge as shown in Figure 12.24.

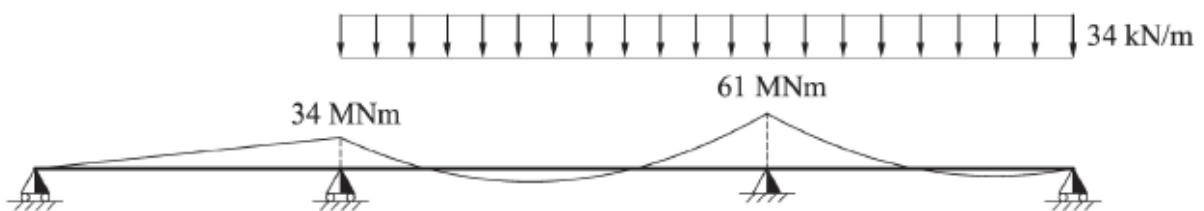
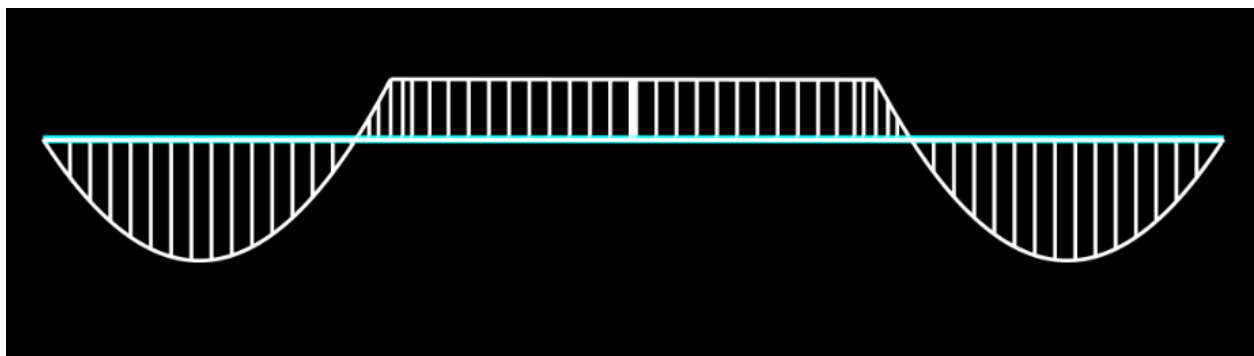


Figure 12.24: Distribution of live loads in phase 3

In Figure 12.25 and Table 12.10 the response of the system is displayed.



(a) Shear force distribution



(b) Bending moment distribution

Figure 12.25: Internal forces diagrams for phase 3 loads

Type	Positive	Negative
Shear force [kN]	2159	-2307

(a) Shear force

Location	Central span	Side span	Pier support
Bending moment [kN.m]	18316	-27898	-59972

(b) Bending moment

Table 12.10: Maximum values

Concentrated loads

Two additional distributions of loads are examined for each live load case, they are related to the testing that takes place after the design process. These tests take place with trucks, weighing 340kN each, that move on the bridge and in the modeling are considered as concentrated loads.

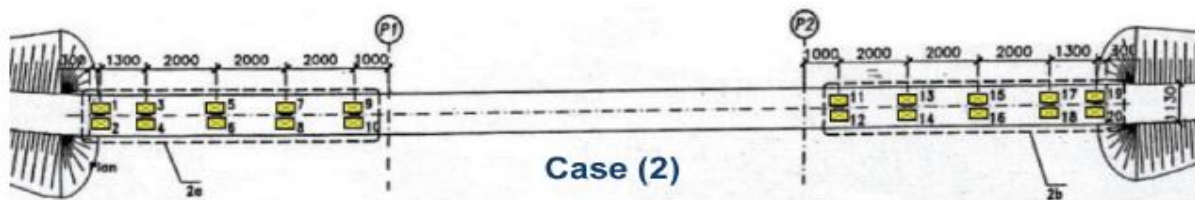
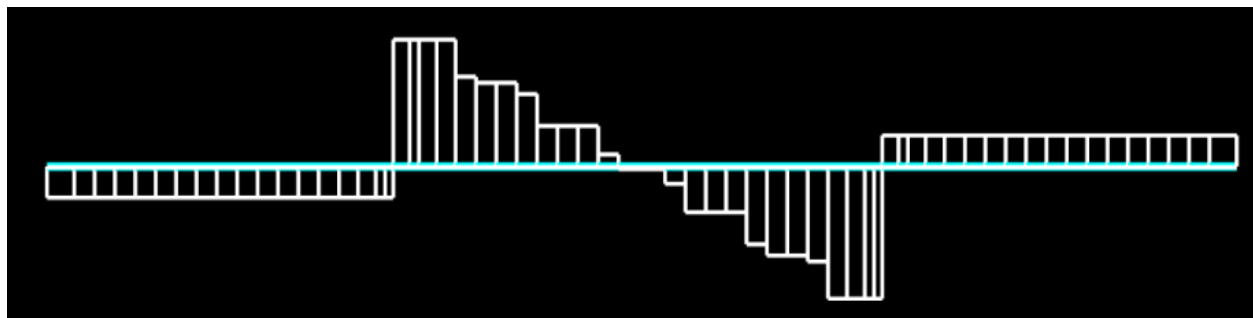


Figure 12.26: Phase 3 testing

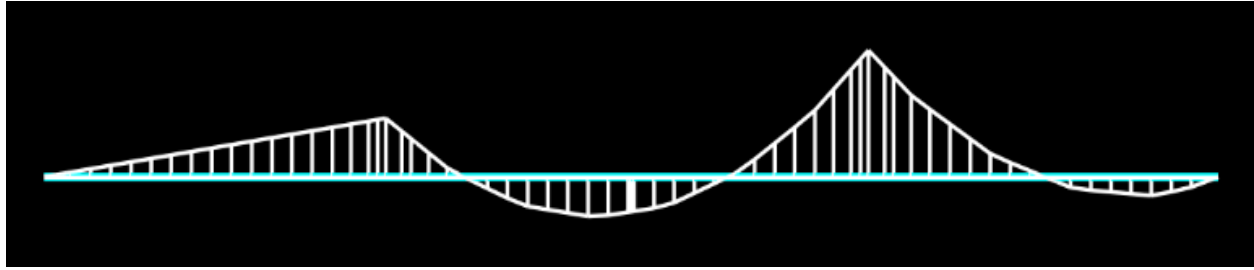
Each phase has 2 different loading conditions:

Phase 3A

In this first case only the first 12 trucks displayed in Figure 12.26 are present on the bridge. The resulting diagrams are shown in this section.



(a) Shear force distribution



(b) Bending moment distribution

Figure 12.27: Internal force diagrams for dead loads

Type	Positive	Negative
Shear force [kN]	2040	-2040

(a) Shear force

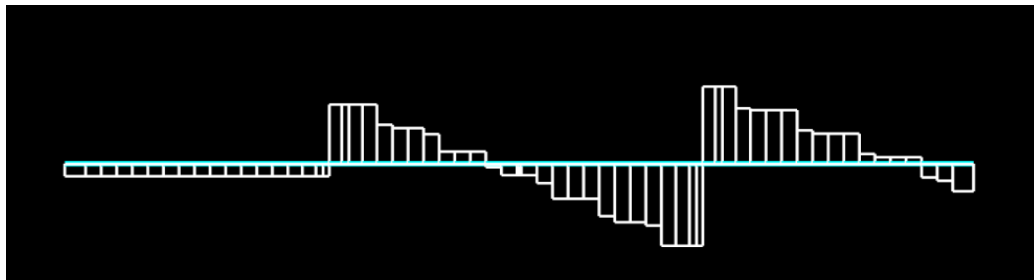
Location	Central span	Side span	Pier support
Bending moment [kN.m]	-41785	25534	-41785

(b) Bending moment

Table 12.11: Maximum values

Phase 3B

In the second case the other 10 are also loaded reaching a total of 22 trucks and a total load of 7480 kN which is equal the distributed load.



(a) Shear force distribution



(b) Bending moment distribution

Figure 12.28: Internal force diagrams for dead loads

Type	Positive	Negative
Shear force [KN]	2240	-2320

(a) Shear force

Location	Central span	Side span	Pier support
Bending moment [kN.m]	21560	9716	-65026

(b) Bending moment

Table 12.12: Maximum values

12.4 Evaluation of the deflections at monitored cross-sections

Table 12.13 contains the values of the deflections obtained with the design properties of the sections in the case of concentrated loads. The values are taken at specific sections of interest as highlighted in Figure 12.29.

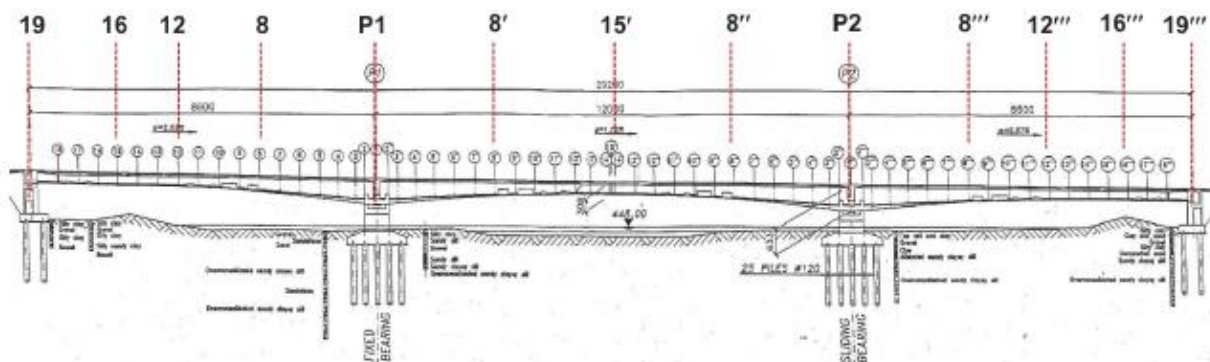


Figure 12.29: Sections of interest

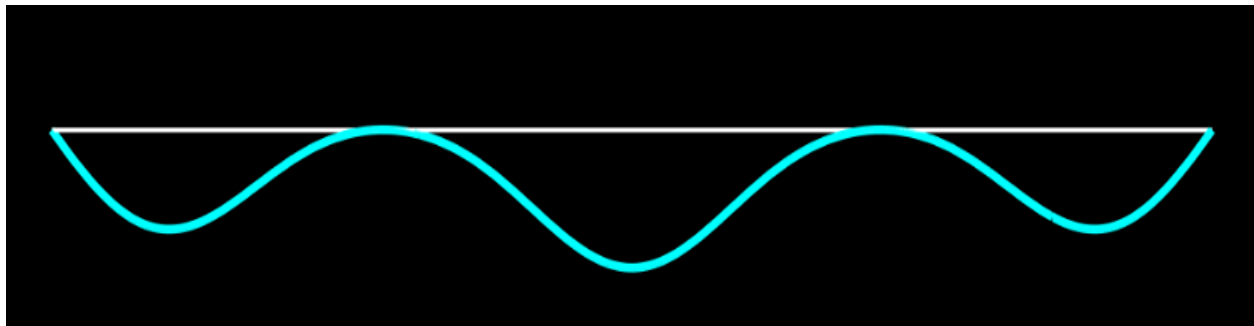
Thanks to the principle of superposition of effects we can consider the solutions for dead and live loads separately to study the real displacements in the structure.

Table 12.13: Displacement at the interested nodes [mm]

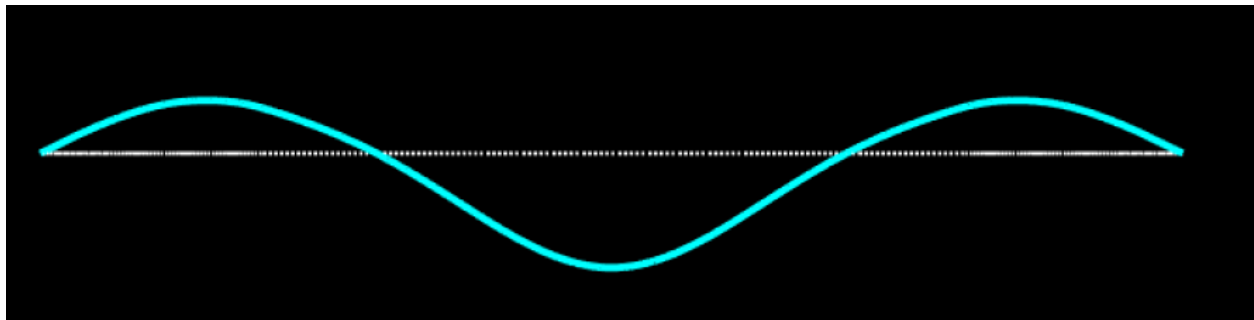
Cross-Section	19	16	12	8	P1	8'	15'	8"	P2	8"	12"	16"	19"
Abscissa[m]	-146	-129.3	-109.3	-89.3	-60	-30.7	0	30.7	60	89.3	109.3	129.3	146
Dead loads	0	-17.1	-20.7	-8.8	0	-14.3	-33.2	-14.3	0	-8.8	-20.7	-17.1	0
Phase 1A	0	9.4	16.5	15.1	0	-21.2	-35.1	-21.2	0	15.1	16.5	9.4	0
Phase 1B	0	18.8	33	30.1	0	-42.4	-70.2	-42.4	0	30.1	33	18.8	0
Phase 2A	0	-23.8	-35.6	-26.4	0	20.6	23.3	11	0	-7.4	-8.1	-4.6	0
Phase 2B	0	-28.4	-43.7	-33.8	0	31.6	46.6	31.5	0	-33.7	-43.5	-28.3	0
Phase 3A	0	18.8	33	30.1	0	-42.4	-70.2	-42.4	0	30.1	33	18.8	0
Phase 3B	0	14.2	24.9	22.7	0	-31.5	-46.9	-21.9	0	3.9	-2.5	-4.9	0

Deflection diagrams

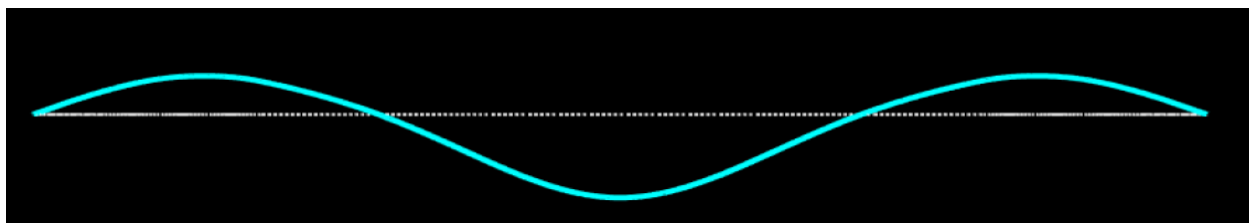
In this section all the deformation diagrams are displayed.



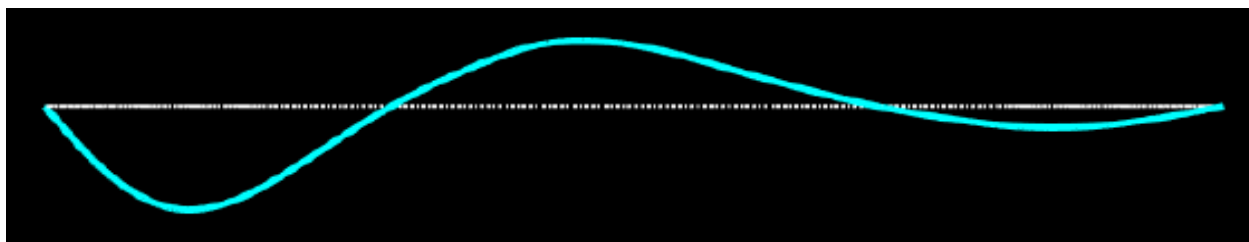
(a) Dead Loads



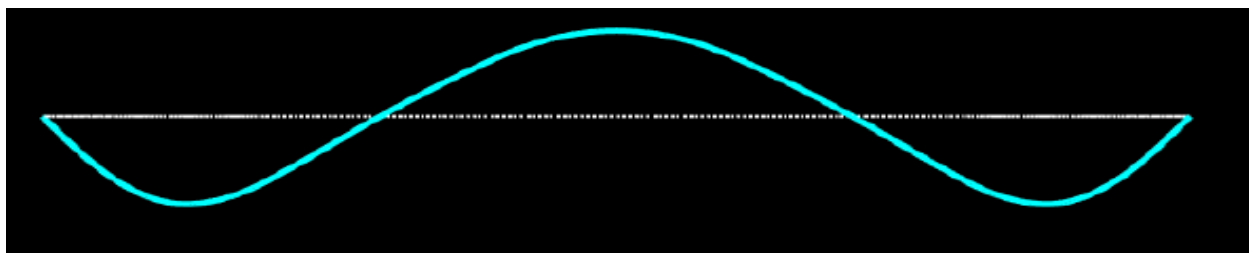
(b) Phase 1A



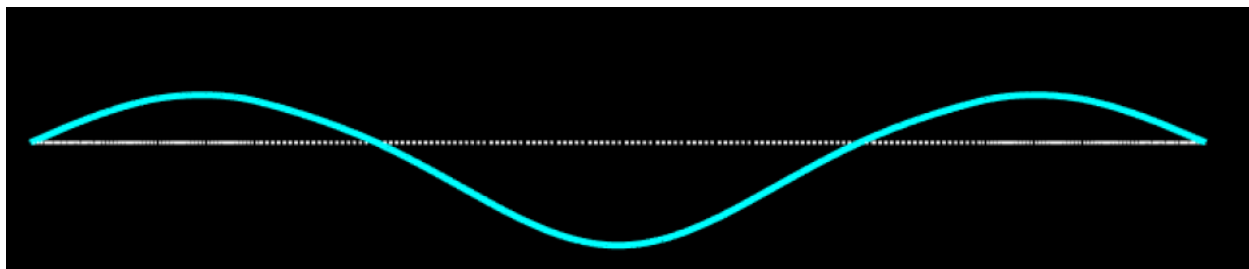
(c) Phase 1B



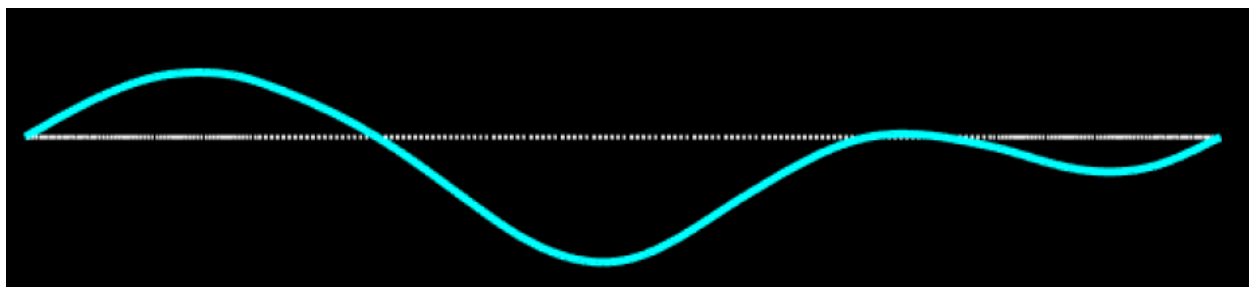
(d) Phase 2A



(e) Phase 2B



(f) Phase 3A



(g) Phase 3B

12.5 Comparison between experimental and numerical results

The main objective of this application is the study of the deflection in the bridge and consequent observations on the difference between observed and expected results from the numerical evaluation

12.5.1 Deflections computed with the design values of the material properties

As mentioned in Section 12.3 the bridge is subject to testing of the displacements at the sections of interest (Figure 12.29). The results are measured through both topographical measurements (Figure 12.14a) and mechanical ones (12.14b).

Cross-Section	19	16	12	8	P1	8'	15'	8"	P2	8"	12"	16"	19"
Abscissa[m]	-146	-129.3	-109.3	-89.3	-60	-30.7	0	30.7	60	89.3	109.3	129.3	146
Phase 1A	0	5	10	9	2	-13	-23	-14	-1	10	11	6	0
Phase 1B	0	13	22	21	1	-28	-50	-30	-1	24	23	13	0
Unloading	0	0	-1	0	-1	2	0	2	-2	0	0	-1	0
Phase 2A	0	-17	-26	-17	1	15	16	7	1	-5	-5	2	0
Phase 2B	0	-21	-32	-24	1	23	33	24	-1	-24	-30	-20	0
Unloading	0	-1	-1	0	1	1	2	2	-1	-1	-1	-1	0
Phase 3A	0	13	22	22	1	-28	-49	-28	-1	20	23	12	0
Phase 3B	0	10	19	17	1	-21	-35	-15	-2	3	-1	-4	0
Unloading	0	-1	0	0	-1	2	1	1	-2	0	0	-1	0

(a) Topographical measurements [mm]

Cross-Section	19	16	12	8	P1	8'	15'	8"	P2	8"	12"	16"	19"
Abscissa[m]	-146	-129.3	-109.3	-89.3	-60	-30.7	0	30.7	60	89.3	109.3	129.3	146
Phase 1A		4.8	10	8.9		-13.9		-11.9		7.7	9.3	5.8	
Phase 1B		11.3	21.6	19.4		-29		-26.9		17.6	20.6	12.2	
Unloading		-1	-0.7	-0.9		0.1		-0.1		-1.5	-0.7	-0.9	
Phase 2A		-15	-23.3	-16.9		12.2		3.2		-3.2	-3.7	-2.2	
Phase 2B		-18.8	-29.9	-22.1		20.4		19.4		-19.4	-25.2	-18.4	
Unloading		-0.9	-1.1	-1		0.5		2		-0.2	0.2	-0.3	
Phase 3A		12.3	22.7	20.3		-26.2		-26.6		18.6	20.4	3.2	
Phase 3B		10.9	29.3	17.2		-44.5		-17.9		4.5	0.8	-2.3	
Unloading		0.9	-2.7	1		-1		-0.5		0.7	-0.6	0.4	

(b) Mechanical measurements [mm]

Table 12.14: Displacement at the interested nodes

Comments:

Comparing these results to the ones obtained in Table 12.13 shows that the design structure is much more flexible than the real bridge. This is due to the factors mentioned in Section 12.1.1, the difference can be ascribed to:

1. Approximation of the cross-sectional properties (varying cross-section);
2. Restrained nodes at the top of the pillars centered in the centroid of the cross section (wrong point of contact)
3. Non-zero values of the displacements after unloading hint at possible plastic deformations or in-situ material settlement (beyond the program's linear theory)
4. Stiffening contribution of reinforcing steel bars and of pre-stressing cables (neglected);
5. Underestimation of the Young's modulus.

Better modelling solutions to combat such problems can be taken:

- To better represent the structure's varying cross-sections and reduce the incidence of problem n. 1 a routine can be implemented that discretizes the length of beams providing linearly varying cross-sections;
- In order to try and represent the rotation and following deflection of the centroid of the sections coinciding with the pillars and avoid problem n. 2, the restraint could be applied to a master node located at the base of the section;
- A correction factor can be multiplied to the design Young's modulus to try to account for the additional stiffness granted by the reinforcing steel.

The last point is expanded upon in the next section.

12.5.2 Error estimate and updated values of material properties

In order to correct the low stiffness considered in the design process a η correction factor can be introduced. Its value can be extrapolated by an analysis of the deflection error and then multiplied to the current design Young's modulus. η is defined as follows:

$$\begin{aligned} E_c^* &= \eta E_c & K^* &= \eta K \\ s^* &= K^{*-1} f = \frac{1}{\eta} K^{-1} f = \frac{1}{\eta} s \end{aligned}$$

The way its value is found is by minimizing the norm of the difference vector between the numerical and experimental results. In such a way we're adjusting the value as closely as possible, a design decision has then to be taken when settling on one single correction factor for all load cases.

$$\min_{\eta > 0} \{ ||\Delta \mathbf{s}|| \} = \min_{\eta > 0} \{ ||\mathbf{s}^*(\eta) - \mathbf{s}_{exp}|| \}$$

To accomplish this minimization problem, the vector of the difference (only topographical results were used) was imported in Mat Lab and the command *fminbnd* was used on its norm. A similar process was done using the solver toolbox in Excel and the results yielded are identical.

Correction factor	MatLab	Excel
η Phase 1A	1.5618	1.5618
η Phase 1B	1.4215	1.4215
η Phase 2A	1.4439	1.4439
η Phase 2B	1.3930	1.3930
η Phase 3A	1.4631	1.4631
η Phase 3B	1.3779	1.3779
$\bar{\eta}$	1.4435	1.4435

Table 12.15: Possible correction factors

The correction factor is therefore selected and the value of the Young's modulus updated.

$$\eta = 1.44 \quad E_c^* = 52910 \text{ MPa}$$

12.5.3 Updated deflection results and comparison

With the newly updated Young's modulus it is possible to go back and compute again the values of the deflections and compare all the results.

Cross-Section	19	16	12	8	P1	8'	15'	8''	P2	8''	12''	16''	19''
Abscissa[m]	-146	-129.3	-109.3	-89.3	-60	-30.7	0	30.7	60	89.3	109.3	129.3	146
Phase 1A	0	6.6	11.5	10.5	0	-14.8	-24.5	-14.8	0	10.5	11.5	6.6	0
Phase 1B	0	13.2	23	21	0	-29.7	-49.1	-29.7	0	21	23	13.2	0
Phase 2A	0	-16.6	-24.9	-18.4	0	14.4	16.3	7.7	0	-5.2	-5.7	-3.2	0
Phase 2B	0	-19.9	-30.6	-23.6	0	22.1	32.6	22	0	-23.5	-30.4	-19.8	0
Phase 3A	0	13.2	23	21	0	-29.7	-49.1	-29.7	0	21	23	13.2	0
Phase 3B	0	9.9	17.4	15.9	0	-22	-	-	0	2.7	-1.7	-3.4	0
							32.8	15.3					

Table 12.16: Displacement at the interested nodes

Before the implementation of the correction factor the errors where as high as 20 mm in the midspan, now they're 10 times lower.

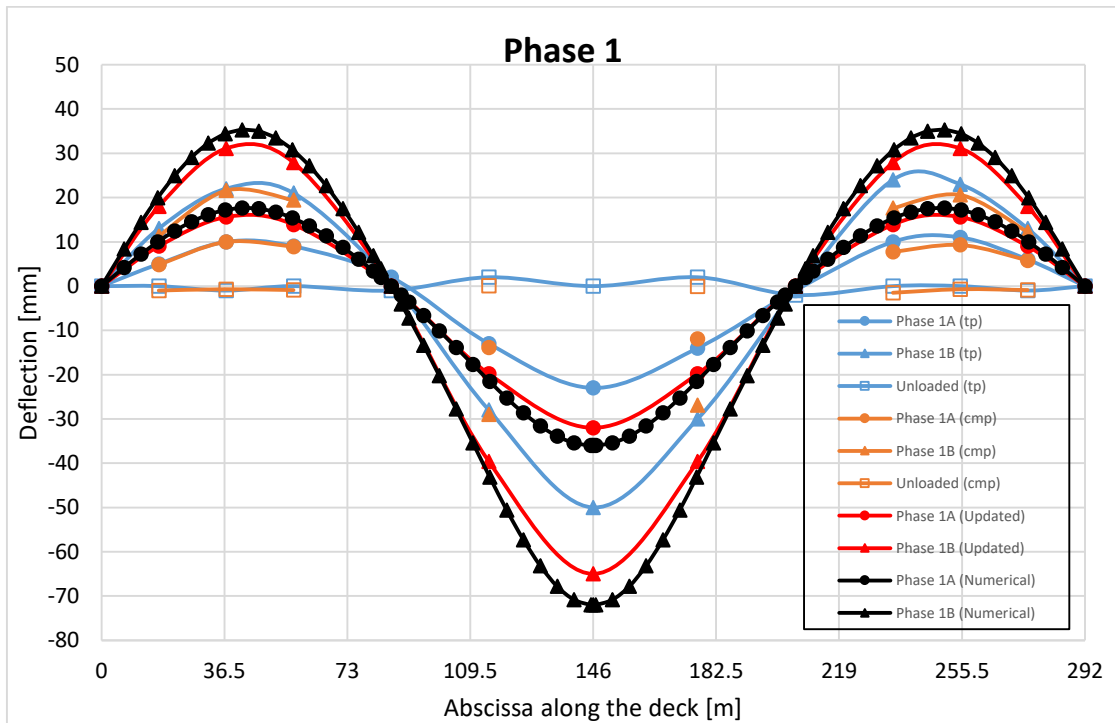


Figure 12.31: Deflection comparison in Phase 1

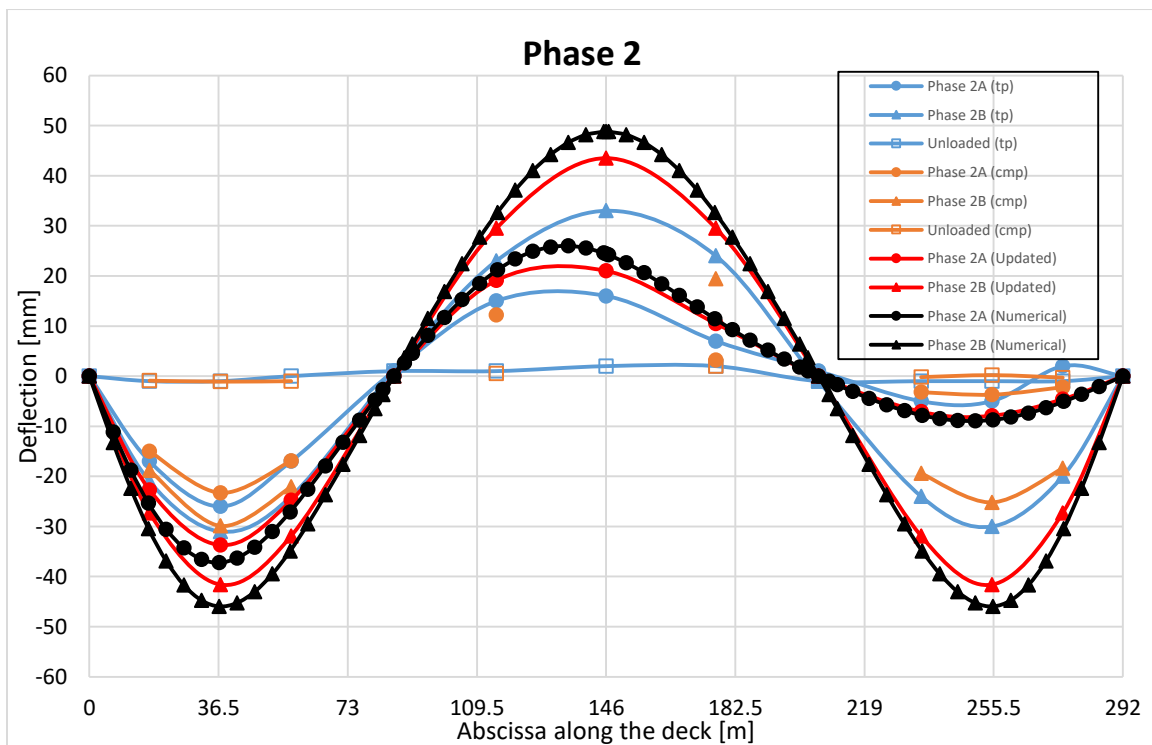


Figure 12.32: Deflection comparison in Phase 2

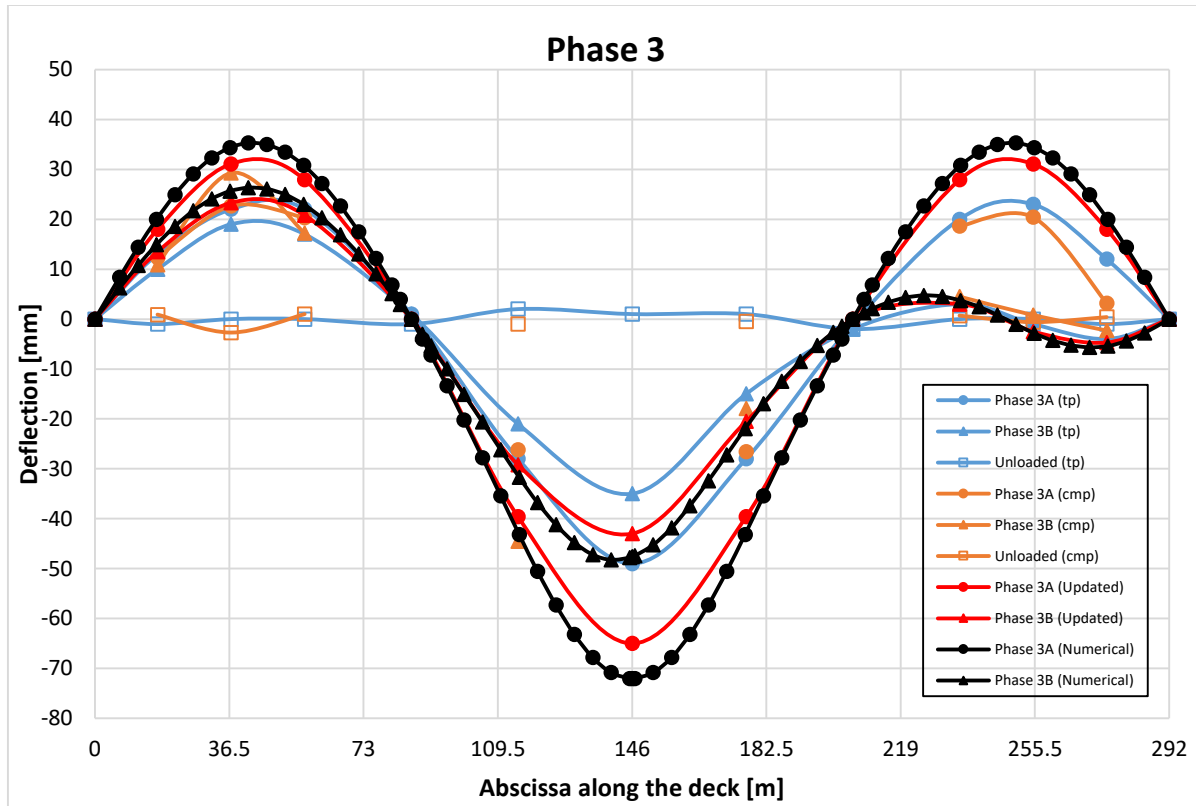
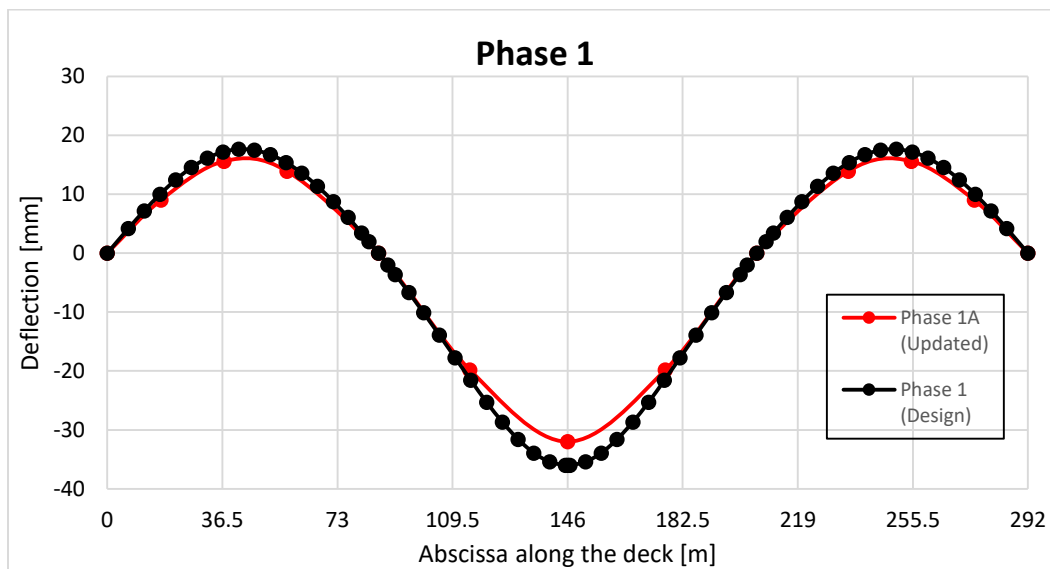


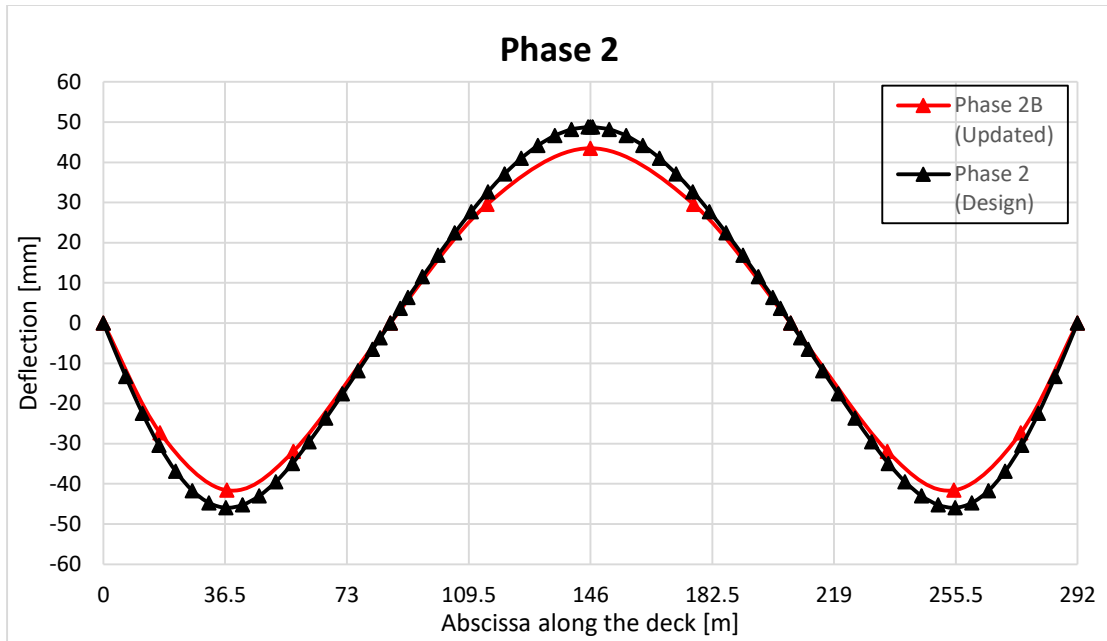
Figure 12.33: Deflection comparison in Phase 3

12.6 Comparison between design and testing loads

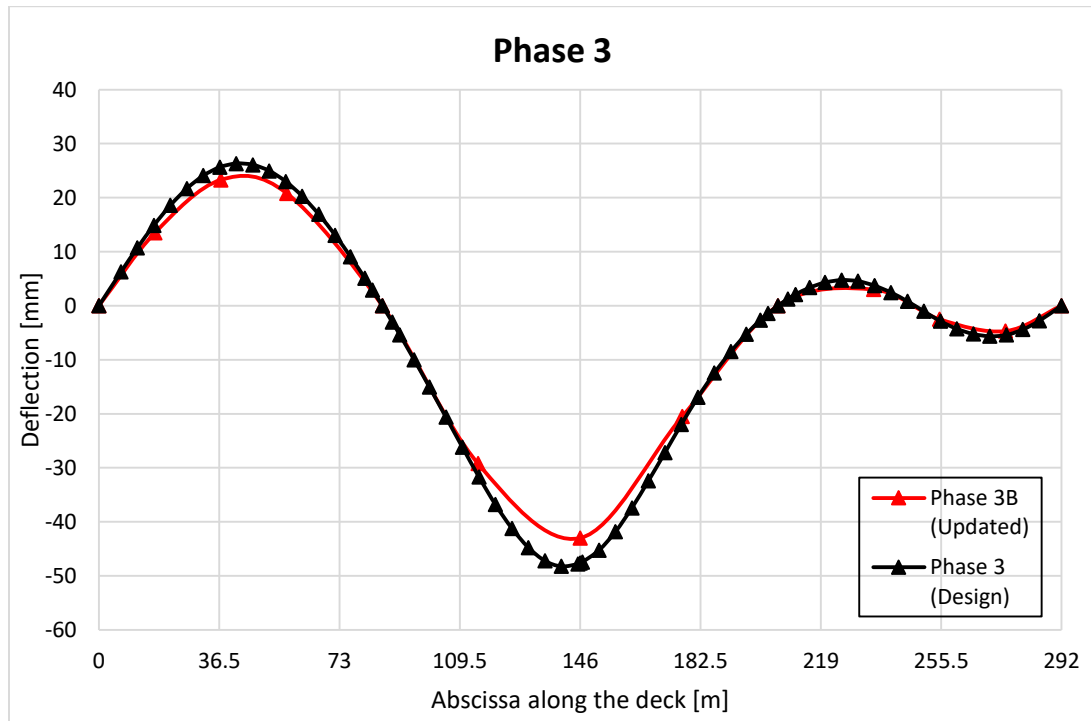
Although the absolute values of the distributed loads in the design stage and the concentrated loads in case B of the testing stage are equal in every phase, their different layout leads to different results in the displacements.



(a) Deflection comparison in Phase 1



(b) Deflection comparison in Phase 2



(c) Deflection comparison in Phase 3

Figure 12.34: Displacements for design and test loads

The testing load case leads to greater displacements every time and the difference can be as high as 10%.

Cross-Section	19	16	12	8	P1	8'	15'	8"	P2	8"	12"	16"	19"
Abscissa[m]	-146	-129.3	-109.3	-89.3	-60	-30.7	0	30.7	60	89.3	109.3	129.3	146
Phase 1A	0	11.9	20.8	19	0	-26.7	-44.3	-26.7	0	19	20.8	11.9	0
Phase 1B	0	13.2	23	21	0	-29.7	-49.1	-29.7	0	21	23	13.2	0
Difference	0	-1.3	-2.2	-2	0	3	4.8	3	0	-2	-2.2	-1.3	0
Phase 2A	0	-18.7	-29	-22.5	0	21.1	31.1	21.1	0	-22.5	-29	-18.7	0
Phase 2B	0	-19.9	-30.6	-23.6	0	22.1	32.6	22	0	-23.5	-30.4	-19.8	0
Difference	0	-1.2	-1.6	-1.1	0	1	1.5	0.9	0	-1	-1.4	-1.1	0
Phase 3A	0	8.8	15.4	14.1	0	-19.4	-28.7	-13	0	1.4	-2.8	-3.7	0
Phase 3B	0	9.9	17.4	15.9	0	-22	-32.8	-15.3	0	2.7	-1.7	-3.4	0
Difference	0	-1.1	-2	-1.8	0	2.6	4.1	2.3	0	-1.3	-1.1	-0.3	0

Table 12.17: Displacement at the interested nodes [mm]

12.7 Comparison among different numerical models

To understand the effect of the design decision it is interesting to evaluate different scenarios and through the code output gauge the importance of these parameters. The relevant results are all summarized in Section 12.7.6, Figure 12.35 and Table 12.18.

12.7.1 Different shear factor

The first comparison is with the evaluation of different shear factors, done against the extreme option of neglecting shear deformation entirely and considering the shear given from an equivalent box section. To do this, the results of the case of dead loads will be compared, the updated value of the Young's modulus will be employed and all other design decision previously discussed will be maintained.

It is evident from the results that neglecting the shear deformation is an important approximation as it underestimates the value of the deflection by about 10% while utilizing a simple approximation as an equivalent box section leads to lower discrepancy.

12.7.2 No offsets

Although the offsets were implemented in the model and every element programmed to account for its effects the results are null due to the bridge being only affected by vertical forces. By the way this structure is designed no axial forces and therefore no bending moments are induced by the geometry. This is because the horizontal movement of the structure is only restrained in 1 point therefore no axial forces can be sustained by the structure. If the model did not take into account any offsets, then the solution wouldn't be any different with additional horizontal restraints, but with the current design, strong moments would be induced that could in some cases be beneficial.

A quick comparison of the displacements shows, in the case of dead loads, a small increase in the central span and a 40% reduction in the side spans. This design decision however cannot be taken

in the numerical modeling phase alone but must be possible through the physical structure of the abutments and pillars.

12.7.3 No rigid ends

The rigid ends constituted only a small part of the structure and are already considered as having more flexural inertia through the section properties. The removal of the rigid ends makes less than 1% difference both in the computation of displacements and internal forces.

12.7.4 Different discretization

In this application the number of elements roughly matches the number of different sections measured so reducing the number of elements would mean losing important information relative to the problem and thus approximating the results significantly.

Reducing the mesh to 12 elements would lead to an overestimation of the displacements of 30% throughout the structure and in some cases even 60% with respect to the design.

Increasing the mesh instead would lead to no significant benefit as the difference in the deflections for a model with 5 times as many elements is less than 1‰.

12.7.5 Exploiting symmetry

For symmetric load cases a case could be made for a lighter mesh that involves half of the elements. Disregarding horizontal displacements, asymmetric due to varying boundary conditions, all other factors can be considered symmetric and therefore the problem could be simplified to just half the structure.

Imposing an additional boundary condition is necessary for this procedure and it has to be implemented on the axis of symmetry, on the mid-span in this case, and it consists on a rotational restraint, as the bridge would have a discontinuous slope otherwise. This approximation leads to no significant error in the displacements as the difference in the nodes of interest is lower than 1‰.

12.7.6 Visual and data comparison

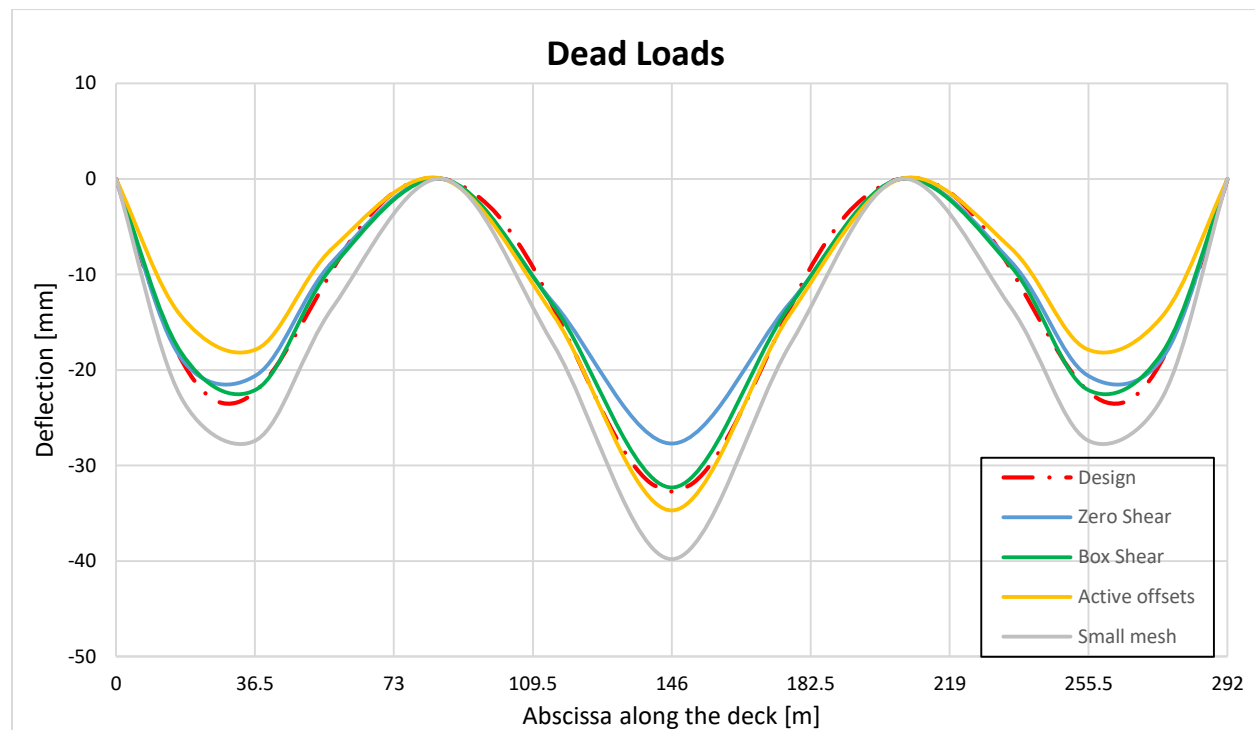


Figure 12.35: Displacements for different numerical models

Cross-Section	19	16	12	8	P1	8'	15'	8"	P2	8"	12"	16"	19"
Abscissa[m]	-146	-129.3	-109.3	-89.3	-60	-30.7	0	30.7	60	89.3	109.3	129.3	146
Design	0	-12	-14.5	-6.2	0	-10	-23.2	-10	0	-6.2	-14.5	-12	0
Zero shear	0	-11.3	-13.6	-5.7	0	-9.1	-21.4	-9.1	0	-5.7	-13.6	-11.3	0
Difference	0	0.7	0.9	0.5	0	0.9	1.8	0.9	0	0.5	0.9	0.7	0
Box shear	0	-11.7	-14.3	-6.1	0	-9.9	-22.8	-9.9	0	-6.1	-14.3	-11.7	0
Difference	0	0.3	0.2	0.1	0	0.1	0.4	0.1	0	0.1	0.2	0.3	0
Active offsets	0	-7.2	-8.8	-2.7	0	-11.4	-24.6	-11.4	0	-2.7	-8.8	-7.2	0
Difference	0	4.8	5.7	3.5	0	-1.4	-1.4	-1.4	0	3.5	5.7	4.8	0
Small mesh	0	-13.7	-17.4	-9.8	0	-12.5	-27.4	-12.5	0	-9.8	-17.4	-13.7	0
Difference	0	-1.7	-2.9	-3.6	0	-2.5	-4.2	-2.5	0	-3.6	-2.9	-1.7	0

Table 12.18: Displacement at the interested nodes [mm]

12.7.7. Improvements and further comments

In structural model additional improvements could be made to the numerical model that could further improve the accuracy:

- More accurate geometric properties: In the end, this was factor that created the greatest discrepancy with the end results. As mentioned previously, one of the inertia values was 89.2% the size with respect to one of the values provided. This was too significant of a deviation. Using a polyline tool on AutoCAD with more points would give it greater accuracy.
- Nodal points directly at the truck loads: Placing the nodal points directly under the trucks would have made the model more accurate, and we would not have had to approximate the nodal loads.
- Dividing the truck loads into separate axle loads: this would make the model even closer to the experimental setup.
- Taking a core of the concrete: Use the exact experimental material parameters

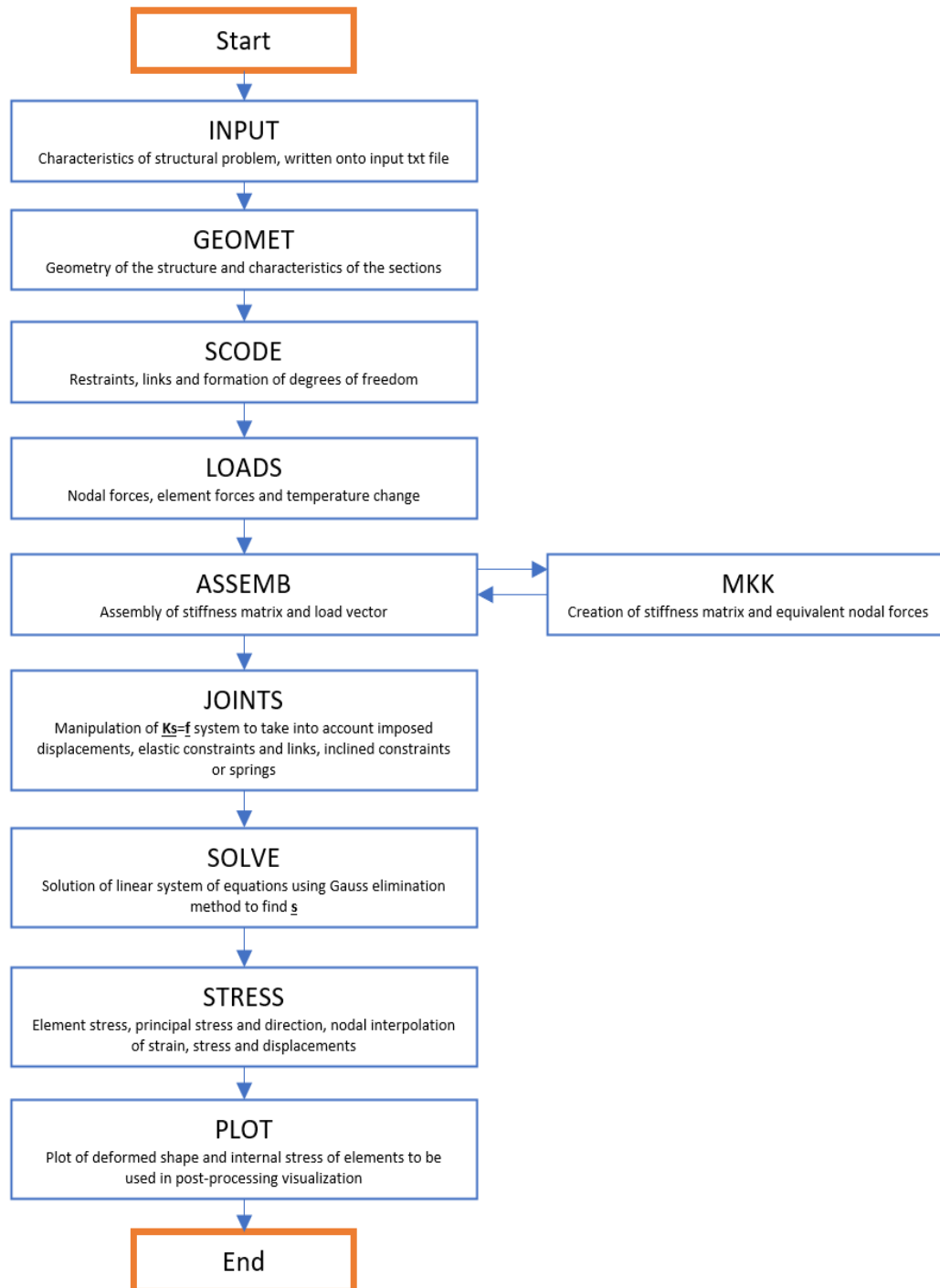
13. References

Biondini, F. (2022). Computational Structural Analysis: Lecture Notes.

Marshal, A. C. (2010). Introduction to Computer Programming using Fortran95.

Annex

A1: Flowchart of the program



A2: User Manual

The user manual presented here serves as a step-by-step guide for generating the results pertaining to the analysis of 2D frame structures. The files required and generated by the program are listed as follows:

- **input.txt:** Contains the description of the structural system
- **output.txt:** Stores the results produced from the analysis
- **plot.dat:** Enables visualisation of the results

The process for developing the input file is comprehensively detailed in the following section.

B.1 Input

The input file provides the program with all the specifications associated to the structural model, and hence plays a critical role in the analysis. To begin with, the user has to specify a title for the problem along with the units of measurement adopted. Consistency in the units of measurements is pivotal for obtaining the correct results. Following this, the focus is laid upon the definition of the physical model and the parameters required are as noted below. It is worth mentioning that for each input line, <·> denotes a new line of input to be provided.

NODES <NNODE> <IDNODE, X, Y>

The user provides the information on the nodes that constitute the structural model. It should be noted that besides the nodes defining the geometry of the structure, additional nodes that are needed to capture specific features from the analysis should be accounted for. Firstly, the total number of nodes are stated and then in subsequent lines, the ID and coordinate of each node is to be specified.

SECTIONS <NSEC> <IDSEC, AREA, INERT IA, DEPTH, SHEAR FACTOR>

The data related to different sections present in the structure is indicated here. The total number of sections are identified first, followed by listing the properties corresponding to a particular section ID. In cases where axial inextensibility is desired, the user should provide a relatively large area for that section. Moreover, if shear deformation is to be neglected, a shear correction factor of 0 should be given as input.

MATERIALS <NMAT> <IDMAT, YOUNG, POISSON, THERMAL, WEIGHT>

The total number of distinct materials to be assigned to the structure are defined first. Then on the next line, the ID representative of each material is allocated, along with the characteristics such as the Young's modulus, the Poisson's ratio, thermal expansion and the self weight can be neglected, if necessary, by providing null values.

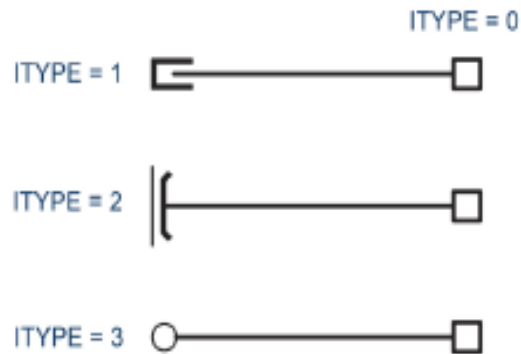
ELEMENTS <NELE> <IDELE, NODE1, NODE2, IDSEC, IDMAT>

The number of elements constituting the structural model are prescribed first. Subsequently, an ID is designated for each element with information on the ID of the two nodes connecting it.

Furthermore, the user now assigns the ID of the section type and the material corresponding to each element.

END RELEASES <NER> <IDELE, ITYPE1, ITYPE2>

The number of end releases to be implemented in the model are noted here. Following this, the user has to mention the ID of the element incorporating the end releases and the type of end release that they wish to employ. The code to be given in the input for various types of end releases are presented in the Figure below.



RIGID ENDS <NRE> <IDELE, D1, D2>

The total number of rigid ends and the element ID pertaining to them are noted here. Besides this, the dimension of the rigid end on either side of each element is provided.

OFFSETS <NOFF> <IDELE, EX1, EY1, EX2, EY2>

The number of offsets recognised in the structural model is specified here with the ID of the element that it is associated to. Moreover, the eccentricities in the x and y direction for either nodes of the element are to be stated in the global reference system.

RESTRAINTS <NRES> <NNODE, DIRECTION>

The total number of restraints imposed in the structure, bearing in mind that each direction to be restricted counts as an individual restraint, is provided here. Next, the restricted node ID and the ID relative to the restricted direction is given.

LINKS <NLINK> <MASTER, SLAVE, DIRECTION>

The number of master slave links connecting certain elements have to be indicated here. Additionally, the ID of the master and the slave node, and the corresponding direction of this link has to be mentioned.

NODAL LOADS <NLOADS> <NODE, DIR, FORCE>

The number of nodal loads that the structure is subjected to is provided first. Then, the associated node ID, direction of load application and its value is stated in the subsequent line. It is worth

noting that the value of the load must be provided in accordance with the sign convention adopted, i.e., positive in the upward direction and the right side.

ELEMENT LOADS <NLOADS> <ELEM, PX, PY1, PY2, DTtop, DTbot>

The total number of element loads acting on the model should be given followed by the element ID in the next line. The code allows for a uniform variation of the element loads in the x direction whereas, even linearly varying loads can be accounted for in the y direction. As a result, the uniform intensity of the applied distribution in the x direction, and the variable (if required) intensity at either ends of the element in the y direction should be specified.

PRESTRESSING LOADS <NPRES> <ELEM, NTEND> <TEND, P, E1, EM, E2>

The number of prestressed elements in the structure and the ID of the associated element, along with the number of tendons composing that element is indicated here. Now, for each tendon an ID is allocated, and the information on the value of the prestressing load and the eccentricities at the begin, the middle, and at the end of the tendon is provided.

ELASTIC RESTRAINTS <NRES> <NODE, DIR, STIFFNESS>

The total number of elastic restraints, such as springs, are given first. Following this, the associated node ID, the direction of the restraint, and the stiffness of the elastic restraint has to be specified.

ELASTIC LINKS <NLINKS> <NODE1, NODE2, DIR, STIFFNESS>

The number of elastic restraints present in the system, along with the ID of the nodes connected to the links, the direction of the connection, and the stiffness value should be noted here.

IMPOSED ABSOLUTE DISPLACEMENT <NADIS> <NODE, DIR, DISPL>

The total number of imposed absolute displacements that the structural system is subjected to is quantified first. Then, the ID of the node that the absolute displacement is imposed on is mentioned, along with the direction, and the value of this displacement.

IMPOSED RELATIVE DISPLACEMENT <NRDIS> <NODE1,NODE2,DIR,DISPL>

The number of imposed relative displacements between two nodes is specified, followed by the associated node IDs, the direction, and the value of the relative displacement.

INCLINED JOINTS <NINCL> <IDNODE,BETA, ITYPE, ADIS>

The total number of inclined joints present in the structural model is noted here. The user has to provide the ID of the associated node, accompanied by information pertaining to the angle of inclination of the constraint, the type of the constraint (0 if elastic and 1 for imposed displacement), and the corresponding value of stiffness or displacement for an elastic constraint and an imposed displacement, respectively.

To illustrate, the input file produced for the benchmark of frame with elastic restraints and temperature change, as discussed in Section 11.3, is presented here.

```

2D FRAME [M] [KN]
NODES <NNODE> <IDNODE, X, Y>
4
1, 0, 0
2, 0, 4
3, 6, 4
4, 6, 0
SECTION <NSEC> <IDSEC, AREA, INERTIA, DEPTH, SHEAR>
2
1, 20000000, 0.01041666, 0.4, 0
2, 20000000, 0.00520830, 0.4, 0
MATERIALS <NMAT> <IDMAT, YOUNG, POISSON, THERMAL, WEIGHT>
1
1, 30000000, 0.2, 0, 0
ELEMENTS <NELE> <IDELE, NODE1, NODE2, IDSEC, IDMAT>
3
1, 1, 2, 2, 1
2, 2, 3, 1, 1
3, 3, 4, 2, 1
END_RELEASES <NER> <IDELE, ITYPE1, ITYPE2>
0
RIGID_ENDS <NRE> <IDELE, D1, D2>
0
OFFSETS <NOFF> <IDELE, EX1, EY1, EX2, EY2>
0
RESTRAINTS <NRES> <NODE, DIR>
6
1, 1
1, 2
1, 3
4, 1
4, 2
4, 3
LINKS <NLINKS> <MASTER, SLAVE, DIR>
0
NODAL LOADS <NLOADS> <NODE, DIRECTION, FORCE>
0
ELEMENT LOADS <NLOADS> <EL, PX, PY1, PY2, DTTOP, DTBOTT>
1
2, 0, -50, -50, 0, 0
PRESTRESSED ELEMENTS <NPRES> <ELEM, NTEND> <TEND, P, E1, Em, E2>
0
ELASTIC RESTRAINTS <NERES> <NODE, DIR, STIFFNESS>
0
ELASTIC LINKS <NELINKS> <NODE1, NODE2, DIRECTION, STIFFNESS>
0
IMPOSED ABSOLUTE DISPLACEMENT <NDISP> <NODE, DIRECTION, DISPLACEMENT>
0
IMPOSED RELATIVE DISPLACEMENT <NCED> <NODE1, NODE2, DIRECTION, DISPL>
0
INCLINED JOINTS <NINCL> <IDNODE, BETA, ITYPE, ADIS>
0

```

A3: Fortran Code

```
module GLOBAL_VAR
implicit double precision (A-H, O-Z)

!GEOMET
integer ::NNODE, NELE, NSEC, NMAT
integer, dimension(:), allocatable,save:: ISEC,IMAT
integer, dimension(:, :), allocatable,save:: IN,ITYPE
real(2), dimension(:, :), allocatable,save:: COORD,CSEC,CMAT, ECC,OFF
real(2), dimension(6,6)::identity

!SCODE
integer::NDOF
integer, dimension(:, :),allocatable,save::IDOF

!LOADS
real(2),dimension(:, :),allocatable,save::ELOADS
real(2),dimension(:, :),allocatable,save::VLOADS

!PRESTRESSING
real(2),dimension(:, :),allocatable,save::PRES

!ASSEMB-MKK
real(2),dimension(:, :),allocatable,save::VK
real(2),dimension(6,6)::ST
real(2),dimension(6)::EQFG

!JOINTS
real(2),dimension(:, :),allocatable,save::CINCL

!SOLVE
Real(2),dimension(:, :),allocatable,save::VDISP
end module GLOBAL_VAR

program FRAME
use GLOBAL_VAR
real(2)::DEE,ETA,SSTAR(9),DELTA(9),SEXP(9)

!OPENING OF INPUT/OUTPUT FILE
character(len=10)::TITLE
write(*,*)"Running program FRAME"
write(*,*)
open(11,file='INPUT_INC.txt',status='UNKNOWN')
open(12,file='OUTPUT_INC.txt',status='UNKNOWN')
open(9,file='MKK.TEMPORARY',status='UNKNOWN')
open(99,file='DEBUG.txt',status='UNKNOWN')
open(13,file='TEXT.txt',status='UNKNOWN')

!CALL THE SUBROUTINES
call GEOMET
call SCODE
```

```

call LOADS
call ASSEMB
call JOINTS
call SOLVE_GAUSS

call STRESS
call plot

close(11)
close(12)
close(9)
close(99)
close(13)
end program FRAME

!SUBROUTINE GEOMET
subroutine GEOMET
use GLOBAL_VAR
integer::IDNODE, IDSEC, IDMAT, IDELE, I, J, IDCODE, NER, ITYPE1, ITYPE2, NRE, NOFF
character(len=80)::REC, TITLE
write(12, '(1X, "2D FRAME ANALYSIS-AUTHOR: HASEEB ALI" )')
write(12, '(1X, "-----" )')
write(12, '(1X, "GEOMETRY/SECTIONS/MATERIALS" )')

!CREATING AN IDENTITY MATRIX
identity=0.
do i=1,6
    identity(i,i)=1.
end do

!READING/WRITING DATA OF NODES
read(11,*) TITLE
read(11,*) REC
read(11,*) NNODE
write(12, '(1X,A30)')TITLE
if(NNODE<2) then
write(12, '(/,1X,"ERROR.UNCORRECTED NO. OF NODES" )')
    stop
    return
end if
write(12, '(/,1X,"NUMBER OF NODES=", 1X, I3)')NNODE
write(12,100) 'NODE', 'COORD X', 'COORD Y'
allocate (COORD(NNODE,2))
do I= 1,NNODE
    read(11,*) IDNODE,COORD(IDNODE,:)
    write(12,101) IDNODE,COORD(IDNODE,:)
end do

!READING/WRITING CROSS-SECTION
read(11,*) REC
read(11,*) NSEC
if(NSEC<1) then
write(12, '(/,1X,"ERROR.UNCORRECTED NO. OF SECTION" )')

```

```

        stop
        return
    end if
    allocate (CSEC(NSEC,4))
    write(12, '(/,1X,"NUMBER OF SECTION=", 1X, I3)')NSEC
    write(12,102) 'SECTION', 'AREA', 'INERTIA', 'DEPTH', 'SHEAR C.F'
    do I= 1,NSEC
        read(11,*) IDSEC,CSEC(IDSEC,:)
        write(12,104) IDSEC,CSEC(IDSEC,:)
!CHECK ON THE INPUT DATA
IDCODE=0
    do J=1,3
        if (CSEC(IDSEC,J)<=0) then
            IDCODE=J
        end if
    end do
    if (CSEC(IDSEC,4)<0) IDCODE=4
    if (IDCODE>0) then
        write(12, '(/,1X,"ERROR IN SECTION PROPERTY",1X,I2)') IDSEC,IDCODE
        stop
        return
    end if
end do

!READING/WRITING DATA OF MATERIALS
read(11,*) REC
read(11,*) NMAT
if (NMAT<1) then
    write(12, '(/,1X,"ERROR.UNCORRECTED NO. OF MATERIALS")')
    stop
    return
end if
allocate (CMAT(NMAT,4))
write(12, '(/,1X,"NUMBER OF MATERIALS=", 1X, I3)')NMAT
write(12,105) 'MATERIAL', 'YOUNG MODULUS', 'POSSION RATIO', 'THERMAL COEFF.', 'SPEC. WEIGHT'
do I= 1,NMAT
    read(11,*) IDMAT,CMAT(IDMAT,:)
    write(12,106) IDMAT,CMAT(IDMAT,:)

!CHECK ON THE INPUT DATA
IDCODE=0
    if (CMAT(IDMAT,1)<=0) IDCODE=1
    if ((CMAT(IDMAT,2)<0) .OR. (CMAT(IDMAT,2)>0.5)) IDCODE=2
    if (CMAT(IDMAT,3)<0) IDCODE=3
    if (CMAT(IDMAT,4)<0) IDCODE=4
    if (IDCODE>0) then
        write(12, '(/,1X,"ERROR IN MATERIAL PROPERTY",1X,I2)') IDMAT,IDCODE
        stop
        return
    end if
end do

```

```

!READING/Writing DATA OF ELEMENTS CONECTIVITY

read(11,*) REC
read(11,*) NELE
if (NELE<1) then
    write(12, '(/,1X,"ERROR.UNCORRECTED NO. OF ELEMNETS")')
    stop
    return
end if

allocate (IN(NELE,2))
allocate (ISEC(NELE))
allocate (IMAT(NELE))
allocate (ITYPE(NELE,2))
allocate (ECC(NELE,2))
allocate (OFF(NELE,4))
write(12, '(/,1X,"NUMBER OF ELEMENTS=", 1X, I3)')NELE
write(12,107) 'ELEMENT', 'NODE 1', 'NODE 2', 'IDSEC', 'IDMAT'

do I= 1,NELE
    read(11,*) IDELE, (IN(IDELE,J), J=1,2), ISEC(IDELE), IMAT(IDELE)
    write(12,108) IDELE, (IN(IDELE,J), J=1,2), ISEC(IDELE), IMAT(IDELE)
end do

! INITIALIZATION OF ITYPE MATRIX FOR END RELEASES
ITYPE(:,:)=0.
! READING NO. OF END RELEASES
read (11 ,*) REC
read (11 ,*) NER
! UPDATING ITYPE MATRIX
if(NER>0) then
do I =1,NER
read(11,*) IDELE, ITYPE1, ITYPE2
ITYPE(IDELE,1)=ITYPE1
ITYPE(IDELE,2)=ITYPE2
end do
end if

! INITIALIZATION OF ECCMATRIX FOR RIGID ENDS
ECC(:,:)=0.
! READING LENGTHS OF RIGID ENDS
read (11,*)REC
read (11,*)NRE
! UPDATING THE ECC MATRIX
if(NRE>0) then
do i=1,NRE
read (11,*) IDELE,D1,D2
ECC (IDELE,1)=D1
ECC (IDELE,2)=D2
end do
end if

! INITIALIZATION OF OFF. MATRIX FOR OFFSETS
OFF (:,:)=0.

```



```

! READING ECCENTRICITIES OF THE OFFSETS AND JOINTS
read (11,*)REC
read (11,*)NOFF
! UPDATING THE OFF MATRIX
if(NOFF>0) then
do i=1,NOFF
read (11,*) IDELE ,EX1,EY1,EX2,EY2
OFF (IDELE,1)=EX1
OFF (IDELE,2)=EY1
OFF (IDELE,3)=EX2
OFF (IDELE,4)=EY2
write (99 , '(I2,2X,F3.1,2X,F3.1,2X,F3.1,2X,F3.1)') IDELE,EX1,EY1,EX2,EY2
end do
end if
return

!FORMAT LIST
100 format(T1,A4,T18,A9,T32,A9)
101 format(I4,T10,F10.1,T24,F10.1)
102 format(T1,A7,T18,A4,T32,A7,T49,A5,T58,A9)
104 format(I4,T14,F10.4,T29,F10.6,T43,F10.1,T53,F10.1)
105 format(T1,A8,T23,A13,T46,A13,T64,A13,T80,A12)
106 format(I4,T23,E10.3,T48,F5.2,T64,F10.6,T78,F10.1)
107 format(T1,A7,T18,A6,T36,A6,T54,A5,T68,A5)
108 format(I2,T18,I2,T37,I2,T55,I2,T68,I2)

end subroutine GEOMET

!SUBROUTINE SCORE
subroutine SCORE
use GLOBAL_VAR
integer::I,J,NRES,NLINKS,IDNODE,JDIR
integer::NMAST,NSLAVE
character(len=80)::REC
allocate (IDOF(NNODE,3))

!INITIALIZATION IDOF
IDOF(:,:)=0
write(12,' (/ ,1X,"-----") ')
write(12,' (/ ,"RESTRAINTS/LINKS") ')

!READING AND WRITING OF RESTRAINTS
read(11,*) REC
read(11,*) NRES
write(12,' (/ ,1X,"NUMBER OF RESTRAINTS=",1X,I3) ')NRES
write(12,101) 'NODE','DIRECTION'
do I=1,NRES
read(11,*) IDNODE,JDIR
write(12,102) IDNODE,JDIR
write(99,*) IDNODE
IDOF(IDNODE,JDIR)=-1
end do

```

```

!READING AND WRITING OF LINKS
read(11,*)REC
read(11,*)NLINKS
if(NLINK<0) then
    write(12,'(1X,"DATA ERROR IN NO. OF LINKS")')
    stop
    return
end if
write(12,'(/,1X,"NUMBER OF LINKS=",1X,I3)')NLINKS
if(NLINKS>0) then
    write(12,103)'MASTER','SLAVE','DIRECTION'
    do I=1,NLINKS
        read(11,*)NMAST,NSLAVE,JDIR
        write(12,104)NMAST,NSLAVE,JDIR
        IDOF(NSLAVE,JDIR)=NMAST
    end do
end if

! NUMBERING OF DOF's
NDOF=0
write(12,'(/,1X,"DOF MATRIX")')
write(12,105)'NODE','Ux','Uy','ROTz'

EXT: do I=1,NNODE
    INT: do J=1,3

        if(IDOF(I,J)<0) cycle INT !RESTRAINT
        if(IDOF(I,J)==0) then !DOF
            NDOF=NDOF+1
            IDOF(I,J)=NDOF
        else !LINK
            IDNODE=IDOF(I,J)
            IDOF(I,J)=IDOF(IDNODE,J)
        end if

    end do INT

    !WRITING DOF MATRIX
    write(12,106)I,IDOF(I,:)
end do EXT
return !END OF THE SUBROUTINE GOES BACK TO MAIN PROGRAM

!FORMAT LIST
101 format(T1,A4,T18,A9)
102 format(T1,I1,T21,I1)
103 format(T1,A6,T18,A5,T32,A9)
104 format(T1,I1,T19,I1,T36,I1)
105 format(T1,A4,T18,A2,T28,A2,T38,A4)
106 format(T1,I2,T18,I2,T28,I2,T40,I2)
end subroutine SCODE

!SUBROUTINE LOADS
subroutine LOADS
use GLOBAL_VAR
integer::I,J,NNL,IDNODE,JDIR,JDOF,NEL,IDELE,NEP,INTEND,NTEND
real(2)::NFORCE,ELOAD(5),PRE(4),FP,E1,EM,E2

```

```

character(len=80)::REC
write(12,'(/,1X"-----")')
write(12,'(1X,"NODAL AND ELEMENT LOADS")')
allocate (VLOADS(NDOF))
allocate (ELOADS(NELE,5))
allocate (PRES(NELE,4))
VLOADS(:)=0.
ELOADS(:,:)=0.
PRES(:,:)=0.

!READING/WRITING NODAL LOADS
read(11,*)REC
read(11,*)NNL
if (NNL<0) then
    write(12,'(1X,"ERROR IN NUMBER OF NODAL LOADS")')
    stop
    return
end if
write(12,'(/,1X,"NUMBER OF NODAL LOADS=",1X,I3)')NNL
if(NNL>0) then
    write(12,101)'NODE','DIR','FORCE'
    do I=1,NNL
        read(11,*) IDNODE,JDIR,NFORCE
        if ((IDNODE<1).OR.(IDNODE>NNODE)) then
            write(12,'(1X,"NODAL LOADS APPLIED ON A WRONG LOCATION")')
            stop
            return
        end if
        if ((JDIR<1).OR.(JDIR>3)) then
            write(12,'(1X,"DIRECTION OF NODAL LOAD IS INCORRECT")')
            stop
            return
        end if
        write(12,102) IDNODE,JDIR,NFORCE
        JDOF=IDOF(IDNODE,JDIR)
        VLOADS(JDOF)=VLOADS(JDOF)+NFORCE
    end do
    write(12,'(/,1X,"VECTOR OF VLOADS=NODAL LOADS")')
    write(12,103)'DOF','FORCE'
do I=1,NDOF
    write(12,104) I,VLOADS(I)
end do
end if

!READING/WRITING FOR ELEMENTS LOADS
read(11,*)REC
read(11,*)NEL
if(NEL<0) then
    write(12,'(1X,"NUMBER OF ELEMENT LOAD IS INCORRECT")')
    stop
    return
end if
write(12,'(/,1X,"NUMBER OF ELEMENTS LOADS=",1X,I3,/)')NEL

```

```

if (NEL>0) then
write (12,105) 'ELEMENT', 'Px', 'Py1', 'Py2', 'DT-Top', 'DT-Bot'
do I=1,NEL
    read(11,*) IDELE,ELOAD
    if ((IDELE<1).OR.(IDELE>NELE)) then
        write(12,'(1X,"NUMBER OF LOADED ELEMENTS IS INCORRECT")')
        stop
        return
    end if
    write(12,106) IDELE,ELOAD
    ELOADS(IDELE,1:5)=ELOADS(IDELE,1:5)+ELOAD(1:5)
end do
end if

!READING/WRITING FOR PRESTRESSING LOADS
read(11,*) REC
read(11,*) NEP
if (NEP<0) then
    write(12,'("INCORRECT NUMBER OF PRESTRESSED ELEMENTS")')
    stop
    return
end if
write(12,'(/,1X,"-----")')
write(12,'(1X,"PRESTRESSING")')
write(12,'(/,1X,"NUMBER OF PRESTRESSED ELEMENTS=",1X,I3)' ) NEP
if (NEP>0) then
do I=1,NEP
    read(11,*) IDELE,NTEND
    if ((IDELE<1).OR.(IDELE>NELE)) then
        write(12,'("ERROR IN THE NUMBER OF PRESTRESSED
ELEMENT")')

        stop
        return
    end if
    FP=0.
    E1=0.
    EM=0.
    E2=0.
    write(12,'(/,1X,"ELEM=",I3)' ) IDELE
    write(12,107) 'TENDON', 'P', 'E1', 'EM', 'E2'
    do J=1,NTEND
        read(11,*) IDTEND,PRE(1:4)
        write(12,108) IDTEND,PRE(1:4)
        FP=FP+PRE(1)
        E1=E1+PRE(1)*PRE(2)
        E2=E2+PRE(1)*PRE(4)
        EM=EM+PRE(1)*PRE(3)
    end do
    PRES(IDELE,1)=FP
    PRES(IDELE,2)=E1/FP
    PRES(IDELE,3)=EM/FP
    PRES(IDELE,4)=E2/FP
    write(12,'(/,1X,"EQUIVALENT TENDON")')

```

```

                                write(12,109) 'EQV','FP','E1','EM','E2'
                                write(12,110) PRES(IDELE,1:4)
                                end do
                                end if
return
!FORMAT LIST
101 format(T1,A4,T18,A3,T30,A5)
102 format(T1,I1,T19,I1,T27,F10.1)
103 format(T1,A3,T25,A5)
104 format(T1,I1,T19,F10.1)
105 format(T1,A7,T18,A2,T28,A3,T38,A3,T48,A6,T58,A6)
106 format(T1,I2,T11,F10.1,T22,F10.1,T34,F10.2,T43,F10.1,T53,F10.1)
107 format(T1,A6,T18,A1,T28,A2,T38,A2,T48,A2)
108 format(T1,I2,T11,F10.4,T22,F10.4,T34,F10.4,T43,F10.4)
109 format(T1,A3,T18,A2,T28,A2,T38,A2,T48,A2)
110 format(T1,I2,T11,F10.4,T22,F10.4,T34,F10.4,T43,F10.4)
end subroutine LOADS

!subroutine ASSEMB
subroutine ASSEMB
use GLOBAL_VAR
integer::I,J,NCODE(6),NE,N1,N2,IC,JC
allocate(VK(NDOF,NDOF))
VK(:,:)=0
do NE=1,NELE
    call MKK(NE)
    N1= IN(NE,1)
    N2= IN(NE,2)
    NCODE(1:3)=IDOF(N1,1:3)
    NCODE(4:6)=IDOF(N2,1:3)

!ASSEMBLY PROCEDURE
    EXT: do I=1,6
        IC=NCODE(I)
        if(IC<0) cycle EXT
        VLOADS(IC)=VLOADS(IC)+EQFG(I)
        INT: do J=1,6
            JC=NCODE(J)
            if(JC<0) cycle INT
            VK(IC,JC)=VK(IC,JC)+ST(I,J)
        end do INT
    end do EXT
end do

write(12,' (/ ,1X,"-----") ')
write(12,' (1X,"VECTOR OF EQUIVALENT LOADS",/) ')
write(12,101) 'DOF','FORCE'
write(12,102) (I,VLOADS(I),I=1,NDOF)
write(12,' (/ ,1X,"-----") ')
write(12,' (1X,"STIFFNESS MATRIX") ')
do J=1,NDOF
    write(12,' (/ ,1X,"ROW",T16,"COLUMN",I3) ') J
do I=1,NDOF

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```

                                write(12,103) I,VK(I,J)
                                end do
                                end do

!FORMATS
101 FORMAT(1X,A3,7X,A5)
102 FORMAT(1X,I2,5X,F8.2)
103 FORMAT(1X,I2,10X,F19.2)
return
end subroutine ASSEMB

!SUBROUTINE MKK
subroutine MKK(NE)
use GLOBAL_VAR
integer::I,J,N1,N2,ISC,IMT,NE
real(2)::DX,DY,AL,CA,SA,AA,AJ,AH,CHI,EMOD,POISSON,ALPHA,WEIGHT,GMOD,PHI,PX,PY1,PY2,DTTOP,DTBOT,
TMP(6,6),T(6,6),EQF(6),QQ,FH,FM,TE(6,6)
real(2)::PG,COEF_A,COEF_B,COEF_C,DELTA_TM,DELTA_TD,FP,E1,EM,E2,TETA_1,TETA_2,CURV,WW,KCV(5),KC(
1,5),KBS(5,5),KB(5,5),FB(5),FBS(5),TD(6,6)
character(len=80)::REC

! GEOMETRIC PROPERTIES
N1=IN(NE,1)
N2=IN(NE,2)
DX=(COORD(N2,1)-COORD(N1,1))
DY=(COORD(N2,2)-COORD(N1,2))
AL=sqrt(DX**2+DY**2)-ECC(NE,1)-ECC(NE,2) !FLEXURAL ELEMENT LENGTH
CA=DX/(AL+ECC(NE,1)+ECC(NE,2)) !COSINE
SA=DY/(AL+ECC(NE,1)+ECC(NE,2)) !SINE

! SECTION AND MATERIALS
ISC=ISEC(NE) !SECTION PROPERTIES OF ELEMENT
IMT=IMAT(NE) !MATERIAL PROPERTIES OF ELEMENT
AA=CSEC(ISC,1) !AREA
AJ=CSEC(ISC,2) !MOMENT OF INERTIA
AH=CSEC(ISC,3) !HEIGHT
CHI=CSEC(ISC,4) !SHEAR FACTOR
EMOD=CMAT(IMT,1) !YOUNG'S MODULUS
POISSON=CMAT(IMT,2) !POISSON'S RATIO
ALPHA=CMAT(IMT,3)
WEIGHT=CMAT(IMT,4)
GMOD=EMOD/(2.*(1.+POISSON)) !SHEAR MODULUS
PHI=12.*CHI*EMOD*AJ/(AL**2.*GMOD*AA) !SHEAR DEFORMABILITY COEFFICIENT

!LOCAL STIFFNESS MATRIX
ST(:, :)=0. !INITIALIZATION
ST(1,1)=EMOD*(AA/AL) !AXIAL
ST(2,2)=12.*EMOD*AJ/((1.+PHI)*AL**3.) !SHEAR
ST(3,2)=6.*EMOD*AJ/((1.+PHI)*AL**2.)
ST(3,3)=(4.+PHI)*EMOD*AJ/((1.+PHI)*AL) !MOMENT
ST(4,1)=-ST(1,1)
ST(4,4)=ST(1,1)
ST(5,2)=-ST(2,2)

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ST(5,3)=-ST(3,2)
ST(5,5)=ST(2,2)
ST(6,2)=ST(3,2)
ST(6,3)=(2.-PHI)*EMOD*AJ/( (1.+PHI)*AL)
ST(6,5)=-ST(3,2)
ST(6,6)=ST(3,3)
!COPY THE ELEMENTS IN THE UPPER HALF(SYMMETRY)
do I=1,6
    do J=I+1,6
        ST(I,J)=ST(J,I)
    end do
end do
write(9,*)ST(:, :)

!TRANSFORMATION MATRIX SHIFTED TO THE END

!INITIALIZATION OF EQUIVALENT NODAL FORCE VECTOR
EQF(:)=0.

!DISTRIBUTED LOADS
PX=ELOADS(NE,1) !CONSTANT AXIAL DISTRIBUTED LOAD
PY1=ELOADS(NE,2) !CONSTANT VERTICAL DISTRIBUTED LOAD VALUE NODE 1
PY2=ELOADS(NE,3) !CONSTANT VERTICAL DISTRIBUTED LOAD VALUE NODE 2
DTTOP=ELOADS(NE,4) !THERMAL LOAD TOP FIBER
DTBOT=ELOADS(NE,5) !THERMAL LOAD BOTTOM FIBER

!SELF-WEIGHT
PG= WEIGHT*AA !WEIGHT PER LENGTH
PX= PX-PG*SA !DISTRIBUTED SELF WEIGHT ON AXIS
PY1= PY1-PG*CA !DISTRIBUTED SELF WEIGHT TRANSVERSAL
PY2= PY2-PG*CA !DISTRIBUTED SELF WEIGHT TRANSVERSAL
QQ= PY2-PY1 !FOR TRIANGULAR LOADS

!PRESTRESSING
FP=PRES(NE,1) !PRESTRESSING FORCE
E1=PRES(NE,2) !ECCENTRICITY FIRST NODE
EM=PRES(NE,3) !ECCENTRICITY MIDPOINT
E2=PRES(NE,4) !ECCENTRICITY SECOND NODE
TETA_1=-1./AL*(3.*E1+E2-4.*EM) !THETA NODE 1
TETA_2=1./AL*(E1+3.*E2-4.*EM) !THETA NODE 2
CURV=4.*(E1+E2-2.*EM)/(AL**2.) !CURVATURE
WW=FP*CURV !CONSTANT UDL DUE TO CURVATURE OF TENDON
PY1=PY1+WW
PY2=PY2+WW

!COMPUTATIONAL OF LONGITUDNAL LOAD
EQF(1)=EQF(1)+PX*AL/2. !EQUIVALENT AXIAL LOAD NODE 1
EQF(4)=EQF(4)+PX*AL/2. !EQUIVALENT AXIAL LOAD NODE 2

!COMPUTATIONAL OF TRANSVERSAL LOAD (CONSTANT PART)
EQF(2)=EQF(2)+PY1*AL/2. !EQUIVALENT TRANSVERSAL LOAD NODE 1
EQF(3)=EQF(3)+PY1*AL**2./12. !EQUIVALENT BENDING MOMENT NODE 1
EQF(5)=EQF(5)+PY1*AL/2. !EQUIVALENT TRANSVERSAL LOAD NODE 2

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EQF(6)=EQF(6)-PY1*AL**2./12. !EQUIVALENT BENDING MOMENT NODE 2

!COMPUTATIONAL OF TRANSVERSAL LOAD (LINEAR PART)
COEF_A=AL/(12.*EMOD*AJ)
COEF_B=CHI/(AL*AA*GMOD)
COEF_C=COEF_B/(COEF_A+COEF_B)
EQF(2)=EQF(2)+9./60.*QQ*AL*(1.+COEF_C/9.)
EQF(3)=EQF(3)+1./30.*QQ*AL**2.*(1.+COEF_C/4.)
EQF(5)=EQF(5)+21./60.*QQ*AL*(1.-COEF_C/21.)
EQF(6)=EQF(6)-1./20.*QQ*AL**2.*(1.-COEF_C/6.)

!COMPUTATION OF THERMAL LOAD (ELONGATION)
DELTA_TM=(DTTOP+DTBOT)/2.
FH=EMOD*AA*ALPHA*DELTA_TM
EQF(1)=EQF(1)-FH
EQF(4)=EQF(4)+FH

!COMPUTATION OF THERMAL LOAD (FLEXURAL BEHAVIOUR)
DELTA_TD=(DTTOP-DTBOT)/2.
FM=2.*ALPHA*DELTA_TD*EMOD*AJ/AH !THERMAL COEFFICIENT FOR FLEXURAL
EQF(3)=EQF(3)+FM !CORRECTION FOR THERMAL LOAD NODE 1
EQF(6)=EQF(6)-FM !CORRECTION FOR THERMAL LOAD NODE 2
!SAVE INFORMATION
write(9,*)EQF(:)

!PRESTRESSING LOADS
EQF(1)=EQF(1)+FP !CORRECTION FOR PRESTRESSING FORCE AXIAL NODE 1
EQF(2)=EQF(2)+FP*TETA_1 !CORRECTION FOR PRESTRESSING FORCE TRANSVERSAL NODE 1
EQF(3)=EQF(3)-FP*E1 !CORRECTION FOR PRESTRESSING FORCE MOMENT NODE 1
EQF(4)=EQF(4)-FP !CORRECTION FOR PRESTRESSING FORCE AXIAL NODE 2
EQF(5)=EQF(5)-FP*TETA_2 !CORRECTION FOR PRESTRESSING FORCE TRANSVERSAL NODE 2
EQF(6)=EQF(6)+FP*E2 !CORRECTION FOR PRESTRESSING FORCE TRANSVERSAL NODE 2

!END RELEASE UPDATE TO STIFFNESS
if(ITYPE (NE,1)>0)then
    select case (ITYPE(NE,1))
    case (1)
        KC(1,:)=ST(1,2:6)
        KB(:, :)=ST(2:6,2:6)
        KBS=KB-matmul(transpose(KC),KC)/ST(1,1)
        FB(:)=EQF(2:6)
        KCV(:)=KC(1,1:5)
        FBS=FB-KCV*EQF(1)/ST(1,1)
        ST(:, :)=0.
        ST(2:6,2:6)=KBS(:, :)
        EQF(:)=0.
        EQF(2:6)=FBS(:)
    case (2)
        KC(1,1)=ST(2,1)
        KC(1,2:5)=ST(2,3:6)
        KB(1,1)=ST(1,1)
        KB(1,2:5)=ST(1,3:6)
        KB(2:5,1)=ST(3:6,1)

```



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KB(2:5,2:5)=ST(3:6,3:6)
KBS=KB-matmul(transpose(KC),KC)/ST(2,2)
FB(1)=EQF(1)
FB(2:5)=EQF(3:6)
KCV(:)=KC(1,1:5)
FBS=FB-KCV*EQF(2)/ST(2,2)
ST(:,:)=0.
ST(1,1)=KBS(1,1)
ST(1,3:6)=KBS(1,2:5)
ST(3:6,1)=KBS(2:5,1)
ST(3:6,3:6)=KBS(2:5,2:5)
EQF(:)=0.
EQF(1)=FBS(1)
EQF(3:6)=FBS(2:5)

case (3)
KC(1,1:2)=ST(3,1:2)
KC(1,3:5)=ST(3,4:6)
KB(1:2,1:2)=ST(1:2,1:2)
KB(1:2,3:5)=ST(1:2,4:6)
KB(3:5,1:2)=ST(4:6,1:2)
KB(3:5,3:5)=ST(4:6,4:6)
KBS=KB-matmul(transpose(KC),KC)/ST(3,3)
FB(1:2)=EQF(1:2)
FB(3:5)=EQF(4:6)
KCV(:)=KC(1,1:5)
FBS=FB-KCV*EQF(3)/ST(3,3)
ST(:,:)=0.
ST(1:2,1:2)=KBS(1:2,1:2)
ST(1:2,4:6)=KBS(1:2,3:5)
ST(4:6,1:2)=KBS(3:5,1:2)
ST(4:6,4:6)=KBS(3:5,3:5)
EQF(:)=0.
EQF(1:2)=FBS(1:2)
EQF(4:6)=FBS(3:5)

case default
WRITE(12, '(/,2X,"INCORRECT END RELEASE TYPE")')
end select

end if
if(ITYPE(NE,2)>0) then
select case(ITYPE(NE,2))
case (1)
KC(1,1:3)=ST(4,1:3)
KC(1,4:5)=ST(4,5:6)
KB(1:3,1:3)=ST(1:3,1:3)
KB(1:3,4:5)=ST(1:3,5:6)
KB(4:5,1:3)=ST(5:6,1:3)
KB(4:5,4:5)=ST(5:6,5:6)
KBS=KB-matmul(transpose(KC),KC)/ST(4,4)
FB(1:3)=EQF(1:3)
FB(4:5)=EQF(5:6)
KCV(:)=KC(1,1:5)
FBS=FB-KCV*EQF(4)/ST(4,4)
ST(:,:)=0.

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        ST(1:3,1:3)=KBS(1:3,1:3)
        ST(1:3,5:6)=KBS(1:3,4:5)
        ST(5:6,1:3)=KBS(4:5,1:3)
        ST(5:6,5:6)=KBS(4:5,4:5)
        EQF(:)=0.
        EQF(1:3)=FBS(1:3)
        EQF(5:6)=FBS(4:5)

    case (2)
        KC(1,5)=ST(5,6)
        KC(1,1:4)=ST(5,1:4)
        KB(5,5)=ST(6,6)
        KB(5,1:4)=ST(6,1:4)
        KB(1:4,5)=ST(1:4,6)
        KB(1:4,1:4)=ST(1:4,1:4)
        KBS=KB-matmul(transpose(KC),KC)/ST(5,5)
        FB(5)=EQF(6)
        FB(1:4)=EQF(1:4)
        KCV(:)=KC(1,1:5)
        FBS=FB-KCV*EQF(5)/ST(5,5)
        ST(:, :)=0.
        ST(6,6)=KBS(5,5)
        ST(6,1:4)=KBS(5,1:4)
        ST(1:4,6)=KBS(1:4,5)
        ST(1:4,1:4)=KBS(1:4,1:4)
        EQF(:)=0.
        EQF(6)=FBS(5)
        EQF(1:4)=FBS(1:4)

    case (3)
        KC(1,:)=ST(6,1:5)
        KB(:, :)=ST(1:5,1:5)
        KBS=KB-matmul(transpose(KC),KC)/ST(6,6)
        FB(:)=EQF(1:5)
        KCV(:)=KC(1,1:5)
        FBS=FB-KCV*EQF(6)/ST(6,6)
        ST(:, :)=0.
        ST(1:5,1:5)=KBS(:, :)
        EQF(:)=0.
        EQF(1:5)=FBS(:)

    case default
        write(12, '(/,2X," INCORRECT END RELEASE TYPE") ')

    end select

end if

!RIGID ENDS

if((ECC(NE,1)>0).OR.(ECC(NE,2)>0)) then !TO REDUCE COMPUTATIONAL EFFORT
    TD=IDENTITY !INITIALIZATION OF TRANSFORMATION MATRIX
    TD(2,3)=ECC(NE,1)
    TD(5,6)=-ECC(NE,2)
    EQF=matmul(transpose(TD),EQF) !NEW EQUIVALENT NODAL LOAD VECTOR
    TMP=matmul(transpose(TD),ST)
    ST=matmul(TMP,TD) !NEW STIFFNESS MATRIX

end if

!SAVE INFO. IN TEMPORARY FILE
write(9,*)ST(:, :)

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!TRANSFORMATION MATRIX
T(:, :)=0.
T(1,1)=ca
T(1,2)=-sa
T(2,1)=sa
T(2,2)=ca
T(3,3)=1.
T(4,4)=ca
T(4,5)=-sa
T(5,4)=sa
T(5,5)=ca
T(6,6)=1.
!SAVE INFORMATION
write(9,*)T(:, :)
!STIFFNESS MATRIX IN THE GLOBAL REFERENCE FRAME (TKT')
TMP=matmul(T,ST)           !MATRIX MULTIPLICATION
ST=matmul(TMP,transpose(T)) !LOCAL STIFFNESS IN GLOBAL FRAME
!TRANSFORMATION T*f'
EQFG=matmul(T,EQF)
!RIGID JOINTS WITH OFFSETS
if((OFF(NE,1)==0).AND.(OFF(NE,2)==0).AND.(OFF(NE,3)==0).AND.(OFF(NE,4)==0)) then !TO REDUCE
COMPUTATIONAL EFFORT SKIP
else
    TE=IDENTITY           !INITIALIZATION OF TRANSFORMATION MATRIX
    TE(1,3)=-OFF(NE,2)
    TE(2,3)=OFF(NE,1)
    TE(4,6)=-OFF(NE,4)
    TE(5,6)=OFF(NE,3)
    EQFG=matmul(transpose(TE),EQFG) !NEW EQUIVALENT NODAL LOAD VECTOR
    TMP=matmul(transpose(TE),ST)
    ST=matmul(TMP,TE)           !NEW STIFFNESS MATRIX
end if
return
end subroutine MKK

!SUBROUTINE JOINNTS
subroutine JOINTS
use GLOBAL_VAR
integer:: I, J, K, JDIR, JDOF, JDOF1, JDOF2
integer:: IDNODE, N1, N2
integer:: NELR, NELL, NTDIS, NRDIS, NINCL
real(2):: STIFF, ADIS, VKJJ
character(LEN=80):: REC
write(12, '(/,1X,"-----")')
write(12, '(1X,"ELASTIC RESTRAINTS-LINKS")')

!ELASTIC RESTRAINTS
read(11,*) REC
read(11,*) NELR
if(NELR<0) then
write(12, '(1X,"ERROR IN NUMBER OF ELEASTIC RESTRAINTS")')
stop
return

```

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end if
write(12, '(/, 1X, "NUMBER OF ELASTIC RESTRAINTS =", 1X, I3)') NELR
if(NELR>0) then
write(12, 101) 'NODE', 'DIRECTION', 'STIFFNESS'
do I=1,NELR
    read(11,*) IDNODE, JDIR, STIFF
    write(12, 102) IDNODE, JDIR, STIFF
    JDOF = IDOF(IDNODE,JDIR)
    VK(JDOF,JDOF) = VK(JDOF,JDOF) + STIFF
end do
end if

! ELASTIC LINKS
read(11,*) REC
read(11,*) NELL
if(NELL<0) then
    write(12, '(1X, "ERROR IN THE NUMBER OF ELASTIC LINKS")')
    stop
    return
end if
write(12, '(/, 1X, "NUMBER OF ELASTIC LINKS =", 1X, I3)') NELL
if(NELL>0) then
write(12, 103) 'N1', 'N2', 'DIRECTION', 'STIFFNESS'
do I=1,NELL
    read(11,*) N1, N2, JDIR, STIFF
    write(12, 104) N1, N2, JDIR, STIFF
    JDOF1 = IDOF(N1,JDIR)
    JDOF2 = IDOF(N2,JDIR)
    VK(JDOF1,JDOF1) = VK(JDOF1,JDOF1) + STIFF
    VK(JDOF2,JDOF2) = VK(JDOF2,JDOF2) + STIFF
    VK(JDOF1,JDOF2) = VK(JDOF1,JDOF2) - STIFF
    VK(JDOF2,JDOF1) = VK(JDOF1,JDOF2)
end do
end if

! IMPOSED ABSOLUTE DISPLACEMENTS
read(11,*) REC
read(11,*) NTDIS
if(NTDIS<0) then
    write(12, '(1X, "ERROR IN THE NUMBER OF IMPOSED DISPLC.")')
    stop
    return
end if

write(12, '(/, 1X, "-----")')
write(12, '(1X, "IMPOSED DISPLACEMENTS")')
write(12, '(/, 1X, "NUMBER OF IMPOSED ABSOLUTE DISPLACEMENTS =", 1X, I3)') NTDIS
if (NTDIS > 0) then
write(12, 105) 'NODE', 'DIRECTION', 'DISPLAC.'
do I=1,NTDIS
    read(11,*) IDNODE, JDIR, ADIS
    write(12, 106) IDNODE, JDIR, ADIS
    JDOF = IDOF(IDNODE,JDIR)
    VKJJ = VK(JDOF,JDOF)

```

```

        VK(JDOF,1:NDOF) = 0.
        VLOADS(1:NDOF) = VLOADS(1:NDOF) - VK(1:NDOF,JDOF) * ADIS
        VK(1:NDOF,JDOF) = 0.
        VK(JDOF,JDOF) = VKJJ
        VLOADS(JDOF) = ADIS * VKJJ
    end do
end if

! IMPOSED RELATIVE DISPLACEMENTS
read(11,*) REC
read(11,*) NRDIS
if(NRDIS<0) then
    write(12,'(1X,"ERROR IN THE NUMBER OF RELATIVE IMPOSED DISPLC.")')
    stop
    return
end if
write(12,'(/, 1X, "NUMBER OF RELATIVE DISPLACEMENTS =", 1X, I3)') NRDIS
if (NRDIS > 0) then
    write(12, 107) 'N1', 'N2', 'DIRECTION', 'DISPLAC.'
    do I=1,NRDIS
        read(11,*) N1, N2, JDIR, ADIS
        write(12, 108) N1, N2, JDIR, ADIS
        if((N1<1).OR.(N2>NNODE)) then
            write(12,'(1X,"ERROR IN THE NODE IDENTIFIER - RELATIVE
DISPLACEMENT")')
            stop
            return
        end if
        if((N2<1).OR.(N1>NNODE)) then
            write(12,'(1X,"ERROR IN THE NODE IDENTIFIER - RELATIVE
DISPLACEMENT")')
            stop
            return
        end if
        if(N1==N2) then
            write(12,'(1X,"ERROR IN THE NODE IDENTIFIER - RELATIVE
DISPLACEMENT")')
            stop
            return
        end if
        if((JDIR<1).OR.(JDIR>3)) then
            write(12,'(1X,"ERROR IN THE DIRECTION IDENTIFIER - RELATIVE
DISPLACEMENT")')
            stop
            return
        end if

        JDOF1 = IDOF(N1,JDIR)
        JDOF2 = IDOF(N2,JDIR)
        if((JDOF1== -1).OR.(JDOF2== -1)) then
            write(12,'(1X,"ERROR IN THE PLACEMENT OF DISPLAC. - RELATIVE
DISPLACEMENT")')
            stop

```

```

        return

    end if
    VKJJ = VK(JDOF1,JDOF1) + VK(JDOF2,JDOF2)
    do K=1,NDOF
        VK(JDOF1,K) = VK(JDOF1,K) + VK(JDOF2,K)
    end do
    VLOADS(JDOF1) = VLOADS(JDOF1) + VLOADS(JDOF2)
    do K=1,NDOF
        VK(JDOF2,K) = 0.
        VK(K,JDOF1) = VK(K,JDOF1) + VK(K,JDOF2)
        VLOADS(K) = VLOADS(K) - VK(K,JDOF2) * ADIS
        VK(K,JDOF2) = 0.
    end do
    VK(JDOF1,JDOF1) = VK(JDOF1,JDOF1) + VKJJ
    VK(JDOF1,JDOF2) = -VKJJ
    VK(JDOF2,JDOF1) = -VKJJ
    VK(JDOF2,JDOF2) = VKJJ
    VLOADS(JDOF1) = VLOADS(JDOF1) - VKJJ * ADIS
    VLOADS(JDOF2) = VKJJ * ADIS
end do

end if

! INCLINED CONSTRAINTS
allocate(CINCL(NNODE))

! INITIALIZATION OF INCLINED CONSTRAINTS VECTOR
CINCL(:)=0.

! READING INCLINED CONSTRAINTS / IMPOSED DISPLACEMENTS
ONE=1.
PIGR2=asin(ONE)
read(11,*)REC
read(11,*)NINCL
if(NINCL>0) then
    do NC =1,NINCL
        read(11,*) IDNODE,BETA,ITYP,ADIS
        ! ITYP =0 -> ELASTIC RESTRAINTS
        ! ITYP =1 -> IMPOSED DISPLACEMENT
        if(ITYP==0) STIFF=ADIS
            I=IDOF(IDNODE,1)
            J=IDOF(IDNODE,2)
            BETA=BETA*PIGR2/90.
            CINCL(IDNODE)=BETA
            SB=sin(BETA)
            CB=cos(BETA)

! UPDATE OF STIFFNESS MATRIX
            VKII=VK(I,I)
            VKIJ=VK(I,J)
            VKJJ=VK(J,J)
EXT3: do NR=1,NDOF
            if(NR==I) then
                VK(I,I)=VKII*CB**2.+2.*VKIJ*SB*CB+VKJJ*SB**2.
                VK(I,J)=-VKII*SB*CB+VKIJ*(CB**2.-SB**2.)+VKJJ*SB*CB

```

```

                cycle EXT3
            end if
            if (NR==j) then
                VK(J,I)=-VKII*SB*CB+VKIJ*(CB**2.-SB**2.)+VKJJ*SB*CB
                VK(J,J)=VKII*SB**2.-2.*VKIJ*SB*CB+VKJJ*CB**2.
            cycle EXT3
            end if

            VKRI=VK(NR,I)
            VKRJ=VK(NR,J)
            VK(NR,I)=VKRI*CB+VKRJ*SB
            VK(I,NR)=VK(NR,I)
            VK(NR,J)=-VKRI*SB+VKRJ*CB
            VK(J,NR)=VK(NR,J)

        end do EXT3
! UPDATE OF LOAD VECTOR
        VLI=VLOADS(I)
        VLJ=VLOADS(J)
        VLOADS(I)=VLI*CB+VLJ*SB
        VLOADS(J)=-VLI*SB+VLJ*CB
! APPLICATION OF RESTRAINTS / IMPOSED DISPLACEMENTS
        if (ITYP==0) then
! ELASTIC RESTRAINTS
                VK(I,I)=VK(I,I)+STIFF
            else
! IMPOSED DISPLACEMENTS
                VKJJ=VK(I,I)
                do K=1,NDOF
                    VK(I,K)=0.
                    VLOADS(K)=VLOADS(K)-VK(K,I)*ADIS
                    VK(K,I)=0.
                end do
                VK(I,I)=VKJJ
                VLOADS(I)=VKJJ*ADIS
            end if
        end do
    end if
    return

!FORMAT LIST
101 FORMAT(3( A9, 4X))
102 FORMAT(T1, I9, T14, I9, T27, ES9.2)
103 FORMAT(4( A9, 4X))
104 FORMAT(T1, I9, T14, I9, T27, I9, T40, ES9.2)
105 FORMAT(3( A9, 4X))
106 FORMAT(T1, I9, T14, I9, T27, ES9.2)
107 FORMAT(4( A9, 4X))
108 FORMAT(T1, I9, T14, I9, T27, I9, T40, ES9.2)
end subroutine JOINTS

!SUBROUTINE SOLVE
subroutine SOLVE_GAUSS
!GAUSS ELIMINATION SOLVER
use GLOBAL_VAR

```

```

integer::I,J,K,JDOF
real(2)::DNOD(3),DMX(3),DMN(3),PVAL,COND(NDOF,NDOF),KA(NDOF),KB(NDOF),DELTA(NDOF),M
allocate(VDISP(NDOF))

! PRINT OF STIFFNESS MATRIX
if(ndof<11) then
    write(12,'(/,1X,"STIFFNESS MATRIX")')
    do I=1,NDOF
        write(12,'(10(2X,F12.2))') VK(I,:)
    end do
end if

! CONDITION OF THE STIFFNESS MATRIX
COND(:,:)=0.          ! INITIALIZATION OF THE D MATRIX
do I =1,NDOF
    KA(I)=VK(I,I)      ! DIAGONAL ith TERM OF THE ORIGINAL MATRIX
    COND(I,I)=1./sqrt(VK(I,I)) ! EDITING THE DIAGONAL OF THE PRECONDITIONER
end do
VLOADS=matmul(COND,VLOADS) ! CONDITIONING OF THE NODAL LOAD VECTOR
VK=matmul(COND,VK)
VK=matmul(VK,COND)        ! CONDITIONING OF THE STIFFNESS MATRIX

! INITIALIZATION OF THE VECTOR OF THE CONSTANT TERM
! LOAD VECTOR -> DISPLACEMENT VECTOR
VDISP=VLOADS

!GAUSSIAN ELIMINATION METHOD
do I=1,NDOF
    KB(I)=VK(I,I) !DIAGONAL ith TERM BEFORE THE ELIMINATION OF DOF
    DELTA(I)=KA(I)/KB(I) !VECTOR TO STORE ALL VALUE OF M
    do K=I+1,NDOF
        VK(I,K)=VK(I,K)/VK(I,I)
    end do
    VDISP(I)=VDISP(I)/VK(I,I)
    VK(I,I)=1.
    do J=I+1,NDOF
        do K=I+1,NDOF
            VK(J,K)=VK(J,K)-VK(J,I)*VK(I,K)
        end do
        VDISP(J)=VDISP(J)-VK(J,I)*VDISP(I)
        VK(J,I)=0.
    end do
end do

M=log10(maxval(DELTA)) ! ESTIMATE OF THE MAXIMUM NUMBER OF SIGNIFICANT DIDGIT THAT COULD BE
LOST DUE TO ILL-CONDITIONING
write(12,'(/,1X,"CONDITION NUMBER=",F4.1') M

!BACKWARD SUBSTITUTION
do I=NDOF,1,-1
    do K=I+1,NDOF
        VDISP(I)=VDISP(I)-VK(I,K)*VDISP(K)
        write(9,*) VDISP(I)
    end do
end do
end do

```



```

! RECONDITIONING OF THE DISPLACEMENT VECTOR
VDISP=matmul(COND,VDISP)

! INCLINED JOINTS STUFF
EXT: do ND=1,NNODE
BETA=CINCL(ND)
if(BETA==0.) cycle ext
I=IDOF(ND,1)
J=IDOF(ND,2)
SB=sin(BETA)
CB=cos(BETA)
VDI=VDISP(I)
VDJ=VDISP(J)
VDISP(I)=VDI*CB-VDJ*SB
VDISP(J)=VDI*SB+VDJ*CB
end do EXT

!WRITING NODAL DISPLACEMENT
write(12, '(/,1X,"-----") ')
write(12, '(1X,"-----") ')
write(12, '(/,1X,"SUMMARY OF THE RESULTS") ')
write(12, '(1X,"-----") ')
write(12, '(1X,"-----") ')
write(12, '(/, "NODAL DISPLACEMENT") ')
write(12,101) 'NODE', 'UX', 'UY', 'ROTZ'
DNOD=0. !INITIALIZATION VECTOR OF NODAL DISPLACEMENT
DMX=-1.E6 !INITIALIZATION OF HIGHEST VALUE
DMN=1.E6 !INITIALIZATION OF THE LOWEST VALUE
do I=1,NNODE
do J=1,3
if(IDOF(I,J)<=0) then !IDENTIFYING RESTRAINTS
DNOD(J)=0. !ADJSUTING FOR RESTRAINTS
else
JDOF=IDOF(I,J)
DNOD(J)=VDISP(JDOF) !UNRESTRAINTS DOF's
end if
PVAL=DMX(J) !IMPORT THE HIGHEST VALUE TO COMPARE WITH CURRENT
DMX(J)=MAX(PVAL,DNOD(J)) !COMPARE FOR HIGHEST VALUE IN NODE I
PVAL=DMN(J) !IMPORT THE LOWEST VALUE TO COMPARE WITH CURRENT
DMN(J)=MIN(PVAL,DNOD(J)) !COMPARE FOR LOWEST VALUE IN NODE I
end do
write(12,102) I, (DNOD(J), J=1,3)
end do
write(12, '(1X,"-----") ')
write(12,103) 'MAX', (DMX(J), J=1,3)
write(12,103) 'MIN', (DMN(J), J=1,3)
write(12, '(1X,"-----") ')
write(12, '(/,1X,"DISPLACEMENTS AT DEGREE OF FREEDOM",/)' )
write(12,104) 'DOF', 'DISPLACEMENT'
do I=1,NDOF
write(12,105) I, VDISP(I)
end do

```

```

return

!FORMAT LIST
101 FORMAT(T1,A4,T15,A2,T37,A2,T49,A4,T63)
102 FORMAT(T1,I2,T11,F10.4,T32,F10.4,T42,F10.4)
103 FORMAT(A3,F15.4,F15.4,F15.4)
104 FORMAT(T1,A3,T15,A12)
105 FORMAT(I1,F25.6)
end subroutine SOLVE_GAUSS

!SUBROUTINE STRESS
subroutine STRESS
use GLOBAL_VAR
integer:: I,J,JDOF,NCODE(6)
real(2):: QS(6),T(6,6),EQF(6),QF(6),QSL(6),TD(6,6),TE(6,6)
rewind(9)
write(12,'(/,1X,"-----")')
write(12,'(/,1X,"FORCE AT THE END NODES",/)' )
write(12,100)'ELEMENT','N1','T1','M1','N2','T2','M2'

!LOOP FOR ALL ELEMENTS
do NE=1,NELE
    N1=IN(NE,1)
    N2=IN(NE,2)
    NCODE(1:3)=IDOF(N1,1:3)
    NCODE(4:6)=IDOF(N2,1:3)
    read(9,*) EQF
    read(9,*) ST
    read(9,*) T
! NODAL DISPLACEMENTS
do I=1,6
    JDOF=NCODE(I)
    if(JDOF==1) then
        QS(I)=0. !FOR RESTRAINTS
    else
        QS(I)=VDISP(JDOF) !FOR UNRESTRAINED
    end if
end do

! RIGID JOINTS WITH OFFSETS STUFF
if((OFF(NE,1)==0).and.(OFF(NE,2)==0).and.(OFF(NE,3)==0).and.(OFF(NE,4)==0)) then !TO REDUCE
COMPUTATIONAL EFFORT SKIP
else
TE=IDENTITY !INITIALIZATION OF TRASNFORMATION MATRIX
TE(1,3)=-OFF(NE,2)
TE(2,3)=OFF(NE,1)
TE(4,6)=-OFF(NE,4)
TE(5,6)=OFF(NE,3)
QS=matmul(TE,QS) !NEW ELEMENT DISPLACEMENT VECTOR IN GLOBAL REFERNCE FRAME
end if
! TRANSFORMATION
write(99,'(6(F8.5,2X),/)' ) QS
do I =1,6
write(99,'(6(F8.5,2X))') T(I,:)

```

```

end do
QSL= matmul(transpose(T),QS) !DISPLACEMENT IN THE LOCAL REFERENCE FRAME
write(99,'(/,6(f8.5,2X),//)') QSL
! RIGID END STUFF
if((ECC(NE,1)>0).or.(ECC(NE,2)>0)) then !TO REDUCE COMPUTATIONAL EFFORT
TD=IDENTITY ! INITIALIZATION OF TRANSFORMATION MATRIX
TD(2,3)=ECC(NE,1)
TD(5,6)=-ECC(NE,2)
QSL=matmul(TD,QSL) !NEW LOCAL DISPLACEMENT VECTOR
end if
write(99,'(/,6(f8.5,2X),//)') QSL
! WRITE THE END FORCES
do I=1,6
    QF(I)=-EQF(I)
    do J=1,6
        QF(I)=QF(I)+ST(I,J)*QSL(J)
    end do
write(99,'(/,6(f8.5,2X),//)') QF
end do
    write(12,101)NE, (QF(J),J=1,6)
end do
return

!FORMAT LIST
100 FORMAT(T1,A7,T19,A2,T33,A2,T48,A2,T62,A2,T76,A2,T90,A2)
101 FORMAT(T1,I2,T11,F10.1,T32,F10.1,T42,F10.1,T56,F10.1,T68,F10.1,T74,F10.1)
end subroutine STRESS

subroutine PLOT
use GLOBAL_VAR
implicit double precision (A-H,O-Z)
integer::I,J,JDOF,NCODE(6)
real(2)::T(6,6),EQF(6),QF(6),QS(6),QSL(6),TD(6,6),TE(6,6)

!REWIND OF THE TEMPORARY FILE
rewind(9)
!OPENING FILE 33(INPUT DATA FOR PLOT.exe)
open(33,file='PLOT_B2.dat',status='UNKNOWN')
!WRITING OF COORDINATES AND INCIDENCE
write(33,*)NNODE,NELE
do I=1,NNODE
    write(33,*) (COORD(I,J),J=1,2)
end do
do NE=1,NELE
    write(33,*) (IN(NE,J),J=1,2)
end do
    !LOOP FOR ALL THE ELEMENTS
do NE=1,NELE
    N1=IN(NE,1)
    N2=IN(NE,2)
    NCODE(1:3)=IDOF(N1,1:3)
    NCODE(4:6)=IDOF(N2,1:3)
    !READING OF THE STIFFNESS MATRIX

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```

    read(9,*) ST
    !READING OF THE TRANSFORMATION MATRIX
    read(9,*) T
    !READING OF THE EQUIVALENT NODAL FORCE
    read(9,*) EQF
    !NODAL DISPLACEMENTS IN THE GLOBAL REF. SYSTEM
    do I=1,6
        JDOF=NCODE(I)
        if(JDOF==1) then
            QS(I)=0.
        else
            QS(I)=VDISP(JDOF)
        end if
    end do
    !NODAL DISPLACEMENTS IN THE LOCAL REF. SYSTEM
    QSL=matmul(transpose(T),QS)
    !FORCES AT THE END NODES
    do I=1,6
        QF(I)=-EQF(I)
        do j=1,6
            QF(I)=QF(I)+ST(I,J)*QSL(J)
        end do
    end do
    !DISPLACEMENTS AND FORCES ALONG THE LENGTH OF THE ELEMENT
    DX=COORD(N2,1)-COORD(N1,1)
    DY=COORD(N2,2)-COORD(N1,2)
    AL=sqrt(DX**2.+DY**2.)
    CA=DX/AL
    SA=DY/AL
    ISC=ISEC(NE)
    IMT=IMAT(NE)
    AA=CSEC(ISC,1)
    WEIGHT=CMAT(IMT,4)
    PX=ELOADS(NE,1)
    PY1=ELOADS(NE,2)
    PY2=ELOADS(NE,3)
    PG=AA*WEIGHT
    PX=PX-PG*SA
    PY1=PY1-PG*CA
    PY2=PY2-PG*CA
    FP=PREs(NE,1)
    E1=PREs(NE,2)
    EM=PREs(NE,3)
    E2=PREs(NE,4)
    CURV=4.*(E1+E2-2.*EM)/(AL**2.)
    WW=FP*CURV
    PY1=PY1+WW
    PY2=PY2+WW
    do I=0,20
        XX=AL*dble(I)/20.
        PY=PY1+(PY2-PY1)*XX/AL
        UU=U(QSL,XX,AL)
        VV=V(QSL,XX,AL)
    end do

```

```

        SSN=-QF(1)-PX*XX
        SST=QF(2)+(PY1+PY)*XX/2.
        SSM=-QF(3)+QF(2)*XX+PY1*XX**2./2.+(PY-PY1)*XX**2./6.
        write(33,*)UU,VV,SSN,SST,SSM
    end do
end do
close(33)
return
end subroutine PLOT

double precision function F(I,AL,X)
implicit double precision (A-H,O-Z)
CSI=X/AL
select case (I)
    case(1)
        F=1.-CSI
    case(4)
        F=CSI
    case(2)
        F=1.-3.*CSI**2.+2.*CSI**3.
    case(3)
        F=AL*(CSI-2.*CSI**2.+CSI**3.)
    case(5)
        F=3.*CSI**2.-2.*CSI**3.
    case(6)
        F=AL*(CSI**3.-CSI**2.)
end select
end function F

double precision function U(QSL,X,AL)
implicit double precision (A-H,O-Z)
dimension::QSL(6)
U=0.
do I=1,4,3
    U=U+F(I,AL,X)*QSL(I)
end do
return
end function U

double precision function V(QSL,X,AL)
implicit double precision (A-H,O-Z)
dimension::QSL(6)
V=0.
do I=2,3
    V=V+F(I,AL,X)*QSL(I)
end do
do I=5,6
    V=V+F(I,AL,X)*QSL(I)
end do
return
end function V

```

